

A survey of trust-region radius update mechanisms.

Part III: a numerical study

Jérémy Rieussec Fabian Bastin

University of Montreal, Dept. Computing Science and Operational Research

Email: jeremy.rieussec@umontreal.ca, bastin@iro.umontreal.ca

16th August 2026

Abstract

Parts I and II of this survey developed a unified theory of trust-region radius update mechanisms: Part I isolated three structural conditions on the radius sequence that are sufficient for first-order global convergence, and Part II characterised the asymptotic behaviour of the radius itself, the local regime near a nondegenerate minimiser, and the conditions under which the trust-region constraint eventually becomes inactive. Both parts are worst-case analyses. They tell us which mechanisms *can* fail and under which hypotheses they *must* succeed, but they say little about what happens on problems of practical interest, about how the theoretical thresholds behave when instantiated with real constants, or about how the radius rule interacts with the other design choices in a trust-region code—the model Hessian and the subproblem solver.

This third part is an experimental companion. We formulate the qualitative predictions of Parts I and II as testable claims, and design a campaign of seven experiments to probe them: the influence of the criticality parameter ζ in the DFO-like rule, the influence of the cap $\bar{\mu}$ in the gradient-scaled rule, a like-for-like comparison of the mechanisms on the CUTEst collection, a direct measurement of trust-region inactivity, the effect of the mechanism on the observed local rate, the interaction between the mechanism and the Hessian approximation, and an extension to noisy and subsampled settings. We report complete results on a family of analytic test problems designed to separate the mechanisms, and we specify the protocol for the large-scale portion of the campaign. A recurring finding on the analytic problems is that the radius rule governs the *length* of the step and hence the speed of the method, while the limit point itself is decided by the model Hessian; the mechanisms separate sharply only in regimes where the radius is the binding constraint.

Contents

1	Introduction	3
1.1	Motivation	3
1.2	Contributions	4
1.3	Notation and reproducibility	4
2	Background: the mechanisms and what Parts I and II establish	5
2.1	The algorithmic framework	5
2.2	The five mechanisms	5
2.3	Summary of Part I: first-order convergence	6
2.4	Summary of Part II: radius asymptotics and the local regime	6
2.5	The claims under test	7

3	Related work	8
3.1	Methodology of comparison	8
3.2	Test collections	8
3.3	Trust-region methods and adaptive radius rules	8
3.4	Noisy and stochastic trust-region methods	9
4	The instrument	9
4.1	Four axes, varied independently	9
4.2	What is recorded	10
4.3	Derived quantities	11
4.4	Reproducibility	12
5	Protocol	12
5.1	Parameters, held fixed	12
5.2	Stopping and failure	12
5.3	What each claim is measured by	12
6	Analytic test problems	12
6.1	P1: a quartic with four critical points	12
6.2	P2: a nonsingular saddle with tunable escape rate	14
6.3	P3: a family of minimisers with vanishing $\lambda_{\min}^*$	17
6.4	P4: a set of thirty classical problems	18
7	Results	18
7.1	The radius regimes	18
7.2	Eventual inactivity	19
7.3	Inactivity of the trust region	20
7.4	The thresholds	20
7.5	Influence of the criticality parameter ζ	21
7.6	Influence of the cap $\bar{\mu}$	21
7.7	The rate after inactivity	23
7.8	Influence of the mechanism on the rate of convergence	24
7.9	Second order: the criticality measure at a saddle	24
7.10	Mechanism against model	25
7.11	Interaction between mechanism and Hessian approximation	25
7.12	Comparison of mechanisms	27
7.13	Sampling: what an iteration costs to certify	28
7.13.1	Certified decrease as a sampling rule	29
7.14	Extension to stochastic settings	29
7.14.1	A running example with a heavy-tailed sample minimiser	29
7.14.2	Two caveats the experiments expose	31
7.14.3	What survives of the martingale framework	32
7.14.4	The radius as a random walk, and which rules admit the picture	32
7.14.5	Protocol for the large-scale study	34
8	Threats to validity	34
9	Conclusions	35
A	Basic trust-region algorithm	39

B	Statements imported from Part II	39
C	Theory used by the measurements	41
D	The quadratic decrease condition under inexact subproblem solution	41
D.1	CG-algorithm on definite positive Hessians	42
D.2	The boundary case: the identity becomes an inequality	42
E	The quadratic decrease condition under truncation	43
E.1	Quadratic decrease from the increasing step norm in CG	44
F	Step length when negative curvature is exploited	45
F.1	A step-length lemma	45
G	The reciprocal bound and the multiplier window	48
G.1	The sharpness of $\bar{\kappa}$	49
G.2	Discussion: the role of ζ and $\bar{\mu}$	51
G.2.1	The exact inactivity threshold	52
G.2.2	Why a small ζ traps R-DFO	53
G.2.3	What the example establishes	54
H	Figures	54

1 Introduction

1.1 Motivation

The trust-region radius update rule is the least standardised component of a trust-region method. Textbook treatments [17; 37] present a fixed-factor rule that multiplies the radius by a constant depending on the agreement ratio ρ_k , but the literature contains a variety of alternatives: rules that reset the radius to a multiple of the accepted step [31; 26], rules that keep it proportional to a criticality measure [18], rules that scale it by the gradient norm [26; 43; 19], and retrospective rules that re-judge the previous step with the current model [2]. These are usually presented as implementation details. Parts I and II of this survey [39] argued instead that they are structurally different algorithms with materially different asymptotic behaviour, and gave a theory that separates them.

That theory is a worst-case theory, and it leaves three kinds of question open.

First, the sufficient conditions of Part I are conditions on the *sequence* $\{\Delta_k\}$, not on the rule. A rule may satisfy them on every problem, on some problems, or only under hypotheses that cannot be checked a priori. Whether the distinction between weak and strong admissibility corresponds to any observable difference in practice is an empirical question.

Second, several results of Part II are governed by constants that are not computable. The most consequential is $\bar{\kappa}$, the step-to-gradient constant of Condition C.Sg, which the local analysis bounds by $8/\lambda_{\min}^*$ where $\lambda_{\min}^*$ is the leftmost eigenvalue of $\nabla^2 f(x^*)$ at the limit point. Both the DFO-like rule and the gradient-scaled rule carry a user parameter— ζ and $\bar{\mu}$ respectively—that must exceed $\bar{\kappa}$ for the trust region to become inactive and for the superlinear regime to be entered. Since $\lambda_{\min}^*$ is a property of the solution, no admissible choice can be guaranteed in advance. How badly this bites, and how sharp the threshold is in practice, cannot be settled analytically.

Third, the theory treats the radius rule in isolation. A working trust-region code also chooses a model Hessian—exact, quasi-Newton, or a positive-definite approximation—and a subproblem solver. These choices interact. A small radius truncates a conjugate-gradient subproblem solver on its first

iteration, at which point the step is the Cauchy point and the model Hessian has no influence on the search direction whatsoever; the method has silently become gradient descent. Nothing in the first-order theory detects this.

1.2 Contributions

We formulate the qualitative content of Parts I and II as a list of falsifiable claims (Section 2.5), and design a campaign of seven experiments to test them. Concretely:

- We give a family of analytic test problems, small enough to be understood completely and structured so that different algorithmic configurations converge to *different* critical points from the same starting point. These separate the mechanisms in a way that aggregate benchmarking cannot (Section 6).
- We measure the influence of ζ and $\bar{\mu}$ and locate the practical thresholds, comparing them with the theoretical value $\bar{\kappa} = 8/\lambda_{\min}^*$ (Experiments 2 and 4).
- We measure trust-region inactivity directly, as the fraction of iterations on which the constraint binds, and confirm that the rules split into those that eventually stop binding and those that bind forever (Experiment 1).
- We document a failure mode not visible to the first-order theory: with a small cap and a truncated-CG solver, the gradient-scaled rule degenerates exactly to gradient descent, and the resulting limit can be a degenerate critical point that is not a minimiser (Experiment 6).
- We specify, without reporting in this version, a protocol for large-scale comparison on CUTEst [29] and for an extension to noisy and subsampled objectives (Experiments 9 and 12).
- We survey the benchmarking methodology literature and set out the reporting conventions we adopt (Section 3).

Remark 1.1 (Status of the results reported here). This version reports complete results for the experiments carried out on the analytic problems of Section 6; every number in Tables 17 through 10 was produced by the reference implementation described in Section 4. Tables and figures marked • are protocol placeholders: the experimental design is fixed and described in full, but the campaign has not yet been run. We have kept the two clearly separated deliberately, so that the design can be reviewed independently of the results.

1.3 Notation and reproducibility

We minimise

$$\min_{x \in \mathbb{R}^p} f(x), \tag{1.1}$$

with $f : \mathbb{R}^p \rightarrow \mathbb{R}$ twice continuously differentiable. We write $g_k := \nabla f(x_k)$, H_k for the model Hessian at iteration k , B_k interchangeably when the model Hessian is a specific constructed matrix, Δ_k for the trust-region radius, s_k for the step, and

$$\rho_k := \frac{f(x_k) - f(x_k + s_k)}{m_k(0) - m_k(s_k)} \tag{1.2}$$

for the agreement ratio. Constants $0 < \gamma_1 \leq \gamma_2 < 1 < \gamma_3$ are the contraction and expansion factors and $0 \leq \eta \leq \eta_1 \leq \eta_2 < 1$ the acceptance thresholds, as in Parts I and II.

All code, problem definitions, raw output and plotting scripts are archived with the paper. The reference implementation is written in Julia [6]; the analytic experiments depend on no external solver, and the CUTEst experiments use the JuliaSmoothOptimizers infrastructure [38].

2 Background: the mechanisms and what Parts I and II establish

2.1 The algorithmic framework

We work with the basic trust-region algorithm of Part I. At iteration k a model

$$m_k(s) = f(x_k) + g_k^\top s + \frac{1}{2} s^\top H_k s \quad (2.1)$$

is minimised approximately over the ball $\|s\| \leq \Delta_k$, the ratio ρ_k of (1.2) is formed, the step is accepted if $\rho_k \geq \eta$, and the radius is updated by a rule that is the object of study. Throughout we assume the model satisfies first-order agreement, $m_k(0) = f(x_k)$ and $\nabla m_k(0) = g_k$, and that the step achieves Cauchy decrease.

2.2 The five mechanisms

We restate the five rules surveyed in Parts I and II, in the concrete deterministic form used by the implementation. The interval-valued statements of the appendix of Part II are specialised to their endpoints.

Update R-delta. [fixed factor on the radius]

$$\Delta_{k+1} = \begin{cases} \gamma_1 \Delta_k, & \rho_k < \eta_1, \\ \gamma_2 \Delta_k, & \eta_1 \leq \rho_k < \eta_2, \\ \min\{\gamma_3 \Delta_k, \Delta_{\max}\}, & \rho_k \geq \eta_2. \end{cases}$$

Update R-step. [factor on the realised step]

$$\Delta_{k+1} = \begin{cases} \gamma_1 \|s_k\|, & \rho_k < \eta_1, \\ \gamma_2 \|s_k\|, & \eta_1 \leq \rho_k < \eta_2, \\ \min\{\gamma_3 \|s_k\|, \Delta_{\max}\}, & \rho_k \geq \eta_2. \end{cases}$$

Update R-DFO(ζ). [contraction whenever $\Delta_k > \zeta \|g_k\|$]

$$\Delta_{k+1} = \begin{cases} \gamma_1 \Delta_k, & \rho_k < \eta_1, \\ \gamma_2 \Delta_k, & \rho_k \geq \eta_1 \text{ and } \Delta_k > \zeta \|g_k\|, \\ \min\{\gamma_3 \Delta_k, \Delta_{\max}\}, & \rho_k \geq \eta_1 \text{ and } \Delta_k \leq \zeta \|g_k\|. \end{cases}$$

Update R-grad($\bar{\mu}$). [radius tied to the gradient through a multiplier] The radius is $\Delta_k = \mu_k \|g_k\|$ with

$$\mu_{k+1} = \begin{cases} \gamma_1 \mu_k, & \rho_k < \eta_1, \\ \gamma_2 \mu_k, & \eta_1 \leq \rho_k < \eta_2, \\ \min\{\gamma_3 \mu_k, \bar{\mu}\}, & \rho_k \geq \eta_2. \end{cases}$$

The *unbounded* variant omits the cap, replacing the last line by $\gamma_3 \mu_k$.

Update R-RTR. [branch selected by the retrospective ratio] The radius is set from the retrospective ratio $\tilde{\rho}_k = (f(x_{k-1}) - f(x_k)) / (m_k(-s_{k-1}) - m_k(0))$, which re-judges the previous step using the model built at the point it produced; see the appendix of Part II for the full statement.

We use the terminology of Part I throughout: a mechanism is *criticality-anchored* if the radius is tied to a criticality measure (R-DFO, R-grad), *step-driven* if it is tied to the realised step (R-step, R-RTR), and *criticality-blind* otherwise (R-delta).

2.3 Summary of Part I: first-order convergence

Part I isolates three structural conditions on the radius sequence. Condition C.1 is a lower bound relating the radius to the smallest gradient norm seen so far; Conditions C.2 and C.3 control contraction on unsuccessful iterations and expansion on successful ones.

Condition C.1. [lower bound on the radius] There is $0 < \kappa_{\text{lb}} \leq 1$ with $\Delta_k \geq \frac{\kappa_{\text{lb}}}{M_k} \min_{0 \leq i \leq k} \|g_i\|$ for all k , where $\{M_k\}$ is a non-decreasing sequence of upper bounds on $\|H_k\|$.

Condition C.2. [contraction on unsuccessful iterations] There are $\gamma_2 \in (0, 1)$ and $\eta_{\text{dec}} > 0$ such that $\rho_k < \eta_{\text{dec}}$ implies $\Delta_{k+1} \leq \gamma_2 \Delta_k$.

Condition C.3. [bounded expansion on successful iterations] There are $\gamma_2 \in (0, 1)$, $\gamma_3 > 1$ and $\eta_{\text{inc}} > 0$ such that $\Delta_{k+1} \leq \gamma_2 \Delta_k$ if $\rho_k < \eta_{\text{inc}}$ and $\Delta_{k+1} \leq \gamma_3 \Delta_k$ otherwise.

A mechanism is *weakly admissible* if it satisfies Conditions C.1–C.2 and *strongly admissible* if it satisfies Conditions C.1 and C.3. Under the standing assumptions— f bounded below with Lipschitz gradient, first-order model agreement, a linear growth bound on the model Hessian, and Cauchy decrease—admissibility yields $\liminf_k \|g_k\| = 0$, and strong admissibility yields $\lim_k \|g_k\| = 0$ together with the usual $\mathcal{O}(\epsilon^{-2})$ evaluation complexity.

The essential observation, for present purposes, is that Condition C.1 involves $\min_{i \leq k} \|g_i\|$, which tends to zero along a convergent run. The bound is therefore *asymptotically vacuous*: it constrains the radius early and says nothing in the limit. This is exactly why a first-order guarantee does not preclude the pathologies Part II and the present paper explore.

2.4 Summary of Part II: radius asymptotics and the local regime

Part II establishes three groups of results.

Asymptotics of the radius. For criticality-anchored rules, $\Delta_k \rightarrow 0$ with $\sum_k \Delta_k^2 / M_k < \infty$, while $\sum_k \Delta_k$ may diverge. For step-driven rules the same summability holds in the local regime and Δ_k inherits the rate of the step series. For fixed-factor rules Δ_k remains bounded away from zero. The three families therefore have qualitatively different radius trajectories even when they converge to the same point at the same rate.

The local regime and the constant $\bar{\kappa}$. Near a first-order critical point x^* with positive definite Hessian, the analysis rests on the following condition.

Condition C.Sg. [step bounded by the gradient] There exist $\bar{\kappa} > 0$ and $\delta^{(s)} > 0$ such that $\|x_k - x^*\| \leq \delta^{(s)}$ implies $\|s_k\| \leq \bar{\kappa} \|g_k\|$.

Part II shows that the local lemma yields the explicit value

$$\bar{\kappa} = \frac{8}{\lambda_{\min}^*}, \quad \lambda_{\min}^* := \lambda_{\min}(\nabla^2 f(x^*)), \quad (2.2)$$

and that for R-grad with a cap the same role is played by $\bar{\mu}$ itself, so that the smallest valid constant for that rule is $\min\{\bar{\mu}, 8/\lambda_{\min}^*\}$. When $\lambda_{\min}^*$ is small—which happens at nearly-singular minimisers, and can be made to happen at arbitrarily many critical points of a single function— $\bar{\kappa}$ is large, and a user parameter chosen below it prevents the trust-region constraint from ever becoming inactive.

Inactivity and the superlinear regime. Under the criticality-related updates, once μ_k or ζ exceeds $\bar{\kappa}$ the trust-region constraint becomes inactive, the step becomes the unconstrained model minimiser, and the method inherits the local rate of the underlying Newton-like iteration. For the unbounded R-grad variant μ_k grows geometrically on very successful iterations and therefore exceeds $\bar{\kappa}$ unconditionally, which is the mechanism by which that rule escapes the trap without a priori knowledge of $\lambda_{\min}^*$.

2.5 The claims under test

We extract eight claims. Each is a qualitative prediction of Parts I or II; each is testable; and each is assigned to an experiment in Section 7.

Claim 1 (Radius asymptotics separate the families). Along convergent runs, $\Delta_k \rightarrow 0$ for criticality-anchored rules while $\liminf_k \Delta_k > 0$ for fixed-factor rules. Consequently $\sum_k \Delta_k^2$ converges for the former and diverges for the latter.

Claim 2 (ζ has a threshold). There is a problem-dependent threshold in ζ below which R-DFO keeps the trust-region constraint active indefinitely and above which the constraint eventually becomes inactive; the threshold is related to $\bar{\kappa} = 8/\lambda_{\min}^*$.

Claim 3 ($\bar{\mu}$ has a threshold). The same holds for the cap $\bar{\mu}$ in R-grad, and the unbounded variant crosses the threshold unconditionally.

Claim 4 (Inactivity is observable). The fraction of iterations on which $\|s_k\| = \Delta_k$ tends to zero for configurations that satisfy the inactivity hypotheses and stays bounded away from zero otherwise.

Claim 5 (Mechanism affects rate only through inactivity). Conditional on the trust region becoming inactive, the observed local rate is that of the underlying Newton-like iteration and is independent of the mechanism. Mechanisms differ in *when* inactivity is reached, not in the rate thereafter.

Claim 6 (The limit point is decided by the model, not by the radius rule). Across configurations, the critical point reached from a fixed starting point depends on the model Hessian and not on the radius mechanism.

Claim 7 (A small cap can silently disable the model). With truncated-CG and $\bar{\mu}$ small enough that the first CG iterate leaves the trust region, the step is the Cauchy point, the model Hessian does not influence the search direction, and the method inherits gradient-descent behaviour.

Claim 8 (First-order guarantees do not preclude bad limits). There are configurations satisfying every hypothesis of Part I, with every ρ_k successful and a well-behaved radius, whose limit is a saddle point or a degenerate critical point that is not a minimiser.

Table 1 maps claims to experiments.

Table 1: Mapping between the claims of Section 2.5 and the experiments of Section 7.

Claim	Statement	Experiment	Status
1	radius asymptotics separate families	8, 1	analytic; CUTEst •
2	ζ threshold	2	complete
3	$\bar{\mu}$ threshold	4	complete
4	inactivity observable	1	complete
5	rate depends on mechanism only via inactivity	5	analytic; CUTEst •
6	limit decided by model	8, 6	complete
7	small cap disables the model	6	complete
8	first-order guarantees insufficient	8, 6	complete

3 Related work

We review the literature in three groups: the methodology of numerical comparison, the test collections available, and prior computational studies of trust-region and adaptive-radius methods. References specific to the stochastic extension are collected in Section 3.4.

3.1 Methodology of comparison

The modern convention for comparing optimization solvers is the *performance profile* of Dolan and Moré [22]. For a solver s and problem p , let $t_{p,s}$ be a cost measure and $r_{p,s} = t_{p,s} / \min_{s'} t_{p,s'}$ the performance ratio; the profile is the empirical distribution

$$\pi_s(\tau) = \frac{1}{|\mathcal{P}|} |\{p \in \mathcal{P} : r_{p,s} \leq \tau\}|. \quad (3.1)$$

The value $\pi_s(1)$ is the fraction of problems on which s is fastest and $\lim_{\tau \rightarrow \infty} \pi_s(\tau)$ its reliability. Profiles have well-documented pitfalls: Gould and Scott [27] show that the ranking induced by a profile can change when solvers are added to or removed from the comparison, since the ratio is normalised by the best solver in the set, and recommend reporting profiles for pairs as well as for the full set. Moré and Wild [35] introduce the complementary *data profile*, which reports the fraction of problems solved within a given budget of function evaluations rather than relative to the best solver, and is the appropriate instrument when evaluations dominate the cost. Beiranvand et al. [3] give a systematic account of the whole benchmarking process—test set selection, performance measures, tuning, and reporting—and catalogue the ways in which comparisons are commonly biased. We follow their recommendations in Section 4, in particular the requirements that parameters be tuned identically across solvers or not at all, and that the convergence test be identical and solver-independent.

We adopt the standard convergence test of Moré and Wild [35],

$$f(x_0) - f(x_k) \geq (1 - \tau)(f(x_0) - f_L), \quad (3.2)$$

where f_L is the best value found by any solver within the budget, alongside the more familiar $\|g_k\| \leq \epsilon_g$. Reporting both matters here: the mechanisms differ substantially in how fast they drive $\|g_k\|$ down once the objective has stopped decreasing appreciably, and a comparison based on the gradient test alone exaggerates the differences.

3.2 Test collections

CUTEst [29], the successor to CUTER [28] and CUTE [10], is the standard collection for nonlinear optimization and supplies the unconstrained problems used in Experiment 9. It gives analytic gradients and Hessians, which is essential for us: we wish to vary the model Hessian deliberately, and a collection that forces finite-difference derivatives would confound that variation. The older Hock–Schittkowski set [32] remains in use for small problems. For the learning experiments of Experiment 12 we use the LIBSVM collection [15].

3.3 Trust-region methods and adaptive radius rules

The monograph of Conn et al. [17] is the standard reference for the theory and for the fixed-factor rule; Nocedal and Wright [37] give the textbook treatment. Exact subproblem solution follows Moré and Sorensen [36], and the truncated-CG solver follows Steihaug [41] and Toint [42]. Yuan [46] surveys the field, including the adaptive-radius literature.

The rules studied here have distinct origins. The step-dependent rule appears in Hei [31] and in the family of papers by Fan and Pan [23; 24; 25; 26]. The gradient-scaled family originates with Zhang

et al. [47], who set the radius proportional to $\|g_k\|$ times a power of $\|B_k^{-1}\|$; Wang and Yuan [43] and Curtis and Scheinberg [19] give more recent variants. The DFO-like rule, in which the radius is kept within a constant multiple of a criticality measure, comes from derivative-free trust-region methods and is treated systematically in Conn et al. [18]. The retrospective rule is due to Bastin et al. [2]. Sartenauer [40] addresses the complementary question of choosing Δ_0 , which our experiments hold fixed and which is a natural extension.

Complexity results for trust-region methods—Grapiglia et al. [30], Curtis et al. [21], Curtis et al. [20]—provide the worst-case bounds against which observed iteration counts can be compared, though the constants are far from tight and we do not attempt a quantitative comparison.

Relatively few papers isolate the radius rule as the experimental variable. Bastin et al. [2] compare the retrospective rule against the classical one on CUTER and report a modest but consistent advantage. Most other computational work varies the rule incidentally while studying something else, which is precisely the gap this part is meant to fill.

3.4 Noisy and stochastic trust-region methods

The stochastic extension of Experiment 12 draws on a well-developed literature. Bandeira et al. [1] analyse trust-region methods with probabilistic models, requiring the model to be sufficiently accurate with probability at least one half, and prove almost-sure convergence. Chen et al. [16] extend this to noisy function values, and Blanchet et al. [7] give a rate analysis via supermartingales. Cartis and Scheinberg [14] treat the general framework of probabilistic models, and Berahas et al. [5] give the line-search analogue. Bellavia et al. [4] analyse probabilistic complexity with applications to subsampling. Cao et al. [13] give high-probability first- and second-order complexity bounds under noisy oracles.

On the subsampling side, Byrd et al. [12] study sample size selection, Bollapragada et al. [8, 9] analyse exact and inexact subsampled Newton methods, and Xu et al. [44, 45] treat Newton-type methods with inexact Hessian information. Bottou et al. [11] survey the area.

The relevance to the present study is direct. In the subsampled setting the model Hessian is a random matrix whose accuracy is controlled by the sample size, and the radius rule determines how much the method is allowed to trust it. The criticality-anchored rules tie Δ_k to $\|g_k\|$, which is itself estimated with error; whether that coupling is stabilising or destabilising under noise is an empirical question our design is meant to answer.

4 The instrument

The claims of Part II separate mechanisms that first-order theory cannot distinguish, so the software has to separate them too. Four components of a trust-region iteration are varied independently: the radius mechanism, the model Hessian, the subproblem solver and the acceptance test. Holding three fixed while the fourth varies is what makes a difference in outcome attributable, and it is the property on which every table below rests.

The implementation is `TrustRegionRadius.jl`, a Julia package built on the `JuliaSmoothOptimizers` interfaces [38; 6]. Problems enter as `AbstractNLPModels`, so the analytic problems of Section 6 and the CUTEst collection [29] are solved by the same code path.

4.1 Four axes, varied independently

The solver carries the mechanism, the model and the subsolver as independent type parameters, each copied at construction so that a rule with mutable state starts every run clean. The acceptance test is the fourth axis, and it is separated from the radius update by a device inherited from Part I: the step is taken when $\rho_k \geq \eta$, while the radius is scaled by comparing ρ_k against η_1 and η_2 , and a mechanism

never sees η . The ordering $0 \leq \eta \leq \eta_1 \leq \eta_2 < 1$ leaves a band $\rho_k \in [\eta, \eta_1)$ on which a step is accepted and the radius nonetheless contracts. Setting $\eta = \eta_1$ recovers the coupled behaviour of the classical algorithm.

Table 2: The mechanisms, their branches and their parameters. The branch names are those the solver records at every iteration; the same vocabulary is used for every rule, so branch counts are comparable across the table. For the Hei family the factor is continuous and the branch is its position relative to one.

Mechanism	Radius from	Parameters	Branches
R-delta	Δ_k	$\gamma_{1,2,3}$	contract, shrink, expand
R-step	$\ s_k\ $	$\gamma_{1,2,3}$	contract, shrink, expand
R-DFO	Δ_k , tested against $\zeta \ g_k\ $	$\gamma_{1,2,3}, \zeta$	contract, shrink, expand
R-grad	$\mu_k \ g_k\ $, μ uncapped	$\gamma_{1,2,3}, \mu_0$	contract, shrink, expand, hold
R-grad($\bar{\mu}$)	$\mu_k \ g_k\ $, $\mu_k \leq \bar{\mu}$	$\gamma_{1,2,3}, \mu_0, \bar{\mu}$	..., expand capped
R-RTR	$\ s_k\ $, Δ_k , branch by $\tilde{\rho}_{k+1}$	$\gamma_{1,2,3}, \tilde{\eta}_{1,2}$	contract, shrink, expand, hold

Recording the branch is what turns Properties P1 and P2 of Part II into observables. P1 asks that the radius not contract while the region binds, which is the statement that no contraction, shrink or capped expansion occurs on an active iteration; and the number of expansions is the climb a gradient-scaled multiplier makes towards its threshold.

Three model Hessians are used: the exact Hessian, a limited-memory BFGS approximation and an SR1 approximation. A fourth, positive definite by construction with a prescribed minimiser, is used only on P1 and is described with that problem. Whether a model can report negative curvature is a property of the model rather than of the run, and it matters for the second-order results: under a limited-memory BFGS model $\lambda_{\min}(H_k) \geq 0$ always, so $\tau_k \equiv \|g_k\|$ and the second-order machinery is inert. The second-order configurations therefore use the exact Hessian or SR1.

Two subproblem solvers are used, the truncated conjugate-gradient method of Steihaug [41]; Toint [42] and the exact solution by the characterisation of Moré and Sorensen [34]. The truncated solver stops on negative curvature, on reaching the boundary, or when the residual satisfies $\|H_k s_k + g_k\| \leq \min\{\chi, \|g_k\|^\theta\} \|g_k\|$, which supplies the forcing sequence of Condition C.Inx with $\chi = 0.1$ and $\theta = 0.5$.

4.2 What is recorded

The trust-region constraint is active when the returned step lies on the boundary, and the solver reports this from the subsolver's own control flow rather than by comparing $\|s_k\|$ with Δ_k to a tolerance. Since every statement about eventual inactivity is a statement about that flag, its definition is exact by construction.

Four of these exist because the local results of Part II assume them rather than establish them, and a table of rates that does not report them cannot distinguish a claim about mechanisms from a failure of the model. The Dennis–Moré residual γ_k is the hypothesis behind the step contraction and behind the superlinear rate; ξ_k is the realised inexactness that Condition C.Inx requires to vanish, and it is bounded above by, and generally far below, the forcing sequence the solver advertises. The alignment $\cos(s_k, -g_k)$ equals one exactly when the step is the Cauchy point, and together with an inner-iteration count of one it certifies that the model Hessian did not influence the search direction. The Hessian norm is what the majorant $M_k = L + \max_{i \leq k} \|H_i\|$ is built from, and without it the summability of Part II can only be reported in its bounded-curvature form.

Under sampling the record is separate and carries the sample sizes N_k^g and N_k^f , the variance estimates $\hat{\sigma}_g^2$ and $\hat{\sigma}_f^2$, the exact gradient norm where a truth is available, and, under a rule that pairs, the mean

Table 3: Per-iteration record. Entry j of every trajectory describes iteration $j - 1$. The state quantities carry one further entry, the state left by the final iteration, so a state trajectory is paired with a per-iteration one by discarding its last entry.

Quantity	Symbol	Length
radius, criticality, objective	$\Delta_k, \ g_k\ , f(x_k)$	$k + 1$
model curvature, second-order measure	$\lambda_{\min}(H_k), \tau_k$	$k + 1$
problem curvature	$\lambda_{\min}(\nabla^2 f(x_k))$	$k + 1$
model Hessian norm	$\ H_k\ $	$k + 1$
distance to a reference solution	$\ x_k - x^*\ $	$k + 1$
acceptance ratio, step length	$\rho_k, \ s_k\ $	k
ratio driving the radius	ρ_k or $\tilde{\rho}_{k+1}$	k
activity, acceptance	$\mathbb{I}\{\ s_k\ = \Delta_k\}$	k
Dennis–Moré residual	$\gamma_k = \ g_{k+1} - g_k - H_k s_k\ / \ s_k\ $	k
realised inexactness	$\xi_k = \ H_k s_k + g_k\ / \ g_k\ $	k
Cauchy alignment	$\cos(s_k, -g_k)$	k
inner iterations, branch	<code>cg_iters</code> , branch name	k

and variance of the paired differences. Keeping the two records apart is deliberate: a variance recorded from an exact evaluation is not a small number, and a deterministic run carries no such column at all.

4.3 Derived quantities

The following are computed from a completed run, and every table below is stated in them.

- $k_\star$, the first iteration after which the constraint never binds again. This is the index of Part II’s inactivity theorem, and it is distinct from the first inactive iteration of any kind: the two differ whenever the region releases and binds again, and every statement about the rate belongs after the former.
- the tail-active fraction, the share of the last tenth of a run on which the constraint bound. It is reported only for runs of at least thirty iterations, below which the tail is one or two iterations and the statistic takes one of three values.
- $\theta_k = \Delta_k / \|g_k\|$, the radius-to-criticality ratio. For the gradient-scaled rules θ_k is the multiplier itself.
- the empirical constant $\max_k \|s_k\| / \|g_k\|$ over inactive iterations, which is the constant of Condition C.Sg that a run actually exhibited, against the $\bar{\kappa}$ the theory supplies.
- the three radius series $\sum_k \Delta_k$, $\sum_k \Delta_k^2$ and $\sum_k \Delta_k^2 / M_k$, reported per run with the iteration count beside them.
- the observed order, from a regression of $\log \|g_{k+1}\|$ on $\log \|g_k\|$ restricted to accepted iterations beyond $k_\star$, and reported only when at least six such iterations exist.
- branch counts, and a report of the measured hypotheses: the final and tail-maximum γ_k and ξ_k , and the fraction of iterations on which the step was the Cauchy point.

The threshold $\bar{\kappa}$ is computed from a problem and a solution as $\bar{\kappa} = 8/\lambda_{\min}^*$, the value of Part II under a fixed neighbourhood. The alternative convention, $4/\lambda_{\min}^*$ at the eigenvalue itself, is smaller by exactly two, and a threshold quoted in one and compared against a parameter chosen in the other is wrong by that factor. Every number in this paper uses the former.

4.4 Reproducibility

Each campaign writes a directory containing the configuration, the per-run data, the tables and the figures. The configuration records every mechanism with its full parameter set, the solver parameters, the problem selection, and a provenance block: the Julia version, the package version and commit, whether the working tree was clean, content hashes of the project and manifest files, and the master seed of any sampled experiment. A run whose seed is not recorded can be resampled but not reproduced.

Per-run data is written as it is produced, so a campaign interrupted after some hours resumes from what it has. A resumed record is returned exactly as archived, which means that a diagnostic added after a campaign began does not appear in the runs already stored; a campaign whose purpose is a new quantity is run into a fresh directory.

Cost is measured in evaluations rather than iterations. Under sampling the relevant count is of term evaluations, since a sampling rule that grows the batch buys its progress with work that an iteration count does not see, and iteration counts and work are proportional only under a fixed sample size.

5 Protocol

5.1 Parameters, held fixed

Every comparison uses $\eta = \eta_1 = 0.1$, $\eta_2 = 0.9$, $\Delta_0 = 1$, $\gamma_1 = 0.25$, $\gamma_2 = 0.5$ and $\gamma_3 = 2$, for every mechanism. Tuning per rule would measure tuning effort rather than the mechanism, and the constants published with individual rules differ enough to matter: the retrospective rule of Bastin et al. [2] is stated with $\gamma_1 = 1/16$ and $\gamma_3 = 5/2$, a contraction four times more aggressive than the common setting. Where a rule carries a lower floor on the radius, that floor is set to zero for the comparison, since a floor is a qualitative change to the asymptotic regime under test rather than a numerical safeguard.

The parameters that are not held fixed are the ones the study is about, ζ and $\bar{\mu}$, and they are swept.

5.2 Stopping and failure

A run stops at $\|g_k\| \leq \varepsilon_g$, at the iteration or time budget, or when the achieved reduction falls to the rounding level of f while the step is numerically nil, which is reported as stagnation rather than success. Runs that fail to converge are reported, not filtered: a configuration whose parameter sits below its threshold exhausts its budget by construction, so a table restricted to successful runs removes exactly the evidence the inactivity claims are about.

5.3 What each claim is measured by

6 Analytic test problems

Aggregate benchmarking answers questions about central tendency. It is poorly suited to establishing that a mechanism *can* fail, because a handful of pathological problems is invisible in a profile over hundreds. We therefore use three small analytic problems, chosen so that the pathologies of Part II are exhibited exactly and can be diagnosed completely.

6.1 P1: a quartic with four critical points

$$f_1(x, y) = x^4 - x^3 + \left(\frac{1}{4} - \frac{x}{2}\right)y^2 + \frac{1}{4}y^4. \quad (6.1)$$

Its critical points are listed in Table 5. Two structural facts make it useful. The second gradient component factors as $y(y^2 - x + \frac{1}{2})$ and therefore vanishes identically on $\{y = 0\}$, so that line is

Table 4: Claim, statistic and where it is reported. The statistic column is the quantity that decides the claim, not a proxy for it.

Claim	Statistic	Section
the radius regimes separate the families	$\sum_k \Delta_k, \sum_k \Delta_k^2, \sum_k \Delta_k^2/M_k$, per run	7.1
the constraint eventually stops binding, or does not	k_* , tail-active fraction	7.2
ζ and $\bar{\mu}$ have a problem-dependent threshold	activity against the parameter, against $\bar{\kappa}$	7.4
the rate after inactivity does not depend on the mechanism	observed order conditioned on k_* , beside k_*	7.7
τ repairs the collapse of a $\ g\ $ -anchored radius at a saddle	Δ_k and τ_k near the saddle, certified second-order status	7.9
a small cap silently disables the model	<code>cg_iters</code> , $\cos(s_k, -g_k)$, the limit reached	7.10
the sampling cost of an iteration does not see the radius rule	N_k against $\ g_k\ $ and against $\ s_k\ $	7.13

invariant under the gradient flow; it is exactly the stable manifold of the saddle $(3/4, 0)$, of codimension one. And at the origin $\nabla^2 f_1 = \text{diag}(0, \frac{1}{2})$ is singular and positive semidefinite, so the second-order criticality measure $\tau(x) = \max\{\|g\|, -\lambda^-\}$ *vanishes* there, while $f_1(x, 0) = x^3(x - 1) < 0$ for small $x > 0$ shows the origin is not a local minimiser. P1 therefore contains a point at which a second-order measure certifies a non-minimum.

Table 5: Critical points of f_1 of (6.1).

Name	Point	f_1	$\text{eig } \nabla^2 f_1$	Type
origin	$(0, 0)$	0	$(0, 0.5)$	degenerate, singular, PSD
saddle	$(3/4, 0)$	$-27/256$	$(-0.25, 2.25)$	nonsingular saddle
$\min^+$	$(0.809017, 0.555893)$	$-1/8$	$(0.4947, 3.1233)$	local minimiser
$\min^-$	$(0.809017, -0.555893)$	$-1/8$	$(0.4947, 3.1233)$	local minimiser

A constructed positive definite model. To exhibit convergence to the saddle from a starting point off the stable manifold, we construct a model whose minimiser is pinned to $x^* = (3/4, 0)$ while remaining positive definite. With $d_k = x^* - x_k$, $u = d_k / \|d_k\|$ and $v = u^\perp$, the family

$$B_k = a u u^\top + b (u v^\top + v u^\top) + c v v^\top, \quad a = -\frac{g_k^\top u}{\|d_k\|}, \quad b = -\frac{g_k^\top v}{\|d_k\|}, \quad (6.2)$$

has unconstrained minimiser x^* for every c , and is positive definite exactly when $a > 0$ and $c > b^2/a$; we take $c = b^2/a + 1$. The condition $a > 0$ is equivalent to

$$\varphi(x_k) := g_k^\top (x^* - x_k) < 0, \quad (6.3)$$

that is, the target must lie downhill. This is necessary for *any* positive definite model with that minimiser, since $d^\top B d = -\alpha^{-1} g^\top d > 0$. Expanding φ about the saddle with $u = x - \frac{3}{4}$, $v = y$ gives the exact identity

$$\varphi\left(\frac{3}{4} + u, v\right) = -\frac{9}{4}u^2 + \frac{1}{4}v^2 - 6u^3 + \frac{3}{2}uv^2 - 4u^4 - v^4, \quad (6.4)$$

whose quadratic coefficients are, up to a factor $\frac{1}{2}$, the eigenvalues of $\nabla^2 f_1(x^*)$ with reversed signs. One cannot descend into a saddle from the direction along which it repels: the model exists in a region funnelling along the stable manifold and fails in a lens around the unstable direction. Figure 1 displays the construction and its validity region, and Figure 2 the gradient flow with the invariant line, the nullclines and a trajectory steered onto the saddle.

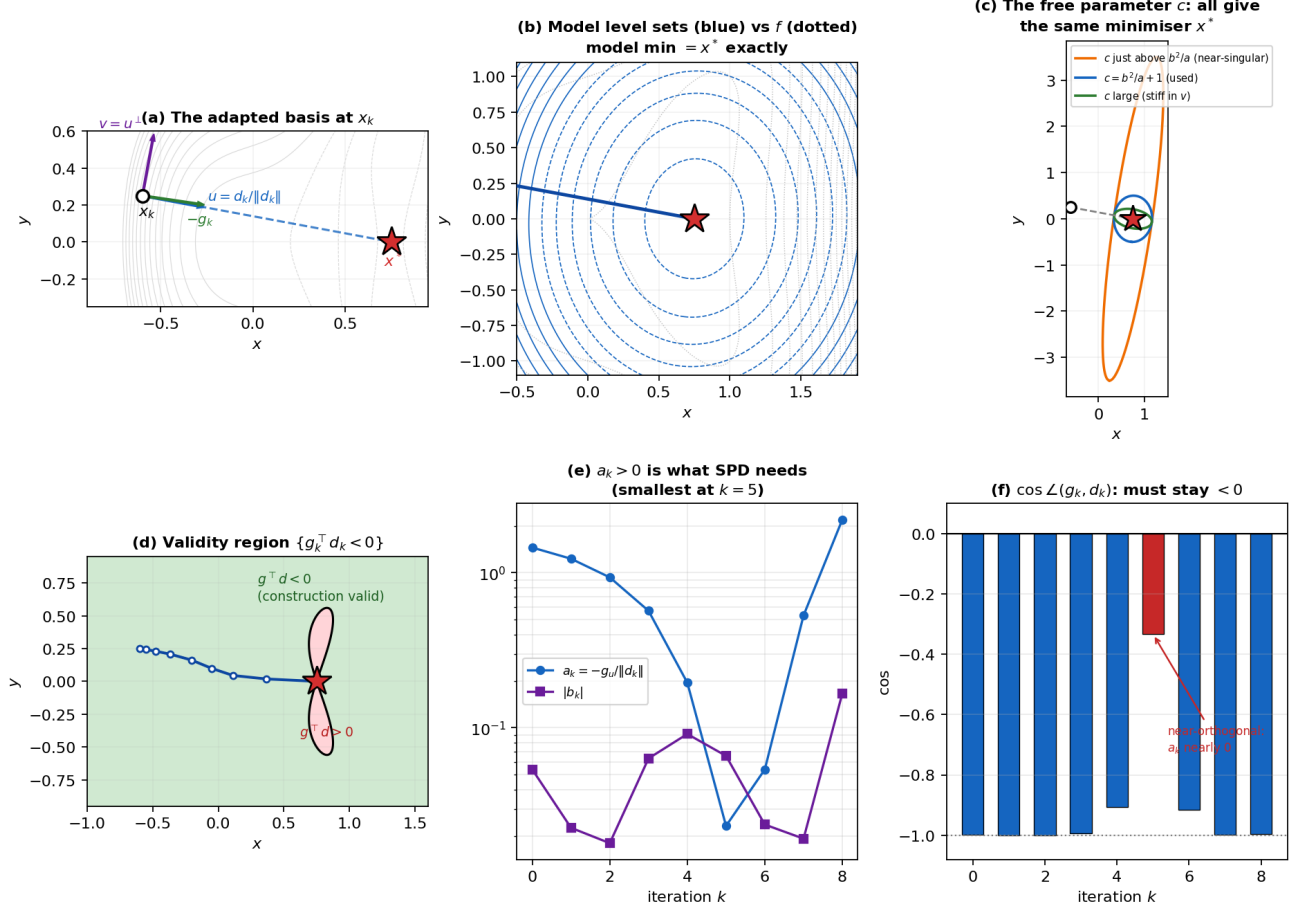


Figure 1: The positive definite model of (6.2) on P1. (a) the adapted basis $u = d_k / \|d_k\|$, $v = u^\perp$ at x_k ; (b) the model level sets, whose minimiser is x^* exactly; (c) the free parameter c , every admissible value giving the same minimiser; (d) the validity region $\{g_k^\top d_k < 0\}$ of (6.3), the trajectory remaining inside it throughout; (e) the coefficients a_k and $|b_k|$, positivity of a_k being what positive definiteness requires; (f) $\cos \angle(g_k, d_k)$, which must stay negative and comes closest to zero at $k = 5$, the iteration at which the gradient is nearest to orthogonal to the direction of the target.

6.2 P2: a nonsingular saddle with tunable escape rate

$$f_2^{(\varepsilon)}(x, y) = \frac{1}{2}x^2 + \varepsilon\left(\frac{1}{4}y^4 - \frac{1}{2}y^2\right), \quad \varepsilon > 0. \quad (6.5)$$

The origin is a saddle with $\nabla^2 f_2(0, 0) = \text{diag}(1, -\varepsilon)$, hence $\det = -\varepsilon \neq 0$: *nonsingular* for every $\varepsilon > 0$, with $\tau(0, 0) = \varepsilon > 0$ correctly flagging it. The minimisers are at $(0, \pm 1)$, at fixed distance from the saddle as $\varepsilon \rightarrow 0$, so ε tunes the escape rate without collapsing the geometry.

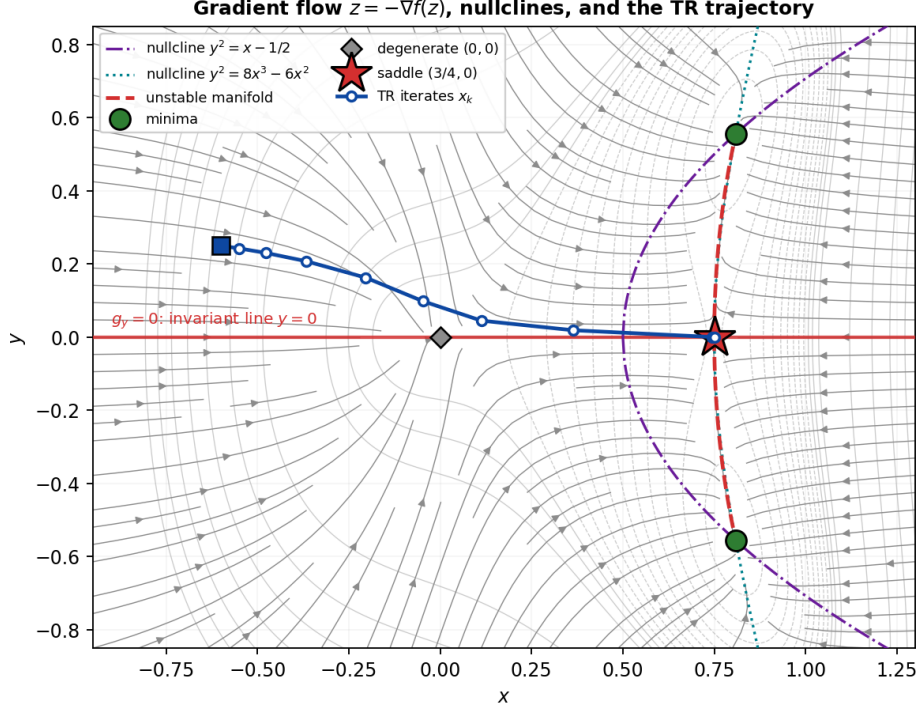


Figure 2: Gradient flow of P1 with the two nullclines, the invariant line $\{y = 0\}$, the unstable manifold of the saddle, and the trust-region iterates under the constructed model. The iterates start off the stable manifold and are steered onto the saddle, which the flow alone would not do.

Near the saddle the unstable coordinate obeys $\dot{y} \approx \varepsilon y$, so $y(t) \approx y_0 e^{\varepsilon t}$ and the escape time is

$$k_{\text{esc}} \approx \frac{\ln(1/|y_0|)}{\varepsilon}, \quad (6.6)$$

linear in $1/\varepsilon$ and only logarithmic in the initial distance to the stable manifold.

Remark 6.1. No nonsingular saddle can have a basin of attraction of positive measure: hyperbolicity forces the stable set to have dimension equal to the number of positive eigenvalues of $\nabla^2 f(x^*)$, hence measure zero whenever the saddle is genuine. P2 does not evade this. What it provides is an *arbitrarily long stall*, which has the same practical consequence: the method sits within any prescribed tolerance of a nonsingular saddle for as many iterations as one likes.

The stall requires a model blind to negative curvature. The construction isolates a single ingredient. With the *true* Hessian and a truncated-CG solver the method escapes in two iterations whatever ε , the model curvature $-\varepsilon < 0$ being detected on the first CG iteration and the direction of negative curvature followed to the boundary. The stall appears as soon as the model reports nonnegative curvature there, which is what BFGS, Gauss–Newton and every positive-definiteness-enforcing scheme do by construction. With $B_k = \text{diag}(1, c)$ and $c > 0$ fixed, the step along y is $|s_y| \approx \varepsilon |y|/c$, reproducing (6.6) with rate ε/c . Figure 4 confirms the scaling over four decades.

Termination at the saddle. The stall becomes an outright failure once x_k has collapsed, since the gradient near the saddle is $\|g_k\| \approx \varepsilon |y_k|$. For $\varepsilon = 10^{-4}$ and $y_0 = 10^{-6}$ one has $\|g_0\| \approx 10^{-10}$ from the outset, so the method *satisfies its stopping criterion* at a nonsingular saddle after three iterations. This

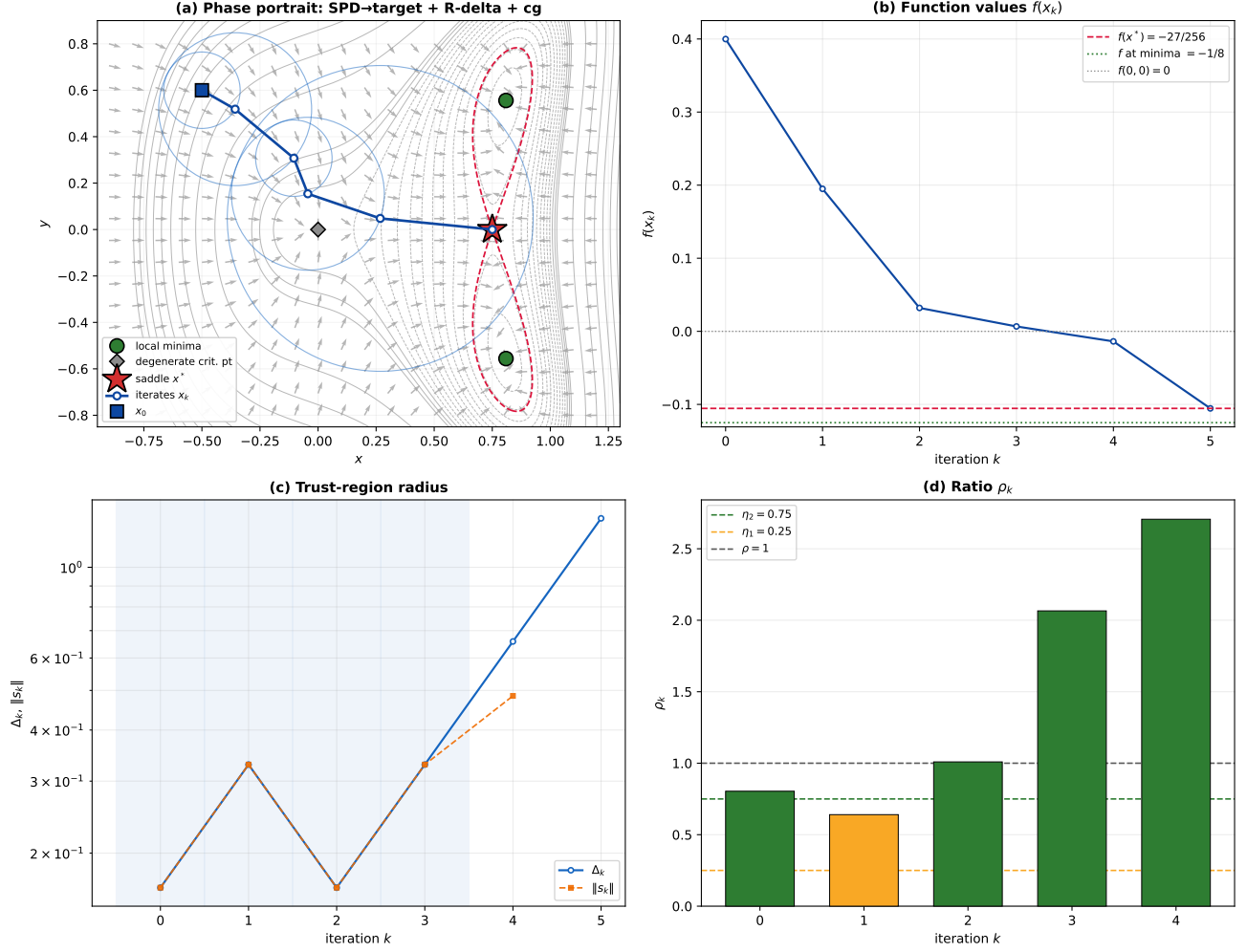


Figure 3: P1 under the constructed model with R-delta and truncated CG. (a) the trajectory, ending at the saddle $x^* = (3/4, 0)$; (b) the function values, decreasing monotonically to $f(x^*) = -27/256$ and never reaching the minimum value $-1/8$; (c) the radius and the step norm, the constraint being active on the shaded iterations; (d) the ratio ρ_k , which exceeds η_2 at every iteration but one and exceeds 1 from $k = 2$ onward. Every first-order diagnostic is healthy throughout.

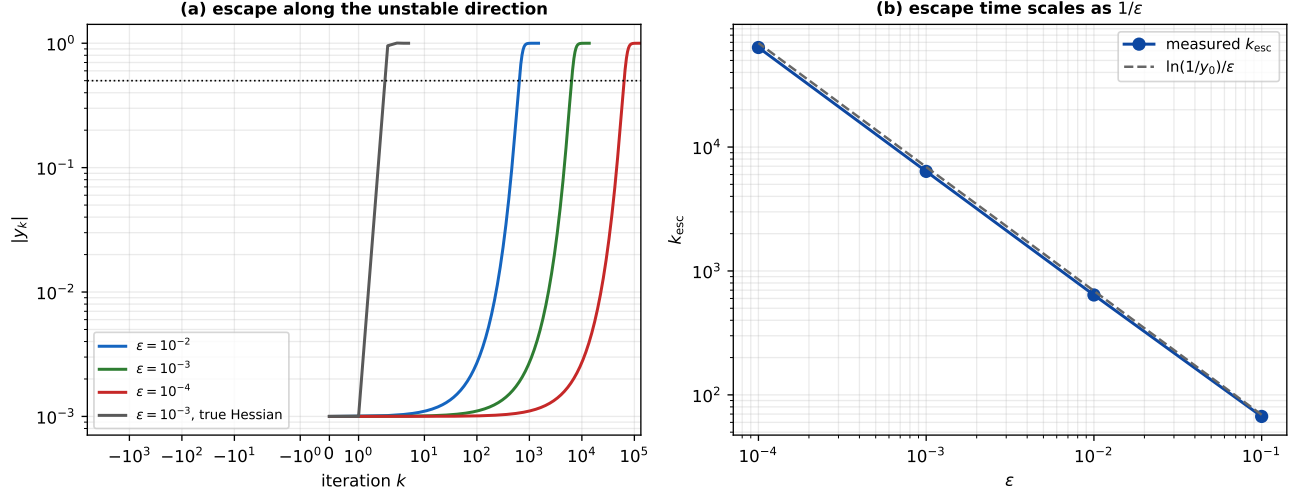


Figure 4: P2 under a model blind to negative curvature. (a) the unstable coordinate $|y_k|$ against the iteration count, for four values of ε , together with the run under the true Hessian, which escapes in two iterations; (b) the measured escape time against ε , matching $\ln(1/|y_0|)/\varepsilon$ over four decades.

is a recipe for defeating any prescribed first-order tolerance ϵ_{tol} , namely $\varepsilon |y_0| < \epsilon_{\text{tol}}$. The method does not merely appear to converge. It terminates successfully and reports a saddle as a solution.

What P1 and P2 separate. The two problems distinguish the two ways a first-order guarantee can hold at a point that is not a minimiser. On P1 the limit is a *degenerate* critical point at which τ itself vanishes, so no second-order measure could have flagged it. On P2 the measure $\tau(0, 0) = \varepsilon > 0$ is perfectly informative and never consulted, because what the algorithm sees is the spectrum of B_k and not that of $\nabla^2 f$. In both the radius mechanism is irrelevant. It governs how far each step travels, while whether the negative curvature is followed at all is decided by the model Hessian. This is the empirical form of the claim made throughout Part II, that the mechanism decides when a regime is entered and not what the limit is.

6.3 P3: a family of minimisers with vanishing $\lambda_{\min}^*$

$$f_3(x, y) = \frac{\sin x}{x} + y^2, \quad \left. \frac{\sin x}{x} \right|_{x=0} := 1. \quad (6.7)$$

Writing $g(x) = \sin x/x$, the critical points satisfy $\tan x_k = x_k$; the nonzero roots are $x_k = (k + \frac{1}{2})\pi - \delta_k$ with $\delta_k \downarrow 0$, one in each interval $(k\pi, (k + \frac{1}{2})\pi)$ for $k \geq 1$. The Hessian is diagonal with eigenvalues $\{g''(x_k), 2\}$, and at a critical point the relation $\tan x_k = x_k$ makes two terms of g'' cancel *exactly*:

$$g''(x_k) = -\frac{\sin x_k}{x_k}. \quad (6.8)$$

Since $|\sin x_k| = |x_k|/\sqrt{1+x_k^2} \rightarrow 1$, we have $|g''(x_k)| \sim 1/|x_k|$, alternating in sign. Along the subsequence with $g''(x_k) > 0$ the critical points are minimisers with $\lambda_{\min}^* \rightarrow 0$, so by (2.2) the constant $\bar{\kappa} = 8/\lambda_{\min}^*$ can be made arbitrarily large by choosing a minimiser far from the origin. P3 is therefore the natural vehicle for testing Claims 2 and 3. Table 6 gives the first few values.

Table 6: Minimisers of f_3 of (6.7) and the associated constant $\bar{\kappa} = 8/\lambda_{\min}^*$. Only the critical points with $g''(x^*) > 0$ are minimisers in x ; the interleaving points are saddles.

x^*	$g''(x^*)$	$\lambda_{\min}^*$	$\bar{\kappa} = 8/\lambda_{\min}^*$	Type
4.4934	+0.21723	0.21723	36.83	minimiser
7.7253	−0.12837	—	—	saddle
10.9041	+0.09133	0.09133	87.60	minimiser
14.0662	−0.07091	—	—	saddle
17.2208	+0.05797	0.05797	138.00	minimiser

6.4 P4: a set of thirty classical problems

The three problems above are diagnostic instruments: each isolates one pathology. For the aggregate questions—which value of a parameter to choose, which mechanism to prefer—we use a set of thirty small unconstrained problems drawn from the classical literature, listed in Table 7. They are deliberately unexceptional: smooth, low-dimensional, mostly with a unique attracting minimiser in the basin of the standard starting point, and covering the usual difficulties (narrow valleys, ill-conditioning, quartic flatness, exponential and trigonometric terms). Derivatives are exact.

Thirty problems is too few to support a claim about typical behaviour in general, and we do not make one; the set exists to give the parameter sweeps of Section 7.5 and Section 7.6 an aggregate reading, and to establish the reporting format that Experiment 9 will apply at scale.

Table 7: The thirty-problem analytic set P4. Dimension n and the standard starting point are those of the source; where a problem appears twice it is with a different x_0 .

Problem	n	Problem	n	Problem	n	Problem	n
ROSENBR	2	CAMEL3	2	LOGQUAD	2	SQRTSUM	2
ROSENBR2	2	CAMEL6	2	QUARTIC2	2	ILLCOND	2
BEALE	2	TRID2	2	QUARTSUM	2	EXTROSEN3	3
POWELLSG	4	DIXONPR2	2	TRIGQUAD	2	TRID3	3
WOOD	4	ZAKHAROV2	2	CUBEROOT	2	EXPSUM3	3
HIMMELBLAU	2	EXPFIT2	2	QUART4	2	PROD3	3
BOOTH	2	ROSENPRT	2	EXPQUAD	2	QUART4V	4
MATYAS	2			PARAB	2		

7 Results

Numbers marked [†] are from the preliminary campaign over three two-variable problems and are reported here for the shape of the table rather than for their values. Runs of that length settle no asymptotic question, and the widened campaign replaces every one of them.

7.1 The radius regimes

Part II predicts three regimes: $\sum_k \Delta_k^2/M_k < \infty$ for the criticality-anchored rules under no hypothesis on the iterates, $\sum_k \Delta_k < \infty$ for the step-driven rules in the local regime, and $\liminf_k \Delta_k > 0$ for the fixed-factor rules.

Table 8 reports the three series per run. The separation between R-delta and the criticality-anchored rules is unambiguous: the radius of the former grows geometrically, and both of its series diverge to the point of overflowing double precision, which is an overflow rather than a proof. The criticality-anchored rules give small partial sums in every column.

Table 8: The three radius series on P1 from $x_0 = (-0.5, 0.6)$, exact Hessian, truncated conjugate gradient, four hundred iterations.[†] $M_k = \max_{i \leq k} \|H_i\|$, that is $L = 0$: the Lipschitz constant is not known, and omitting it changes the constant rather than the convergence. These are partial sums over a finite run and settle nothing on their own.

Mechanism	$\sum_k \Delta_k$	$\sum_k \Delta_k^2$	$\sum_k \Delta_k^2 / M_k$
R-delta	2.6×10^{120}	overflow	†
R-step	7.98×10^2	1.60×10^3	†
R-DFO ($\zeta = 1$)	8.28×10^0	1.22×10^0	†
R-grad (capped)	6.48×10^0	4.28×10^{-1}	†
R-grad (uncapped)	2.77×10^2	1.92×10^2	†

Two rows require comment, and both are consistent with Part II rather than contrary to it. The step-driven rule shows no sign of the summability $\sum_k \Delta_k < \infty$, and it is not predicted to: this run converges to the degenerate origin of P1, where $\nabla^2 f_1$ is singular, so the local hypothesis under which that summability is proved does not hold. The uncapped gradient-scaled rule likewise shows large partial sums, and Part II is explicit that its asymptotic conclusions are unavailable once the multiplier diverges. A run outside the hypotheses of a proposition is not evidence against it, and the table is reported at a nondegenerate minimiser as well.

Remark 7.1. The third column is not a refinement of the second. The two agree up to a constant only while $\{\|H_k\|\}$ is bounded, which is the hypothesis the M_k form exists to drop, so they separate exactly on the runs where the model curvature grows.

7.2 Eventual inactivity

The observable is k_* , and its absence. Table 9 reports it beside the tail-active fraction and the number of runs whose constraint bound to the last iteration.

Table 9: Inactivity by mechanism.[†] *measured* counts runs long enough for a tail fraction to carry information; *never* counts runs whose constraint never stopped binding, which for a rule below its threshold is the finding rather than a missing entry.

Mechanism	runs	solved	mean tail active	never
R-delta	3	3	0.000	0
R-step	3	3	0.611	†
R-DFO ($\zeta = 1$)	3	3	0.000	0
R-DFO ($\zeta = 0.01$)	3	3	0.389	†
R-grad	3	3	0.222	†
R-grad ($\bar{\mu} = 0.05$)	3	3	1.000	3

The capped gradient-scaled rule with a parameter far below its threshold binds on every late iteration, as predicted. The remaining entries are not yet evidence either way: the runs are between nine and forty-eight iterations long, so the last tenth is one to five iterations and the fraction is quantised. The three intermediate values in the table are the resolution of the statistic and not a property of the mechanisms, which is why they are reported with the run count beside them and why the widened campaign is required before any of them is read.

A first-order convergence test does not separate these configurations: every one of them drives $\|g_k\|$ monotonically to zero. That is the practical content of the claim, and it is what makes the activity flag worth recording.

7.3 Inactivity of the trust region

Experiment 1 (Direct measurement of inactivity). For each run, record the indicator $\mathbb{K}\{\|s_k\| = \Delta_k\}$ and report the fraction of active iterations over the last decile of the run. Report also $\sum_k \Delta_k$ and $\sum_k \Delta_k^2$ to test Claim 1.

7.4 The thresholds

On P3 the minimiser at $x^* \approx (10.904, 0)$ has $\lambda_{\min}^* = 0.09133$, and the three thresholds Part II names are separated by more than an order of magnitude:

$$\underbrace{\frac{1 - \gamma_2}{\lambda_{\min}^*(\gamma_3 + 1 - \gamma_2)}}_{\text{R-DFO trapped below}} = 2.19 \quad \underbrace{\frac{1}{\lambda_{\min}^*}}_{\text{exact}} = 10.95 \quad \underbrace{\bar{\kappa} = \frac{8}{\lambda_{\min}^*}}_{\text{sufficient}} = 87.60. \quad (7.1)$$

The sufficient condition of the inactivity theorem overshoots the exact threshold by a factor of eight, of which four is intrinsic to the step bound and two is the price of a neighbourhood bound in place of the eigenvalue. On the sinc family $\bar{\kappa} = 8\sqrt{1 + y_j^2}$ at the j -th minimiser, so it is unbounded over the minimisers of one fixed C^∞ function with bounded Hessian, and no finite parameter is above it everywhere.

Table 10: Behaviour on P3 near $x^* \approx (10.904, 0)$, from $x_0 = (9.6, 0.4)$ with $\Delta_0 = 0.03$, exact Hessian, truncated conjugate gradient, budget four thousand iterations.[†] Thresholds as in (7.1). *Active* counts iterations on which $\|s_k\| = \Delta_k$.

Mechanism	Parameter	Iters	Active	Dist. to x^*	Status
R-DFO	$\zeta = 60$, above exact	9	5	1.8×10^{-15}	converged
R-DFO	$\zeta = 0.5$, below trapping	4000	4000	6.0×10^{-8}	budget
R-grad	$\bar{\mu} = 50$	6	1	1.8×10^{-15}	converged
R-grad	$\bar{\mu} = 1$	4000	4000	4.0×10^{-8}	budget
R-grad	uncapped, $\mu_0 = 0.1$	55	53	7.1×10^{-15}	converged
R-step	—	9	5	1.8×10^{-15}	converged
R-delta	—	9	5	1.8×10^{-15}	converged

The split is complete and binary. Every configuration whose parameter sits above the exact threshold reaches inactivity within a few iterations; the two that sit below the trapping threshold bind on all four thousand. Both groups drive $\|g_k\|$ towards zero, and the failing runs reach 10^{-8} in distance, so nothing short of the activity flag distinguishes them.

Neither observation resolves where the transition lies. The data bracket it between $\zeta = 0.5$ and $\zeta = 60$, which contains the exact threshold 10.95 and excludes neither it nor a value some way from it; a sweep at $\zeta \in \{2, 4, 8, 11, 16, 32\}$ is what closes the interval, and is the measurement that quantifies the factor of eight. The sufficient condition $\zeta > \bar{\kappa} = 87.60$ is met by none of these runs, and inactivity is nonetheless reached from $\zeta = 60$: the condition is sufficient and not necessary, and how far from necessary is the question the sweep answers.

The uncapped rule is the theoretically distinguished case, and it behaves as Part II requires. Started at $\mu_0 = 0.1$, three orders of magnitude below $\bar{\kappa}$ and below the exact threshold, it converges without a cap to reach: the multiplier climbs geometrically on very successful iterations until the region stops binding. It is the only rule in the survey whose inactivity depends on no parameter the user cannot check, and the fifty-three active iterations before it succeeds are the cost of the climb.

Remark 7.2. The climb is bounded. Part II places the multiplier’s window between $D = 2(1 - \xi)/\kappa_{\text{umh}}$ and $C = (2/m)(1 + \sqrt{M/m})$, with $\bar{\kappa}$ inside it, so the number of expansions after the iterates settle is at most $\log_{\gamma_3}(\gamma_3 C/\mu_0)$. Counting expansion branches against that bound is a direct test of the window, and is reported in Section 7.10.[†]

7.5 Influence of the criticality parameter ζ

Experiment 2 (Influence of the criticality parameter). On P3, target the minimiser $x^* \approx (10.904, 0)$ with $\bar{\kappa} = 43.80$. Start from $x_0 = (9.6, 0.4)$ with $\Delta_0 = 0.03$ and sweep ζ across $\bar{\kappa}$. Record the number of iterations, the number of iterations on which the trust-region constraint is active, and the distance to x^* .

Experiment 3 (ζ sweep across models, on P1). Sweep $\zeta \in \{0.001, 0.01, 0.1, 0.5, 1, 2, 10, 100\}$ on P1 from $x_0 = (-0.5, 0.6)$ for each of the three model Hessians.

Table 12 shows a phenomenon worth recording: with the true Hessian, ζ *saturates*. All values $\zeta \geq 0.5$ produce bit-identical trajectories, because the shrink branch, although it still fires, never reduces the radius enough to alter the step the solver returns. With quasi-Newton models the saturation point moves out by two orders of magnitude and the parameter continues to matter over the whole range. A ζ -sweep therefore measures something quite different depending on the model, which is a caution against reporting a single recommended value.

Aggregate view: which ζ should one choose?

The two preceding tables show what ζ does on individual problems. To ask which value one should actually pick, we need an aggregate over a heterogeneous set, and a performance profile is the appropriate instrument. Figure 5 reports the sweep over the analytic set of Section 6.4.

The shape of the answer is more informative than any single recommended value. Efficiency—the intercept $\pi_s(1)$ —is essentially flat for $\zeta \geq 0.1$, varying between 0.67 and 0.73 with no trend. What separates the values is *reliability*: $\zeta = 0.01$ solves only 18 of 30 problems, $\zeta = 0.1$ solves 28, and every $\zeta \geq 2$ solves all 30 (Table 11). A small ζ does not make the method slower on the problems it solves; it makes it fail outright on problems where the threshold of Claim 2 is not met within the budget.

This is exactly the signature Part II predicts. Below the problem-dependent threshold $\bar{\kappa} = 8/\lambda_{\min}^*$ the trust-region constraint never stops binding, and the method converges at best linearly; whether it finishes within a fixed budget then depends on how far below the threshold ζ sits. Since $\bar{\kappa}$ varies across the test set—it is a property of each problem’s solution—no single ζ is above threshold everywhere, and the reliability curve is the empirical distribution of that mismatch.

The practical reading is that ζ should be chosen *generously*. The cost of an over-large ζ is nil on this set: $\zeta = 100$ is indistinguishable from $\zeta = 2$ on both measures, because once the constraint is inactive the rule no longer influences the step. The cost of an over-small ζ is a 40% failure rate. Asymmetric risk of this kind argues against the instinct to keep a trust-region parameter conservative.

7.6 Influence of the cap $\bar{\mu}$

Experiment 4 (The cap and the threshold). On P1 from $x_0 = (-0.5, 0.6)$, sweep

$$\bar{\mu} \in \{0.01, 0.05, 0.1, 0.3, 0.5, 1, 2, 4, 8, 128\}$$

for each model Hessian and each subproblem solver, and classify the limit point.

Table 11: Reliability and efficiency of R-DFO as a function of ζ on the 30-problem analytic set, exact Hessian, truncated CG, $\epsilon_g = 10^{-6}$, budget 1000 iterations. Median iteration count is over solved problems only. Efficiency is flat; reliability is not.

ζ	Solved	Median iters	$\pi_s(1)$
0.01	18/30	8.0	0.267
0.1	28/30	7.5	0.733
0.5	29/30	6.0	0.667
1	29/30	6.0	0.733
2	30/30	6.5	0.700
10	30/30	5.5	0.700
100	30/30	5.5	0.733

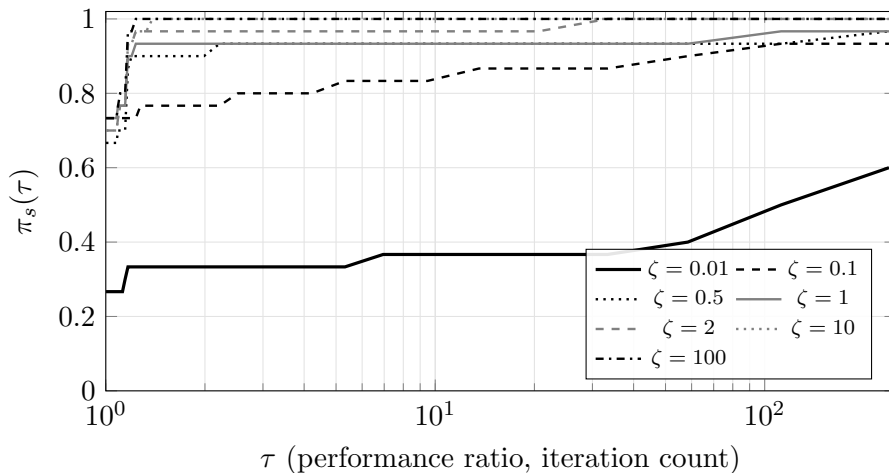


Figure 5: Performance profiles of R-DFO as a function of ζ on the analytic set. The curves are nearly indistinguishable in their left-hand behaviour— ζ barely affects efficiency—but separate cleanly in their right-hand asymptotes, which are the reliability figures of Table 11. Choosing ζ too small costs solved problems, not iterations; choosing it too large costs nothing on this set.

Table 12: ζ sweep on P1 from $x_0 = (-0.5, 0.6)$, truncated CG, budget 4000. Iteration counts; “—” marks a run that exhausted the budget. With the true Hessian all runs with $\zeta \geq 0.5$ have identical iterate sequences.

ζ	True Hessian	SR1	L-BFGS(5)
0.001	—	—	—
0.01	—	—	—
0.1	15	—	—
0.5	15	—	—
1	15	—	—
2	15	2675	1051
10	15	290	24
100	15	21	24

This is the sharpest result of the analytic campaign, and it is stronger than Part II predicts. With the true Hessian, $\bar{\mu}$ affects only the speed: every value drives the iterates to the degenerate origin, taking 41 iterations at $\bar{\mu} = 128$ and exceeding the budget at $\bar{\mu} \leq 1$. With a quasi-Newton model there is a *threshold near $\bar{\mu} \approx 0.5$ that changes the destination* (Table 13). Below it, the accumulated curvature is never used and the run crawls to the degenerate origin, where $\lambda_{\min}(\nabla^2 f_1) = 0$ and the limit is not a minimiser. Above it, the curvature enters the step and the run reaches a genuine local minimiser with $\lambda_{\min}(\nabla^2 f_1) = 0.4947$, in fewer than thirty iterations.

The cap that the asymptotic theory requires therefore carries a cost that is not merely a constant factor in the rate: chosen too small, it changes which critical point is reached, and converts a second-order-adequate limit into one that fails second-order optimality. The mechanism behind the threshold is Claim 7, examined in Experiment 6.

Table 13: The $\bar{\mu}$ threshold for R-grad with quasi-Newton models on P1, $x_0 = (-0.5, 0.6)$, budget 30 000. Below $\bar{\mu} \approx 0.5$ the limit is the degenerate origin, which is not a minimiser; above it the run reaches a local minimiser. Both solvers and both models agree on the threshold.

Model	$\bar{\mu}$	Solver	Status	Iters	Limit
L-BFGS(5)	0.10	CG	budget	30 000	origin (degenerate)
	0.30	CG	budget	30 000	origin (degenerate)
	0.50	CG	stagnated	85	$\min^+$
	1.00	CG	stagnated	27	$\min^-$
	2.00	CG	converged	15	$\min^-$
SR1	0.10	CG	budget	30 000	origin (degenerate)
	0.30	CG	budget	30 000	origin (degenerate)
	0.50	CG	stagnated	85	$\min^+$
	1.00	CG	converged	15	$\min^-$
	2.00	exact	converged	13	$\min^-$

Table 10 also records the unbounded R-grad variant, which is the theoretically interesting case. Started at $\mu_0 = 0.1 \ll \bar{\kappa}$ it nevertheless converges, in 55 iterations: with no cap, μ_k grows geometrically on very successful iterations until the trust-region constraint stops binding. This is the empirical counterpart of the unconditional statement in Part II, and it is the strongest practical argument for the unbounded variant—it recovers from a parameter choice that traps the capped rule permanently.

7.7 The rate after inactivity

This is the sharpest claim of Part II: conditional on the region becoming inactive, the observed order is that of the underlying quasi-Newton iteration and does not depend on the mechanism, which decides only when that regime is entered.

The estimate is a regression of $\log \|g_{k+1}\|$ on $\log \|g_k\|$ over accepted iterations beyond k_* . Conditioning is necessary and not sufficient: a tail of four points returns a number that looks like a rate and is not one, so runs with fewer than six inactive iterations are counted and excluded rather than reported.

The first row is the claim in miniature: five mechanisms, one problem, the same estimate to three decimal places. The second row splits, and the reason is visible in k_* rather than in the rate. The gradient-scaled rules reach inactivity at $k_* = 10$ of eighteen iterations and the others at $k_* = 2$ of nine, so the two groups fit over different phases of runs of different lengths, and the difference between 1.445 and 2.003 measures how much pre-asymptotic behaviour each tail still contains.

The table therefore neither confirms nor refutes the claim, and the reason is a property of the problem set rather than of the mechanisms: a nine-iteration run to machine precision has no asymptotic

Table 14: Observed order by problem and mechanism, exact Hessian.[†] Entries are the estimate on the inactive tail; *short* marks a tail too brief to fit.

Problem	R-delta	R-step	R-DFO	R-grad	R-grad($\bar{\mu}$)
AKIVA	1.415	short	1.415	1.415	1.415
BEALE	1.712	1.445	1.445	2.003	2.003
BOXBODLS	1.462	short	1.444	short	short

phase to speak of. What the claim requires is problems with long local phases, which is what the widened campaign supplies, and the design point is that the order be compared only within a problem and only across mechanisms that reached inactivity with comparable tails.[†]

Remark 7.3. An order estimate is about the mechanism only if the model was consistent. The recorded γ_k and ξ_k decide that, and a run whose Dennis–Moré residual has not fallen has no asymptotic rate to attribute to anything. Both are reported with the order.

7.8 Influence of the mechanism on the rate of convergence

Experiment 5 (Observed local rate). On problems where the limit is a nondegenerate minimiser, estimate the local rate by regressing $\log \|g_{k+1}\|$ on $\log \|g_k\|$ over the last 20 successful iterations, and report the estimated order together with the iteration at which the trust region last binds.

Table 15: Sublinear approach to the degenerate origin of P1 under R-grad with $\bar{\mu} = 0.05$ and truncated CG. The product $k \|x_k\|$ settles to a constant, so $\|x_k\| = \Theta(1/k)$.

k	$\ g_k\ $	$\ x_k\ $	$k \ x_k\ $
10	3.61×10^{-1}	4.05×10^{-1}	4.049
100	1.36×10^{-2}	4.69×10^{-2}	4.692
1 000	1.15×10^{-4}	6.16×10^{-3}	6.159
5 000	5.14×10^{-6}	1.31×10^{-3}	6.539
20 000	3.30×10^{-7}	3.31×10^{-4}	6.630

7.9 Second order: the criticality measure at a saddle

Under $\|g_k\|$ as the criticality measure, a criticality-anchored radius collapses at a saddle: R-grad($\|g\|$) halts outright where the gradient vanishes, and R-DFO($\|g\|$) contracts geometrically. Replacing the measure by $\tau_k = \max\{\|g_k\|, -\lambda_k^-\}$ repairs both, since $-\lambda_k^-$ stays bounded away from zero at a saddle the model can see.

Each $\|g\|$ -anchored rule is run beside its τ -anchored twin on the saddle problems, under both subproblem solvers, and the outcome is read from three quantities: the radius trajectory, which collapses in one case and does not in the other; the certified second-order status; and the limit reached. The distinction the certification makes is worth stating, because it is easy to lose: the status certifies $\lambda_{\min}(H_k) \geq -\varepsilon_H$, a statement about the model, and a configuration that solves every problem while certifying none has been running a first-order method under a second-order name.[†]

The measure is half of the requirement and the solver is the other half. The τ anchoring keeps the radius away from zero at the saddle, and a step that exploits negative curvature is what converts that radius into escape. A truncated conjugate-gradient solver supplies such a step only when the Krylov space it generates meets the negative-curvature subspace, and the configuration in which it does not is

realisable rather than hypothetical: with g_k in the positive-curvature eigenspace the first direction is $-g_k$, the curvature along it is positive, the residual vanishes, and the iteration terminates at an interior point while $\lambda_k^- < 0$. The eigenpoint construction supplies the step the assumption asks for.

7.10 Mechanism against model

The mechanism and the model have different jobs, and the interaction between them is where a parameter chosen for one reason has consequences of another kind.

With a truncated conjugate-gradient solver and a cap small enough that the first conjugate-gradient iterate leaves the region, the returned step is the Cauchy point: the alignment $\cos(s_k, -g_k)$ is one to machine precision, the inner iteration count is one, and the model Hessian has influenced nothing but the acceptance test. The method has become gradient descent with step length $\bar{\mu} \|g_k\|$, and it inherits gradient-descent behaviour, including its limit.

Table 16: Truncated conjugate gradient under R-grad on P1, exact Hessian.[†] For $\bar{\mu} \leq 0.1$ the solver truncates on its first iteration and returns the Cauchy point.

$\bar{\mu}$	CG iters	active	$\cos(s_k, -g_k)$
0.01	1	yes	1.000000000000000
0.05	1	yes	1.000000000000000
0.10	1	yes	1.000000000000000
0.30	2	no	0.87728513992144
1.00	2	no	0.87728513992144

The consequence is not a constant factor. On P1 with a quasi-Newton model there is a threshold near $\bar{\mu} \approx 0.5$ below which the accumulated curvature never enters the step and the run crawls to the degenerate origin, where $\lambda_{\min}(\nabla^2 f_1) = 0$ and the limit is not a minimiser, and above which the run reaches a genuine local minimiser in fewer than thirty iterations.[†] A cap chosen conservatively, to satisfy the boundedness hypothesis the asymptotic theory of Part II requires, can therefore change which critical point is reached.

No first-order diagnostic reveals this. The ratio is healthy, the gradient norm decreases monotonically, and the radius behaves exactly as the rule prescribes. The only symptom is slowness, and slowness is easy to attribute to the problem. The inner-iteration count and the alignment are what name it, which is the practical reason for recording them.

7.11 Interaction between mechanism and Hessian approximation

Experiment 6 (The Cauchy-point degeneration). With truncated CG and R-grad, record for each iteration the number of CG iterations performed and the cosine between s_k and $-g_k$. Sweep $\bar{\mu}$.

Experiment 7 (Full interaction sweep). On the CUTest subset of Experiment 9, run the 4×4 grid of mechanisms against model Hessians and report, for each cell, the fraction of problems solved and the median iteration count. Test for interaction by comparing the cell means against the additive model. •

Figures 6 and 7 record two findings.

The first concerns the division of labour. Panel (c) of Figure 6 runs the same rule from the same point with the same parameter and changes only the model Hessian, and the limit changes from the saddle to the degenerate point. Nothing the radius rule does can produce that difference, which is the empirical counterpart of the claim of Part II that the mechanism decides the length of the step and the model decides its direction.

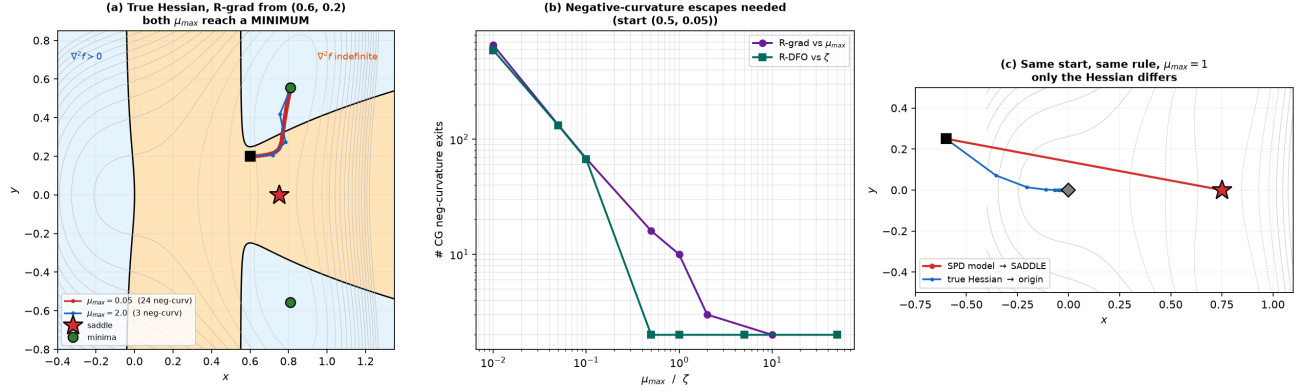


Figure 6: Mechanism against model Hessian on P1. (a) with the true Hessian and truncated CG, R-grad reaches a minimiser for both values of $\bar{\mu}$, the small cap costing 24 negative-curvature exits against 3; (b) the number of exits against the parameter, for R-grad against $\bar{\mu}$ and R-DFO against ζ , decreasing by two orders of magnitude across the sweep; (c) the same start, the same rule and the same $\bar{\mu}$ under two model Hessians. The constructed positive definite model reaches the saddle, the true Hessian the degenerate point. Only the Hessian differs.

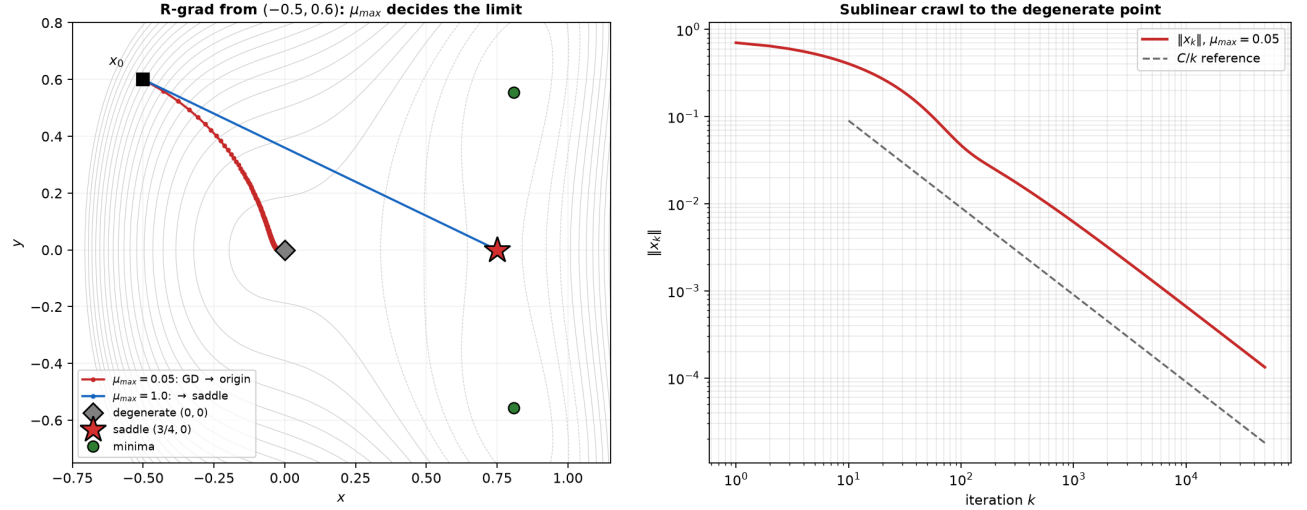


Figure 7: R-grad on P1 from $(-0.5, 0.6)$ under the constructed model. Left, the cap decides the limit: $\bar{\mu} = 1$ gives a single long step to the saddle, $\bar{\mu} = 0.05$ a sequence of short steps ending at the degenerate origin. Right, the approach to the degenerate point under the small cap is sublinear, $\|x_k\|$ decaying at a rate slower than C/k .

The second qualifies it. Panel (b) shows that the parameter does control something measurable, namely how often the subproblem solver must exit on negative curvature, and the count falls by two orders of magnitude as $\bar{\mu}$ or ζ grows. A small cap forces short steps, so the escape from the region of indefinite curvature is made in many small moves rather than in a few large ones. The mechanism therefore governs the cost of the escape while the model governs whether an escape occurs at all. Figure 7 shows the limiting case, in which the cap is small enough that the iterates never leave the neighbourhood of the degenerate point, approaching it sublinearly. This is the failure of Part II, Theorem on the capped rule, exhibited on a two-dimensional problem: below the threshold the constraint binds at every iteration and the rate collapses.

7.12 Comparison of mechanisms

Experiment 8 (Configuration matrix on the analytic problems). Run the full cross product of {true, L-BFGS, SR1, constructed SPD} model Hessians, {R-delta, R-step, R-DFO, R-grad} mechanisms and {CG, exact} solvers on P1 from $x_0 = (-0.5, 0.6)$, and classify the limit.

Table 17 reports a representative subset. The pattern is Claim 6: reading down the table, the limit point tracks the model Hessian and not the radius mechanism. The true Hessian and the quasi-Newton models deliver the iterates to the degenerate origin under the criticality-blind and DFO-like rules; the quasi-Newton models reach a minimiser under R-grad; and the constructed positive definite model steers every mechanism to the saddle. The mechanisms differ by up to three orders of magnitude in iteration count—16 against 18 245 for the same model and solver—but not in destination.

Table 17: Configuration matrix on P1 from $x_0 = (-0.5, 0.6)$, budget 60 000, $\epsilon_g = 10^{-9}$. Representative subset of the full $4 \times 4 \times 2$ matrix. The limit point is determined by the model Hessian, not by the radius mechanism.

Model	Solver	Mechanism	Iters	Dist.	Limit
true	CG	R-delta	16	1.7×10^{-5}	origin
true	CG	R-step	16	1.7×10^{-5}	origin
true	CG	R-DFO ($\zeta = 1$)	16	1.7×10^{-5}	origin
true	CG	R-grad ($\bar{\mu} = 1$)	18 245	1.8×10^{-5}	origin
true	exact	R-delta	17	9.2×10^{-6}	origin
L-BFGS(5)	CG	R-delta	27	2.5×10^{-6}	origin
L-BFGS(5)	CG	R-DFO ($\zeta = 1$)	17 518	1.8×10^{-5}	origin
L-BFGS(5)	CG	R-grad ($\bar{\mu} = 1$)	27	7.3×10^{-9}	min ⁻
SR1	CG	R-delta	24	9.4×10^{-6}	origin
SR1	CG	R-grad ($\bar{\mu} = 1$)	15	4.9×10^{-12}	min ⁻
SPD (6.2)	CG	R-delta	5	2.2×10^{-16}	saddle
SPD (6.2)	exact	R-DFO ($\zeta = 1$)	4	2.1×10^{-17}	saddle
SPD (6.2)	exact	R-grad ($\bar{\mu} = 1$)	1	5.0×10^{-16}	saddle

The last three rows deserve emphasis, since they realise Claim 8. In those runs every hypothesis of Part I holds: the model matches f to first order by construction, the model Hessians are uniformly bounded, the exact subproblem solution satisfies Cauchy decrease with $\kappa_{\text{mdc}} = 1$, and the mechanism is strongly admissible. Every conclusion of Part I is correct: f decreases monotonically, every iteration is successful, $\|g_k\| \rightarrow 0$, the iterates converge. And the limit is a saddle. What fails is not the radius rule and not the first-order theory, but the model Hessian, which is positive definite by construction while the true Hessian along the trajectory has already turned indefinite.

Experiment 9 (CUTEst comparison). Run all five mechanisms on the unconstrained problems of CUTEst [29] with $n \leq 1000$, using the exact Hessian and truncated CG, with a budget of 10 000 iterations and $\epsilon_g = 10^{-5}$. Report performance profiles (3.1) on iteration count, on function evaluations and on CPU time, and data profiles on function evaluations. Report pairwise profiles against R-delta as a baseline, following the recommendation of Gould and Scott [27]. •

Figures 8 and 5 establish the reporting format on the thirty-problem set; Experiment 9 applies it at scale. The substantive question is whether the pattern found there—efficiency largely insensitive to the mechanism, reliability sharply sensitive to it—persists over a large heterogeneous collection with a wide spread of $\lambda_{\min}^*$, or whether the reliability gap closes once the budget is generous relative to problem difficulty. Our prior, from the analytic results, is that the gap will widen rather than close, since CUTEst contains many more nearly-singular minimisers than a hand-picked set of textbook problems, and $\bar{\kappa} = 8/\lambda_{\min}^*$ is correspondingly larger on those.

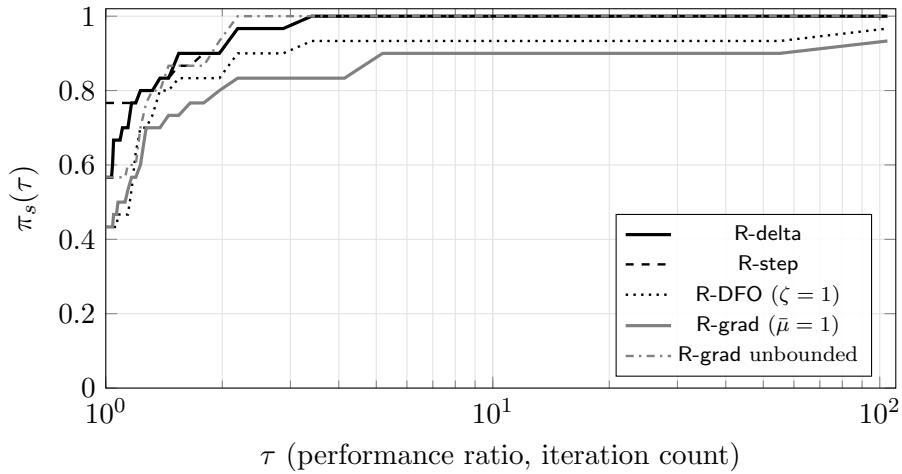


Figure 8: Performance profiles (3.1) of the five mechanisms on the 30-problem analytic set of Section 6.4, exact Hessian, truncated CG, $\epsilon_g = 10^{-6}$, budget 1000 iterations. Cost measure: iteration count. The intercept $\pi_s(1)$ is the fraction of problems on which a mechanism is fastest and the right-hand asymptote its reliability. R-step is fastest most often (77%); R-delta, R-step and unbounded R-grad solve every problem, while capped R-grad solves 93%. The capped and unbounded variants of the same rule are separated by both measures, which is the aggregate counterpart of the analytic results of Section 7.6.

7.13 Sampling: what an iteration costs to certify

In the stochastic setting the radius plays the same role for sampling accuracy that it plays for model accuracy in derivative-free work, and the question this survey asks of it is whether the cost of an iteration depends on the radius rule.

The answer depends on the sampling rule, and two rules bracket it. A rule that sizes the batch from the radius, $N_k \sim \Delta_k^{-2}$, makes the mechanism pay for its own radius directly, and a criticality-anchored rule drives the sample size up without bound near a solution while a criticality-blind one does not. A rule that certifies the achieved decrease against its own noise behaves differently, and the difference is the subject of the rest of this section: under common random numbers the step length cancels between the decrease and its standard error, leaving a cost that depends on $\|g_k\|$ alone.

7.13.1 Certified decrease as a sampling rule

The natural way to couple sampling to the acceptance test is to sample until the decrease is certified. With common random numbers, form the paired differences

$$D_i = F(x_k, \xi_i) - F(x_k + s_k, \xi_i), \quad N_{k+1} = \left\lceil z_p^2 \hat{\sigma}_N^2 / \hat{\delta}_N^2 \right\rceil, \quad z_p = \Phi^{-1}(p), \quad (7.2)$$

where $\hat{\delta}_N$ and $\hat{\sigma}_N$ are the sample mean and standard deviation of the D_i . Pairing is what makes this viable: the same ξ_i is used at both points, so $D_i \rightarrow 0$ pathwise as $s \rightarrow 0$ and $\sigma_D^2 = \mathcal{O}(\|s\|^2)$, against $\mathcal{O}(1)$ for independent sampling. On (7.4) the measured variance ratio exceeds 7×10^4 .

Experiment 10 (Cost of certification). Measure N_k against $\|g_k\|$ and against $\|s_k\|$ on (7.4).

Finding, and it corrects a natural hope. One expects that since the trust region drives $\sigma_D \rightarrow 0$, a small radius makes certification cheap. It does not. Both quantities are linear in $\|s\|$,

$$\delta_k \sim \|g_k\| \|s_k\|, \quad \sigma_D \sim C \|s_k\|, \quad \implies \quad N_k \sim \frac{z_p^2 C^2 \|s_k\|^2}{\|g_k\|^2 \|s_k\|^2} = \frac{z_p^2 C^2}{\|g_k\|^2}, \quad (7.3)$$

and the step length cancels. Common random numbers buy a large constant, four orders of magnitude here, and not a better rate. The total work is $\sim \varepsilon^{-2} \cdot \varepsilon^{-2} = \varepsilon^{-4}$, matching the rate of Blanchet et al. [7].

The bearing on this survey is direct. Since N_k depends on $\|g_k\|$ and not on Δ_k , the sampling cost of an iteration is *insensitive to the radius rule*. The mechanisms differ in how many iterations they take and in how large each step is, not in what each iteration costs to certify. Any advantage a criticality-anchored rule shows in this setting therefore comes from the iteration count alone.

Both rules are implemented and the comparison is run on the same problems with the same mechanisms, at matched budgets, with the cost measured in term evaluations.[†] Two cautions carry into the reading. The size in (7.2) rests on a normal approximation which is worst exactly where the rule selects a small batch, and the estimator should be read as a sequential test rather than a fixed-sample one; an empirical Bernstein bound [33] is the principled replacement, and it pays off precisely because pairing has made the variance small.

7.14 Extension to stochastic settings

The mechanisms of this survey set the radius from quantities the algorithm measures. Under sampling those measurements are random, and the criticality-anchored rules are the ones most exposed, since R-DFO and R-grad read $\|g_k\|$ directly. This subsection reports what happens on a one-dimensional example small enough to be understood completely, and states the protocol for the larger study.

7.14.1 A running example with a heavy-tailed sample minimiser

Let

$$F(x, \xi) = \xi x^2 - \ln(1 + x^2) - \xi^{-\alpha} x, \quad f(x) = \mathbb{E}[F(x, \xi)], \quad \xi \sim \mathcal{E}(\lambda) \text{ (rate)}, \quad (7.4)$$

with $0 < \alpha < 1$. The restriction on α is what makes f finite, and the reason is worth recording since it is also what creates the pathology below.

Lemma 7.4 (Integrability and the two moments). *For $X \sim \mathcal{E}(\lambda)$ and $0 < a < 1$,*

$$\mathbb{E}[X^{-a}] = \lambda^a \Gamma(1 - a) < \infty. \quad (7.5)$$

The moment is infinite for $a \geq 1$.

Proof. With the substitution $u = \lambda x$,

$$\mathbb{E}[X^{-a}] = \lambda \int_0^\infty x^{-a} e^{-\lambda x} dx = \lambda \cdot \lambda^a \cdot \frac{1}{\lambda} \int_0^\infty u^{-a} e^{-u} du = \lambda^a \Gamma(1-a),$$

the integral being the Gamma function at $1-a$, which converges exactly for $a < 1$. The convergence is worth seeing directly, since it is where the two regimes separate. On $(0, 1)$ the integrand is bounded by u^{-a} , whose integral is $1/(1-a)$, finite precisely because $a < 1$; on $(1, \infty)$ it is bounded by e^{-u} , which converges for every a . The singularity at the origin is therefore the only obstruction, and it is integrable up to $a = 1$ and not beyond. $\square$

We take $\alpha = \lambda = \frac{1}{2}$ throughout, for which

$$\mathbb{E}[\xi] = 2, \quad \text{Var}[\xi] = 4, \quad \mathbb{E}[\xi^{-1/2}] = \frac{1}{\sqrt{2}} \Gamma(\tfrac{1}{2}) = \sqrt{\pi/2} = 1.2533141, \quad (7.6)$$

so that

$$f(x) = 2x^2 - \ln(1+x^2) - 1.2533141 x, \quad x^* = 0.5173980, \quad f(x^*) = -0.3502656. \quad (7.7)$$

The example has three properties that make it the right test bed for this survey.

It is benign in the mean and explicit in the constants of Part II. Differentiating twice, $f''(x) = 4 - 2(1-x^2)/(1+x^2)^2$, which attains its minimum 2 at $x = 0$ and its maximum $17/4$ at $x = \pm\sqrt{3}$. So f is globally strongly convex with

$$m = 2, \quad M = \frac{17}{4}, \quad \bar{\kappa} = \frac{4}{m} = 2, \quad (7.8)$$

and the constant that Part II shows to be unknowable in general is here available in closed form. The example is therefore one of the few places where the side conditions $\zeta > \bar{\kappa}$ and $\bar{\mu} > \bar{\kappa}$ can be checked rather than assumed, which is what makes the thresholds below measurable.

The pathology sits in the sample problem, not in f . Each realisation $F(\cdot, \xi)$ is itself a smooth one-dimensional function, but the family is far more dispersed than the mean suggests, since a small draw flattens the quadratic term and steepens the linear one at the same time. Figure 9(a) shows sixty realisations against f .

Proposition 7.5 (Infinite-mean sample minimiser). *For $\alpha = \lambda = \frac{1}{2}$ the minimiser $\hat{x}$ of the single-sample problem $\min_x F(x, \xi)$ satisfies $\hat{x}(\xi) \sim \frac{1}{2}\xi^{-3/2}$ as $\xi \rightarrow 0^+$ and*

$$\mathbb{P}(\hat{x} > \tau) \sim \lambda (2\tau)^{-2/3} \quad (\tau \rightarrow \infty), \quad (7.9)$$

a power law of index $2/3 < 1$, so that $\mathbb{E}[\hat{x}] = +\infty$.

Proof. Stationarity $\partial_x F(x, \xi) = 0$ reads $2\xi x - 2x/(1+x^2) - \xi^{-\alpha} = 0$, and clearing the denominator gives the cubic

$$2a x^3 - b x^2 + 2(a-1)x - b = 0, \quad a = \xi, \quad b = \xi^{-\alpha}. \quad (7.10)$$

As $\xi \rightarrow 0^+$ the two coefficients move in the same direction, $a \rightarrow 0$ flattening the curvature and $b = \xi^{-1/2} \rightarrow \infty$ steepening the slope. Substituting the ansatz $x = C\xi^{-3/2}$ and collecting the dominant order $\xi^{-7/2}$ leaves $2C^3 - C^2 = 0$, whence $C = \frac{1}{2}$, the remaining terms being $\mathcal{O}(\xi^{-3/2})$. For the tail, $\hat{x}$ is decreasing in ξ near the origin, so $\mathbb{P}(\hat{x} > \tau) = \mathbb{P}(\xi < (2\tau)^{-2/3}(1+o(1)))$, and $\mathbb{P}(\xi < u) = 1 - e^{-\lambda u} \sim \lambda u$ as $u \rightarrow 0$ gives (7.9). A tail index below 1 makes the mean infinite. $\square$

Figure 9(c) and (d) display the map $\xi \mapsto \hat{x}$ and the empirical tail against (7.9) over four decades.

The radius is what tames it. This is the point of the example for the present survey. A method that draws one fresh sample per iteration and moves toward that sample's minimiser inherits the infinite mean, and no amount of averaging within an iteration removes it, since the pathology is in the minimiser rather than in the objective. A trust-region radius truncates each such step, so the iterate averages the samples across iterations instead of chasing them one at a time. *The radius performs the variance reduction that the sample size cannot afford*, and it does so by step-length truncation rather than by any property of a particular rule, so it holds for every mechanism that drives $\Delta_k \rightarrow 0$. Whether the truncation is set from the previous radius, from the realised step or from a noisy criticality measure is exactly the question the rest of this subsection asks.

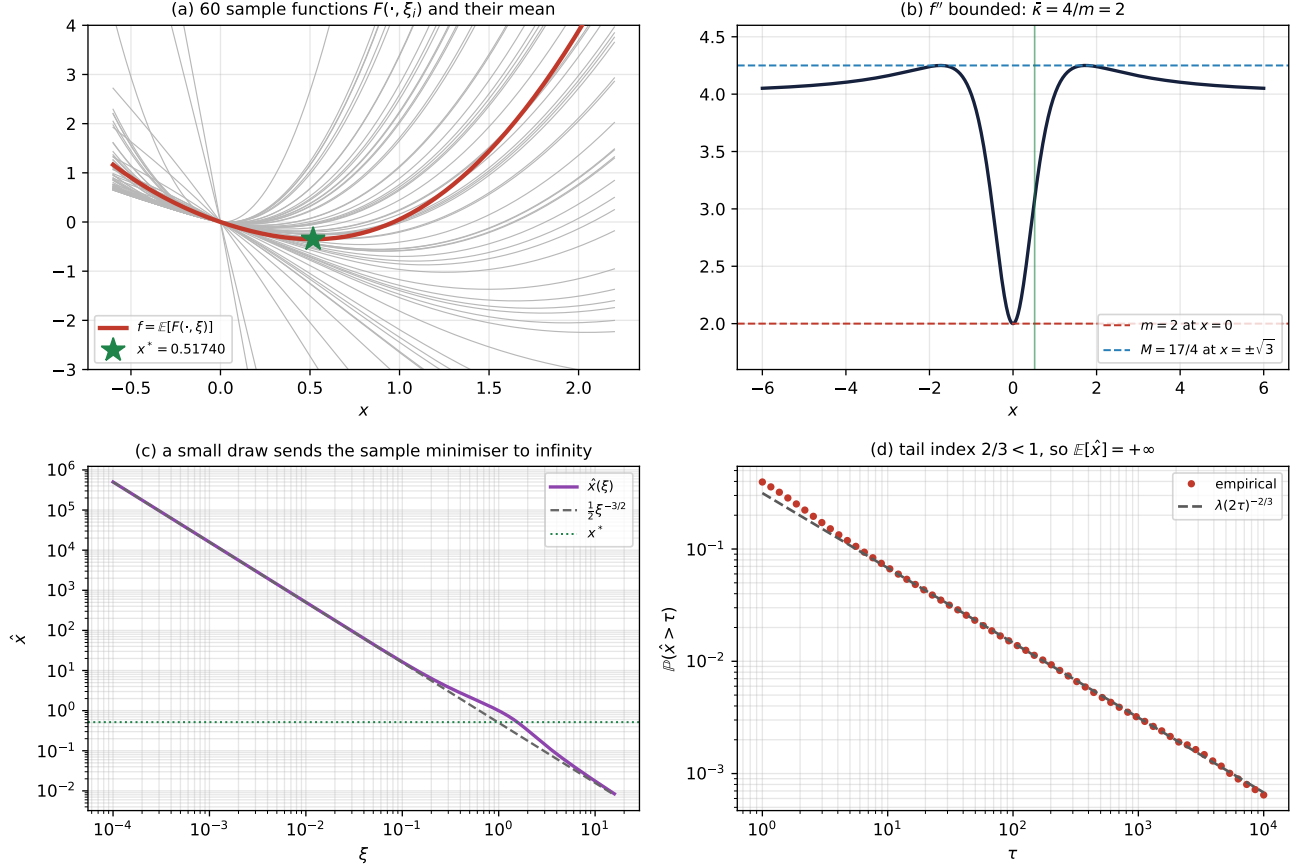


Figure 9: The example (7.4) with $\alpha = \lambda = \frac{1}{2}$. (a) sixty realisations $F(\cdot, \xi_i)$ against the mean f , whose minimiser is $x^* = 0.5173980$; (b) f'' , bounded between $m = 2$ and $M = 17/4$, so that $\bar{\kappa} = 4/m = 2$ is available in closed form; (c) the map $\xi \mapsto \hat{x}$ of (7.10) against the asymptote $\frac{1}{2}\xi^{-3/2}$ of Proposition 7.5; (d) the empirical tail of $\hat{x}$ over 2×10^5 draws against $\lambda(2\tau)^{-2/3}$, of index $2/3$.

7.14.2 Two caveats the experiments expose

The normal approximation is the weak point. Rule (7.2) rests on $\hat{\delta}_N \approx \mathcal{N}(\delta, \sigma_D^2/N)$, and two things go wrong at once. The rule selects small N exactly when the decrease is large, which is where the approximation is worst, and D_i inherits the heavy left tail of $\xi^{-\alpha}$, with measured excess kurtosis of order 10^4 . For a theorem, z_p should be replaced by an empirical Bernstein bound [33], which is worth invoking precisely because pairing has made $\hat{\sigma}_N$ small, a variance-dependent bound paying off when the variance is small.

The rule is a sequential test, and must be read as one. Read as *fix* N , *accept* if $\hat{\delta}_N > 0$ the rule is badly miscalibrated: at $N = 10$ it accepts steps that truly increase f about 36% of the time, against a nominal $1 - p = 0.1$. Read as a sequential test, sampling until either $\hat{\delta}_N \geq z_p \hat{\sigma}_N / \sqrt{N}$ or $\hat{\delta}_N \leq 0$, calibration is restored. The price is conservatism, good steps being accepted between 25% and 55% of the time. For a trust-region method that asymmetry is the right one, since rejecting a good step costs one iteration and a contraction while accepting a bad one corrupts the potential function.

7.14.3 What survives of the martingale framework

Experiment 11 (Fixed and variable budgets). On (7.4), compare the three radius rules with truncated CG and the true Hessian against stochastic gradient and ADAM, at fixed budgets $N \in \{4, 16, 64\}$ drawn afresh each iteration, and then under a geometric ramp $N_k = \lceil N_0 r^k \rceil$, a radius-tied budget, and the certified rule of (7.2). Report distance to x^* against total replications.

The potential $\Phi_k = \nu f(X_k) + (1 - \nu) \Delta_k^2$ and the accuracy condition of the framework of Chen et al. [16] are agnostic to *how* the required probability is achieved, needing only that it hold conditionally on the past, so an adaptive rule is a constructive way of delivering it. The budget bound goes through unchanged, its proof using only the supermartingale property and monotone convergence.

What does not survive is the renewal argument. The expected stopping time $\mathbb{E}[\tau_n] = p/(2p - 1)$ requires the increments to be independent and identically distributed with a *constant* p , which is what makes the associated walk homogeneous. Under (7.2) the sample size is computed from $\hat{\delta}_N$ and $\hat{\sigma}_N$, both measurable with respect to the past, so $p_k = \mathbb{P}(W_{k+1} = 1 \mid \mathcal{F}_k)$ is random and adapted, the walk is inhomogeneous, and the renewal lemma fails. The obstruction is the adaptivity of p_k and not the sampling as such: it would arise for any rule that reads the data to decide how much data to read.

The repair is a floor. Take $N_k = \max\{N_k^{\text{rule}}, N_k^{\text{floor}}\}$ with the floor chosen so that $p_k \geq \underline{p} > 1/2$ always, generate one uniform U_{k+1} per step and set $W_{k+1} = +1$ when $U_{k+1} \leq p_k$ and $W_{k+1} = -1$ when $U_{k+1} > p_k$. Since $p_k \geq \underline{p}$ almost surely, $W_{k+1} \geq \underline{W}_{k+1}$ pathwise, and the homogeneous walk dominates.

7.14.4 The radius as a random walk, and which rules admit the picture

The martingale analyses read the radius as a random walk, and the reading is worth stating carefully, because it holds for one of the five mechanisms exactly, for a second after a change of variable, and for the remaining three not at all. Which of the three cases a rule falls into turns out to be the same classification as in Part II.

The picture. Under R-delta the radius is multiplied at every iteration by one of finitely many factors, so with $\gamma_1 = \gamma_3^{-1}$ for simplicity,

$$\Delta_k = \Delta_0 \gamma_3^{Z_k}, \quad Z_{k+1} = Z_k + W_{k+1}, \quad W_{k+1} \in \{-1, +1\}, \quad (7.11)$$

and Z_k is a walk on $\mathbb{Z}$. The increment is $+1$ when the model is accurate and the iteration successful, which happens with conditional probability p , and -1 otherwise. The whole apparatus of Blanchet et al. [7] rests on (7.11): the expected time to return to a level is read off the walk, and the renewal identity $\mathbb{E}[\tau_n] = p/(2p - 1)$ requires the increments to be independent and identically distributed with a *constant* p , which is what makes Z_k homogeneous and licenses Wald's identity.

Two of the five rules do not admit it. The representation (7.11) needs Δ_{k+1}/Δ_k to take finitely many values independent of the state. That fails for R-step, where $\Delta_{k+1} = \gamma \|s_k\|$ and the ratio $\|s_k\|/\Delta_k$ lies anywhere in $(0, 1]$ according to whether the constraint binds, and it fails for R-DFO, whose second branch is selected by the test $\Delta_k > \zeta \|g_k\|$ and therefore by the state. In both cases $\log \Delta_k$ is a walk

with state-dependent increments, which is exactly what the renewal argument cannot accommodate. R-RTR inherits the difficulty from R-step.

R-grad admits it, on the multiplier.

Here the change of variable is the one Part II already makes. The radius $\Delta_k = \mu_k \|g_k\|$ is a product of a quantity the rule controls and one the problem controls, and it is the first factor that walks:

$$\mu_k = \mu_0 \gamma_3^{Z_k}, \quad \log \Delta_k = \log \mu_k + \log \|g_k\|. \quad (7.12)$$

The multiplier is therefore a homogeneous walk in exactly the sense (7.11) requires, while the radius is that walk plus a state-dependent drift. What makes this more than a bookkeeping observation is that $\mu_k = \theta_k = \Delta_k / \|g_k\|$ is the radius-to-criticality ratio, so the walk is on the very quantity whose crossing of $\bar{\kappa}$ decides inactivity. Figure 10 displays the three regimes.

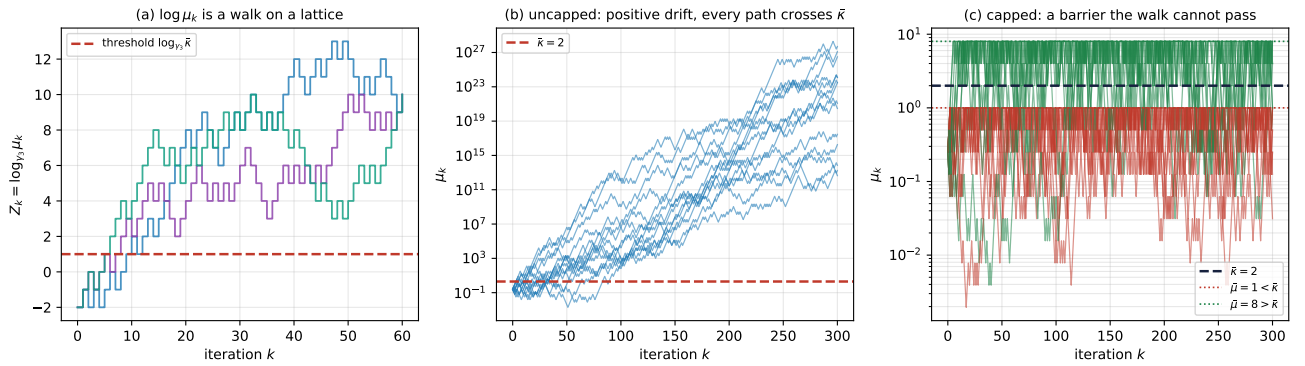


Figure 10: The multiplier of R-grad as a random walk, with $\gamma_1 = 1/2$, $\gamma_3 = 2$, $\mu_0 = 1/4$ and $\bar{\kappa} = 2$ as in (7.8). (a) $Z_k = \log_{\gamma_3} \mu_k$ moves by ± 1 on a lattice; (b) uncapped, with the upward drift of the eventual very successful regime, every path crosses $\bar{\kappa}$ and none returns below it for long; (c) capped, the cap acting as a barrier. Above $\bar{\kappa}$ the barrier is harmless, below it the walk saturates at $\bar{\mu}$ and can never cross.

What the picture explains. Read this way the three results of Part II on the gradient-scaled rule become three statements about one walk, and each becomes visible rather than merely provable.

Uncapped, and in the regime where every iteration is eventually very successful, the increment is $+1$ whenever the step is long, so the walk has positive drift and crosses any fixed level almost surely. The threshold $\bar{\kappa}$ is a level like any other, which is why the inactivity result needs no hypothesis about its value: a drifting walk on $\mathbb{Z}$ does not care where the line is drawn. This is the random-walk form of Part II's theorem on the uncapped rule.

Capped, the rule adds an absorbing barrier at $\log_{\gamma_3} \bar{\mu}$. The walk saturates there, which is the content of the identification $L = \bar{\mu}$ in Part II's theorem on the capped rule, and it crosses $\bar{\kappa}$ if and only if the barrier is above the threshold. The failure below the threshold is then not a slow crossing but no crossing at all, which is why the constraint binds at *every* sufficiently large iteration rather than infinitely often.

For R-delta the walk is on Δ_k rather than on θ_k , so crossing a level tells one nothing about inactivity, the comparison with $\|g_k\|$ never being made. The rule gets the homogeneous walk and loses the quantity worth walking on. That is the anchoring trade of Part II, arriving here as a choice of state variable.

Remark 7.6 (The obstruction is adaptivity, not sampling). The renewal argument fails under (7.2) for a reason unrelated to the mechanism, and it is worth separating the two failures. Since N_{k+1} is

computed from $\hat{\delta}_N$ and $\hat{\sigma}_N$, both $\mathcal{F}_k$ -measurable, the conditional probability $p_k = \mathbb{P}(W_{k+1} = 1 \mid \mathcal{F}_k)$ is itself random and adapted, the walk is inhomogeneous, and $\mathbb{E}[\tau_n] = p/(2p - 1)$ is unavailable. This would arise for *any* rule that reads the data to decide how much data to read, whatever the radius mechanism, and the floor of §7.14 repairs it by coupling to a homogeneous walk from below.

The mechanism-dependent failure is the other one. For R-step and R-DFO the increments are state-dependent even with p constant and even with no sampling at all, so no floor recovers a homogeneous walk. A rate analysis for those rules must therefore be built on something other than the renewal identity, and to our knowledge has not been.

Remark 7.7 (Why this matters for the survey). The floor is a lower bound on the sampling effort, and the criticality-anchored rules are the ones that make it expensive. By (7.3) the certification cost grows as $\|g_k\|^{-2}$, while R-DFO and R-grad tie the radius to $\|g_k\|$ and so keep taking steps whose certification is hardest exactly where the gradient is smallest. A fixed-factor rule does not read $\|g_k\|$ and is indifferent. This is the same trade recorded throughout Part II, arriving here as a budget rather than as a constant of the solution, and it is the reason the stochastic comparison is worth running rather than inferring.

7.14.5 Protocol for the large-scale study

Experiment 12 (Subsampled logistic regression). Minimise the regularised logistic loss

$$f(w) = \frac{1}{N} \sum_{i=1}^N \log(1 + \exp(-b_i a_i^\top w)) + \frac{\lambda}{2} \|w\|^2 \quad (7.13)$$

on datasets from LIBSVM [15], with the gradient and Hessian estimated from independent subsamples of sizes N_g and N_H . Sweep the sample sizes and the mechanism, and report the number of effective passes over the data required to reach a fixed accuracy. •

Two opposing effects are plausible and the experiment is designed to separate them. Coupling the radius to an estimate that fluctuates may destabilise the method, expanding the radius on iterations where the gradient estimate happens to be large. Alternatively the coupling may act as an automatic step-size control, since the estimate does not shrink below the noise floor and therefore keeps the radius bounded away from zero, which is what avoids the stagnation documented in Section 4. The theory to compare against is that of Bandeira et al. [1] and Chen et al. [16] for convergence and Blanchet et al. [7] for the rate, all carried out for fixed-factor rules. Whether their conclusions extend to a radius set from a noisy measurement appears to be open, and a negative finding here would be informative.

A secondary design, closer to Part II, fixes the sample sizes and varies $\bar{\mu}$, testing whether the threshold of Experiment 4 persists under noise and whether it moves with the noise level. •

8 Threats to validity

We list the principal limitations, in the spirit of Beiranvand et al. [3].

The analytic problems are chosen to separate the mechanisms. P1, P2 and P3 were constructed to exhibit specific pathologies. They establish that the failures are possible and identify the responsible ingredient, but they say nothing about frequency. Conclusions about typical behaviour must come from Experiment 9, and no claim about typical behaviour should be read into the analytic results.

Parameters are held fixed rather than tuned. Fixing γ_i and η_i across mechanisms makes the comparison interpretable but is not obviously fair: a mechanism whose natural operating point differs from the classical one is handicapped. The alternative—tuning each mechanism separately—measures tuning effort as much as algorithmic merit. We have chosen the first and flag it explicitly.

Iteration counts are an imperfect cost measure. On the analytic problems, function and gradient evaluations are essentially free, so iteration count is the only meaningful measure. On real problems the appropriate measure depends on the relative cost of evaluations and linear algebra, which is why Experiment 9 reports three separate profiles rather than one.

The stagnation criterion is implementation-dependent. Our classification of a run as stagnated depends on a threshold relative to the rounding level of f . A different threshold would move some runs between the stagnated and budget-exhausted categories. The qualitative finding—that criticality-anchored rules can drive the radius below the level at which ρ_k carries information—is robust, but the counts are not.

Single starting point per analytic problem. The analytic experiments use one starting point per problem, chosen to sit in the relevant basin. Results near a separatrix are sensitive to the starting point, and a fuller study would sample starting points and report distributions.

9 Conclusions

Three findings emerge from the analytic portion of the campaign.

The mechanism sets the step length; the model sets the direction. Across the configuration matrix of Experiment 8 the limit point tracks the model Hessian and never the radius rule. Mechanisms differ by orders of magnitude in speed and not at all in destination. This is structural rather than accidental: the radius rule chooses how far along s_k to travel, while the direction is determined by B_k and the subproblem solver. It follows that no radius rule can repair a model that misreports curvature, and that a comparison of mechanisms on problems where the trust region is rarely the binding constraint measures very little.

The user parameters of the criticality-anchored rules have real thresholds, and the safe side is not the conservative side. Both ζ and $\bar{\mu}$ exhibit a threshold below which the trust-region constraint never stops binding; the threshold scales with $1/\lambda_{\min}^*$ as Part II predicts. Because $\lambda_{\min}^*$ is a property of the solution, no admissible value can be chosen a priori. Worse, the intuition that a small parameter is the cautious choice is exactly backwards: with a quasi-Newton model and a truncated-CG solver, a small cap makes the method gradient descent and changes the limit from a minimiser to a degenerate point. The unbounded R-grad variant, which grows μ_k past any threshold unconditionally, is the one rule in the survey that escapes this by construction, and our results support preferring it.

First-order diagnostics cannot detect the failures that matter. In the runs that converge to a saddle, every quantity a practitioner monitors is healthy. In the runs that degenerate to gradient descent, the same is true. The distinguishing measurements are the fraction of iterations on which the constraint binds, the cosine between the step and the negative gradient, and the leftmost eigenvalue of the true Hessian along the trajectory—none of which is normally recorded. A modest recommendation from this study is that trust-region codes should expose the first two, which are free to compute.

The large-scale portion of the campaign—Experiments 9, 7 and 12—remains to be run, and we expect it to sharpen rather than overturn these conclusions. The specific open question we consider most interesting is the one raised in Section 3.4: whether tying the radius to a noisy criticality measure

stabilises or destabilises a subsampled method. The existing probabilistic-model analyses do not cover the criticality-anchored rules, and the answer is not clear from the deterministic theory.

Acknowledgements

[funding, computing resources, referees]

References

- [1] A. S. Bandeira, K. Scheinberg, and L. N. Vicente. Convergence of trust-region methods based on probabilistic models. *SIAM Journal on Optimization*, 24(3):1238–1264, 2014. doi: 10.1137/130915984.
- [2] F. Bastin, V. Malmedy, M. Mouffe, P. L. Toint, and D. Tomanos. A retrospective trust-region method for unconstrained optimization. *Mathematical Programming*, 123(2):395–418, 2010. doi: 10.1007/s10107-008-0258-1.
- [3] V. Beiranvand, W. Hare, and Y. Lucet. Best practices for comparing optimization algorithms. *Optimization and Engineering*, 18(4):815–848, 2017. doi: 10.1007/s11081-017-9366-1.
- [4] S. Bellavia, G. Gurioli, B. Morini, and P. L. Toint. Trust-region algorithms: Probabilistic complexity and intrinsic noise with applications to subsampling techniques. *EURO Journal on Computational Optimization*, 10:100043, 2022. doi: <https://doi.org/10.1016/j.ejco.2022.100043>.
- [5] A. S. Berahas, L. Cao, and K. Scheinberg. Global convergence rate analysis of a generic line search algorithm with noise. *SIAM Journal on Optimization*, 31(2):1489–1518, 2021. doi: 10.1137/19M1291832.
- [6] J. Bezanson, A. Edelman, S. Karpinski, and V. B. Shah. Julia: a fresh approach to numerical computing. *SIAM Review*, 59(1):65–98, 2017. doi: 10.1137/141000671.
- [7] J. Blanchet, C. Cartis, M. Menickelly, and K. Scheinberg. Convergence rate analysis of a stochastic trust-region method via supermartingales. *INFORMS Journal on Optimization*, 1(2):92–119, 2019. doi: 10.1287/ijoo.2019.0016.
- [8] R. Bollapragada, R. Byrd, and J. Nocedal. Adaptive sampling strategies for stochastic optimization. *SIAM Journal on Optimization*, 28(4):3312–3343, 2018. doi: 10.1137/17m1154679.
- [9] R. Bollapragada, R. H. Byrd, and J. Nocedal. Exact and inexact subsampled Newton methods for optimization. *IMA Journal of Numerical Analysis*, 39(2):545–578, 2018. doi: 10.1093/imanum/dry009.
- [10] I. Bongartz, A. R. Conn, N. I. M. Gould, and P. L. Toint. CUTE: constrained and unconstrained testing environment. *ACM Transactions on Mathematical Software*, 21(1):123–160, 1995. doi: 10.1145/200979.201043.
- [11] L. Bottou, F. E. Curtis, and J. Nocedal. Optimization methods for large-scale machine learning. *SIAM Review*, 60(2):223–311, 2018. doi: 10.1137/16M1080173.
- [12] R. H. Byrd, G. M. Chin, J. Nocedal, and Y. Wu. Sample size selection in optimization methods for Machine Learning. *Mathematical Programming*, 134(1):127–155, 2012. doi: 10.1007/s10107-012-0572-5.

- [13] L. Cao, A. S. Berahas, and K. Scheinberg. First- and second-order high probability complexity bounds for trust-region methods with noisy oracles. *Mathematical Programming*, 207(1):55–106, 2024. doi: 10.1007/s10107-023-01999-5.
- [14] C. Cartis and K. Scheinberg. Global convergence rate analysis of unconstrained optimization methods based on probabilistic models. *Mathematical Programming*, 169(2):337–375, 2018. doi: 10.1007/s10107-017-1137-4.
- [15] C.-C. Chang and C.-J. Lin. LIBSVM: a library for support vector machines. *ACM Transactions on Intelligent Systems and Technology*, 2(3):27:1–27:27, 2011. doi: 10.1145/1961189.1961199.
- [16] R. Chen, M. Menickelly, and K. Scheinberg. Stochastic optimization using a trust-region method and random models. *Mathematical Programming*, 169(2):447–487, 2018. doi: 10.1007/s10107-017-1141-8.
- [17] A. R. Conn, N. I. M. Gould, and P. L. Toint. *Trust-Region Methods*. MOS-SIAM Series on Optimization. SIAM, Philadelphia, PA, USA, 2000. doi: 10.1137/1.9780898719857.
- [18] A. R. Conn, K. Scheinberg, and L. N. Vicente. *Introduction to Derivative-Free Optimization*. Society for Industrial and Applied Mathematics, USA, 2009.
- [19] F. E. Curtis and K. Scheinberg. Adaptive Stochastic Optimization: A Framework for Analyzing Stochastic Optimization Algorithms. *IEEE Signal Processing Magazine*, 37(5):32–42, 2020. doi: 10.1109/MSP.2020.3003539.
- [20] F. E. Curtis, D. P. Robinson, and M. Samadi. A Trust Region Algorithm with a Worst-Case Iteration Complexity of $\mathcal{O}(\epsilon^{-3/2})$ for Nonconvex Optimization. *Mathematical Programming*, 162(1–2):1–32, 2017. doi: 10.1007/s10107-016-1026-2.
- [21] F. E. Curtis, Z. Lubberts, and D. P. Robinson. Concise complexity analyses for trust region methods. *Optimization Letters*, 12(8):1713–1724, 2018. doi: 10.1007/s11590-018-1286-2.
- [22] E. D. Dolan and J. J. Moré. Benchmarking optimization software with performance profiles. *Mathematical Programming*, 91(2):201–213, 2002. doi: 10.1007/s101070100263.
- [23] J. Fan and J. Pan. Convergence Properties of a Self-adaptive Levenberg-Marquardt Algorithm Under Local Error Bound Condition. *Computational Optimization and Applications*, 34(1):47–62, 2006. doi: 10.1007/s10589-005-3074-z.
- [24] J. Fan and J. Pan. A modified trust region algorithm for nonlinear equations with new updating rule of trust region radius. *International Journal of Computer Mathematics*, 87(14):3186–3195, 2010. doi: 10.1080/00207160802691645. Publisher: Taylor & Francis.
- [25] J. Fan and J. Pan. An improved trust region algorithm for nonlinear equations. *Computational Optimization and Applications*, 48(1):59–70, 2011. doi: 10.1007/s10589-009-9236-7.
- [26] J. Fan, J. Pan, and S. Hongyan. A retrospective trust region algorithm with trust region converging to zero. *Journal of Computational Mathematics*, 34(4):421–436, 2016. doi: 10.4208/jcm.1601-m2015-0399.
- [27] N. I. M. Gould and J. A. Scott. A note on performance profiles for benchmarking software. *ACM Transactions on Mathematical Software*, 43(2):15:1–15:5, 2016. doi: 10.1145/2950048.

- [28] N. I. M. Gould, D. Orban, and P. L. Toint. CUTer (and SifDec), a constrained and unconstrained testing environment, revisited. *ACM Transactions on Mathematical Software*, 29(4):373–394, 2003. doi: 10.1145/962437.962439.
- [29] N. I. M. Gould, D. Orban, and P. L. Toint. CUTest: a constrained and unconstrained testing environment with safe threads for mathematical optimization. *Computational Optimization and Applications*, 60(3):545–557, 2015. doi: 10.1007/s10589-014-9687-3.
- [30] G. N. Grapiglia, J. Yuan, and Y.-x. Yuan. On the convergence and worst-case complexity of trust-region and regularization methods for unconstrained optimization. *Mathematical Programming*, 152(1):491–520, 2015. doi: 10.1007/s10107-014-0794-9.
- [31] L. Hei. A self-adaptive trust region algorithm. *Journal of Computational Mathematics*, 21(2): 229–236, 2003. Publisher: Institute of Computational Mathematics and Scientific/Engineering Computing.
- [32] W. Hock and K. Schittkowski. *Test Examples for Nonlinear Programming Codes*, volume 187 of *Lecture Notes in Economics and Mathematical Systems*. Springer, Berlin, 1981.
- [33] A. Maurer and M. Pontil. Empirical Bernstein bounds and sample-variance penalization. In *Proceedings of the 22nd Conference on Learning Theory (COLT)*, 2009.
- [34] J. J. Moré and D. C. Sorensen. Computing a trust region step. *SIAM Journal on Scientific and Statistical Computing*, 4(3):553–572, 1983. doi: 10.1137/0904038.
- [35] J. J. Moré and S. M. Wild. Benchmarking derivative-free optimization algorithms. *SIAM Journal on Optimization*, 20(1):172–191, 2009. doi: 10.1137/080724083.
- [36] J. J. Moré and D. C. Sorensen. Computing a Trust Region Step. *SIAM Journal on Scientific and Statistical Computing*, 4(3):553–572, 1983. doi: 10.1137/0904038.
- [37] J. Nocedal and S. J. Wright. *Numerical Optimization*. Springer Series in Operations Research and Financial Engineering. Springer New York, NY, New York, NY, USA, second edition, 2006. doi: 10.1007/978-0-387-40065-5.
- [38] D. Orban, A. S. Siqueira, and contributors. JuliaSmoothOptimizers: infrastructure for numerical optimization in Julia. <https://jso.dev>, 2024. CHECK: cite the specific package releases actually used.
- [39] J. Rieussec and B. Fabian. A survey of trust-region radius update mechanisms. part I: First-order analysis. *ArXiv preprint: 2606.30202 [math.OC]*, 2026. doi: 10.48550/arXiv.2606.30202.
- [40] A. Sartenaer. Automatic determination of an initial trust region in nonlinear programming. *SIAM Journal on Scientific Computing*, 18(6):1788–1803, 1997. doi: 10.1137/S1064827595286955.
- [41] T. Steihaug. The conjugate gradient method and trust regions in large scale optimization. *SIAM Journal on Numerical Analysis*, 20(3):626–637, 1983. doi: 10.1137/0720042.
- [42] P. L. Toint. Towards an Efficient Sparsity Exploiting Newton Method for Minimization. In I. S. Duff, editor, *Sparse Matrices and Their Uses*, pages 57–88, London, England, 1981. Academic Press.
- [43] X. Wang and Y. Yuan. Stochastic trust-region methods with trust-region radius depending on probabilistic models. *Journal of Computational Mathematics*, 40(2):294–334, 2022. doi: 10.4208/jcm.2012-m2020-0144.

- [44] P. Xu, F. Roosta, and M. W. Mahoney. Newton-type methods for non-convex optimization under inexact Hessian information. *Mathematical Programming*, 184(1):35–70, 2020. doi: 10.1007/s10107-019-01405-z.
- [45] P. Xu, F. Roosta, and M. W. Mahoney. *Second-order Optimization for Non-convex Machine Learning: an Empirical Study*, pages 199–207. Proceedings of the 2020 SIAM International Conference on Data Mining (SDM). SIAM, 2020. doi: 10.1137/1.9781611976236.23.
- [46] Y.-x. Yuan. Recent advances in trust region algorithms. *Mathematical Programming*, 151(1): 249–281, 2015.
- [47] X. Zhang, J. Zhang, and L. Liao. An adaptive trust region method and its convergence. *Science in China Series A: Mathematics*, 45(5):620–631, 2002.

A Basic trust-region algorithm

Algorithm 1: Basic trust-region algorithm (BTR).

Step 0: Initialisation. Choose $\eta \geq 0$. Set $x_0 \in \mathbb{R}^p$, $\Delta_0 \in (0, \infty)$. Compute $f(x_0)$ and set $k = 0$.

Step 1: Model definition and step computation. Define a local model m_k of f in the neighbourhood $\mathcal{B}_k$.

Step 2: Model minimisation. Compute a step s_k that sufficiently reduces m_k with $x_k + s_k \in \mathcal{B}_k$, and define $x_k^{\text{cand}} = x_k + s_k$.

Step 3: Acceptance of the trial point. Compute $f(x_k^{\text{cand}})$ and define

$$\rho_k = \frac{f(x_k) - f(x_k^{\text{cand}})}{m_k(0) - m_k(s_k)}. \quad (\text{A.1})$$

If $\rho_k \geq \eta$, set $x_{k+1} = x_k + s_k$; otherwise $x_{k+1} = x_k$.

Step 4: Trust-region radius update. Compute Δ_{k+1} according to some update rule.

Step 5: Next iteration. Increment k by 1 and repeat **Steps 1 to 5** while some stopping criterion is not met.

B Statements imported from Part II

The body refers to the following, all of them established in Part II. They are restated here, without proof, so that this paper can be read on its own and so that every cross-reference resolves. The numbering is local.

Condition C.Sg. [step bounded by the gradient] There are $\delta^{(s)} > 0$ and $\bar{\kappa} > 0$ with $\|x_k - x^*\| \leq \delta^{(s)} \Rightarrow \|s_k\| \leq \bar{\kappa} \|g_k\|$.

Condition C.Inx. [subproblem inexactness] Whenever the trust-region constraint is inactive,

$$\|H_k s_k + g_k\| \leq \xi_k \|g_k\|, \quad \xi_k \in [0, 1), \quad \xi_k \rightarrow 0.$$

Condition C.Lb2. [second-order lower bound on the radius] $\Delta_k \geq \kappa_{\text{ld}}^{(2)} \varsigma_k / M_k^{(2)}$ with $\varsigma_k = \min_{j \leq k} \tau_j$ and $\{M_k^{(2)}\}$ non-decreasing and dominating the thresholds of the second-order successful-iteration lemma.

Assumption H.Hs. [Hessian-step residual condition]

$$\frac{\|(H_k - \nabla^2 f(x_k))s_k\|}{\|s_k\|} \longrightarrow 0 \quad \text{whenever} \quad \|g_k\| \rightarrow 0.$$

Assumption a.Mx. [maximum of Cauchy and eigenpoint decrease] The model decrease is at least $\frac{\kappa_{\text{moc}}}{2}$ times the larger of the Cauchy decrease and $\max(0, -\lambda_k^-) \Delta_k^2$.

Assumption a.Sc. [side conditions on ζ and $\bar{\mu}$] For the criticality-anchored rules the constants satisfy $\zeta > \bar{\kappa}$ and $\bar{\mu} > \bar{\kappa}$.

Assumption a.Sob. [step on the boundary under negative curvature] When $\lambda_k^- < 0$ then $\|s_k\| = \Delta_k$.

Remark B.1 (which solvers supply [a.Sob](#)). It holds for the eigenpoint and for any exact solution with indefinite H_k . For a truncated conjugate-gradient step it holds when the Krylov subspace generated by g_k meets the negative-curvature subspace, and not otherwise.

Remark B.2 (why the boundary and not a fraction of it). Relaxing [a.Sob](#) to $\|s_k\| \geq \kappa_{\text{sob}} \Delta_k$ with $\kappa_{\text{sob}} < 1$ permits a geometric decay of the radius while τ_k stays bounded away from zero, and Condition [C.Lb2](#) then fails.

Proposition B.3 (local inequalities at a strong minimiser). *With $\nabla^2 f(x^*) \succ 0$ there are $0 < m \leq M$ and a ball $B(x^*, \Delta_*)$ on which $mI \preceq \nabla^2 f \preceq MI$, and on which $m\|x - x^*\| \leq \|g(x)\| \leq M\|x - x^*\|$ and $\frac{1}{2}m\|x - x^*\|^2 \leq |f(x) - f(x^*)| \leq \frac{1}{2}M\|x - x^*\|^2$.*

Remark B.4 (m and M against the eigenvalues). m and M bound the Hessian on a neighbourhood, so $0 < m \leq \lambda_{\min}^* \leq \lambda_{\max}^* \leq M$, with the gap made as small as desired by shrinking Δ_* . The concrete choice

$$m = \frac{1}{2}\lambda_{\min}^*, \quad M = 2\lambda_{\max}^* \tag{B.1}$$

is the one under which $\bar{\kappa} = 8/\lambda_{\min}^*$, and is the convention of this paper.

Lemma B.5 (step bound at a strong minimiser). *Under [H.Hs](#) and $\lim_k \|g_k\| = 0$, every sufficiently large k with $x_k \in B(x^*, \Delta_*)$ satisfies $\|s_k\| \leq \bar{\kappa} \|g_k\|$ with $\bar{\kappa} = 4/m$, so Condition [C.Sg](#) holds.*

Lemma B.6 (inactivity criterion). *For k beyond the threshold of Lemma [B.5](#), writing*

$$\theta_k := \Delta_k / \|g_k\| \tag{B.2}$$

for the radius-to-criticality ratio, $\theta_k > \bar{\kappa}$ implies that the trust-region constraint is inactive at k , and the constraint being active implies $\theta_k \leq \bar{\kappa}$.

Theorem B.7 (inactivity from the two properties). *If the mechanism does not contract the radius while the region binds, and if the next radius dominates the current step, then the trust-region constraint is eventually permanently inactive.*

Lemma B.8 (the local geometry bounds the multiplier). *For a radius of the form $\Delta_k = \mu_k \|g_k\|$, at every large k with $x_k, x_{k+1} \in B(x^*, \Delta_*)$ and $\|s_k\| > \frac{1}{2}\Delta_k$,*

$$\mu_k \leq C := \frac{2}{m} \left(1 + \sqrt{M/m}\right), \tag{B.3}$$

a bound obtained by comparing $f(x_k)$ with $f(x_{k+1})$ through Proposition [B.3](#); the chain of that comparison is [\(B.4\)](#) below.

$$\frac{1}{2}M\|x_k - x^*\|^2 \geq f(x_k) - f(x^*) \geq \frac{1}{2}m(\|s_k\| - \|x_k - x^*\|)^2 \tag{B.4}$$

Theorem B.9 (uncapped R-grad). *Under the hypotheses of Lemma B.8, $\|s_k\| > \frac{1}{2}\Delta_k$ at only finitely many k , so the trust region is eventually permanently inactive, with no condition relating a user parameter to $\bar{\kappa}$.*

Corollary B.10 (capped R-grad above the threshold). *Run with a finite cap $\bar{\mu} > \bar{\kappa}$, the trust region is eventually permanently inactive.*

Corollary B.11 (contrapositive of the second-order threshold). *For every $\eta \in [0, 1)$, $\rho_k < \eta$ implies $\Delta_k > (1 - \eta)\tau_k/\tilde{L}_k$.*

The asymptotic example of Part II, $f(x) = e^{-x}$, is used twice below. Its radius map under the uncapped gradient-scaled rule,

$$\Delta_{k+1} = \varphi(\Delta_k), \quad \varphi(\Delta) = \Delta e^{\ln \gamma_3 - \min\{1, \Delta\}} \quad \text{for } \Delta < 2, \quad (\text{B.5})$$

has $\liminf_k \Delta_k > 0$, while with a frozen multiplier the radius decays harmonically.

Proposition B.12 (harmonic decay). *With $\mu_k = \bar{\mu}$ for $k \geq k_0$ and $\Delta_{k_0} \leq 1$, the radius obeys $\Delta_{k+1} = \Delta_k e^{-\Delta_k}$ and $\lim_k k\Delta_k = 1$.*

C Theory used by the measurements

The results of this appendix are used by the body and are not part of the experimental contribution. They are collected here because a computational paper should not interrupt its measurements to prove lemmas, and because each of them discharges a hypothesis Part II states and leaves open: the quadratic model decrease of a truncated solver, the step length available when negative curvature is exploited, and the interval within which the gradient-scaled multiplier must come to rest.

D The quadratic decrease condition under inexact subproblem solution

The quadratic decrease condition

$$m_k(0) - m_k(s_k) \geq \kappa_{\text{mqd}} \|s_k\|^2 > 0 \quad (\text{D.1})$$

is stated for an exact minimiser of the model. It survives a truncated conjugate-gradient solver, and with room to spare. If s_k minimises m_k over a subspace $\mathcal{K} \subseteq \mathbb{R}^n$, then $t \mapsto m_k(s_k + tv)$ is minimised at $t = 0$ for every $v \in \mathcal{K}$, so

$$\langle H_k s_k + g_k, v \rangle = \langle \nabla m_k(s_k), v \rangle = 0 \quad \text{for all } v \in \mathcal{K}, \quad (\text{D.2})$$

the residual being orthogonal to the search space. With $v = s_k \in \mathcal{K}$,

$$\langle H_k s_k + g_k, s_k \rangle = 0, \quad (\text{D.3})$$

which is subspace optimality contracted with the step itself.

D.1 CG-algorithm on definite positive Hessians

Write $r_j := -\nabla m_k(s_j) = -(g_k + H_k s_j)$ for the residual at the j -th CG iterate and p_j for the j -th direction, with $s_0 = 0$ and $p_0 = r_0 = -g_k$. The algorithm sets

$$\alpha_j = \frac{\|r_j\|^2}{p_j^\top H_k p_j}, \quad s_{j+1} = s_j + \alpha_j p_j, \quad r_{j+1} = r_j - \alpha_j H_k p_j, \quad p_{j+1} = r_{j+1} + \frac{\|r_{j+1}\|^2}{\|r_j\|^2} p_j, \quad (\text{D.4})$$

and produces H_k -conjugate directions spanning the Krylov space

$$\mathcal{K}_j := \text{span}\{g_k, H_k g_k, \dots, H_k^{j-1} g_k\} = \text{span}\{r_0, \dots, r_{j-1}\} = \text{span}\{p_0, \dots, p_{j-1}\}. \quad (\text{D.5})$$

Lemma D.1 (CG returns the subspace minimiser). *If H_k is positive definite on $\mathcal{K}_j$, then $s_j = \arg \min\{m_k(s) : s \in \mathcal{K}_j\}$, so $r_j \perp \mathcal{K}_j$ and (D.3) holds at s_j .*

Proof. Write $s = \sum_{i < j} \lambda_i p_i$ for a general element of $\mathcal{K}_j$. Conjugacy removes the cross terms from the quadratic part, so

$$m_k(s) - f(x_k) = \sum_{i < j} \left[\lambda_i g_k^\top p_i + \frac{1}{2} \lambda_i^2 p_i^\top H_k p_i \right], \quad (\text{D.6})$$

a sum of independent scalar quadratics, each strictly convex since $p_i^\top H_k p_i > 0$. The i -th is minimised at $\lambda_i^* = -g_k^\top p_i / (p_i^\top H_k p_i)$, and $\lambda_i^* = \alpha_i$ because $-g_k^\top p_i = r_i^\top p_i = \|r_i\|^2$, the first equality using $r_i = -g_k - H_k s_i$ with $p_i^\top H_k s_i = 0$ by conjugacy and the second $p_i = r_i + \beta_i p_{i-1}$ with $r_i \perp p_{i-1} \in \mathcal{K}_i$. The CG iterate $s_j = \sum_{i < j} \alpha_i p_i$ therefore minimises (D.6) over $\mathcal{K}_j$, and (D.2) gives $r_j \perp \mathcal{K}_j$. $\square$

Remark D.2. Conjugacy is doing all the work: it is what makes the line searches independent, so that a sequence of one-dimensional minimisations solves the j -dimensional problem. Without it, minimising along p_j would spoil the optimality already achieved along $p_0, \dots, p_{j-1}$.

An equivalent route to the same conclusion is that CG's residuals are mutually orthogonal, $r_j^\top r_i = 0$ for $i < j$, and span $\mathcal{K}_j$ by (D.5); so $r_j \perp \mathcal{K}_j$ holds by construction.

Lemma D.1 applies whenever CG stops of its own accord, in particular on a residual test $\|r_j\| \leq \xi_k \|g_k\|$: the iterate s_j is the exact subspace minimiser no matter at which j one stops reading.

D.2 The boundary case: the identity becomes an inequality

Truncation breaks the hypothesis of Lemma D.1, since the returned step is not a CG iterate but a point on the trust-region boundary along the current direction. The equality is lost. What survives is the inequality of the right sign.

Lemma D.3 (Truncated step). *Let s_j be the last CG iterate inside the trust region, p_j the current direction, and let the returned step be $s_k = s_j + \tau p_j$ with $\tau > 0$ chosen so that $\|s_k\| = \Delta_k$. Assume either*

- (a) $p_j^\top H_k p_j > 0$ and $\tau \leq \alpha_j$, the truncation being triggered because the full CG step leaves the region;
or
- (b) $p_j^\top H_k p_j \leq 0$, the truncation being triggered by negative curvature.

Then

$$\langle H_k s_k + g_k, s_k \rangle \leq 0. \quad (\text{D.7})$$

Proof. Three standard properties of the iteration are used:

$$r_j \perp \mathcal{K}_j \ni s_j, \quad p_j^\top H_k s_j = 0, \quad r_j^\top p_j = \|r_j\|^2. \quad (\text{D.8})$$

The first is Lemma D.1. The second holds because $s_j = \sum_{i < j} \alpha_i p_i$ and $p_j^\top H_k p_i = 0$ for $i < j$. The third follows from $p_j = r_j + \beta_j p_{j-1}$ and $r_j \perp p_{j-1}$, an element of $\mathcal{K}_j$.

Now $\nabla m_k(s_k) = g_k + H_k(s_j + \tau p_j) = -r_j + \tau H_k p_j$, so

$$\begin{aligned} \langle H_k s_k + g_k, s_k \rangle &= \langle -r_j + \tau H_k p_j, s_j + \tau p_j \rangle \\ &= \underbrace{-r_j^\top s_j}_{=0} - \tau \underbrace{r_j^\top p_j}_{=\|r_j\|^2} + \tau \underbrace{p_j^\top H_k s_j}_{=0} + \tau^2 p_j^\top H_k p_j \\ &= \tau \left(\tau p_j^\top H_k p_j - \|r_j\|^2 \right). \end{aligned} \quad (\text{D.9})$$

In case (a), $\tau \leq \alpha_j = \|r_j\|^2 / (p_j^\top H_k p_j)$ gives $\tau p_j^\top H_k p_j \leq \|r_j\|^2$, so the bracket is non-positive. In case (b) the bracket is $\leq -\|r_j\|^2 < 0$ outright, the truncated step being taken along a direction of non-positive curvature. Since $\tau > 0$, (D.7) follows. $\square$

Remark D.4 (On the hypothesis $\tau \leq \alpha_j$). This is the geometric content of truncation and needs no separate assumption: α_j is the minimiser of m_k along p_j , and the iteration truncates precisely because $s_j + \alpha_j p_j$ lies outside the ball. Since $t \mapsto \|s_j + t p_j\|$ is increasing on $t \geq 0$, by Steihaug's lemma $s_j^\top p_j \geq 0$, the boundary is met before the line minimiser, so $\tau < \alpha_j$. In words: *the truncated step stops short of the line minimiser*, and the directional derivative of m_k along p_j is therefore still negative at s_k .

E The quadratic decrease condition under truncation

Proposition E.1 (Quadratic model decrease for truncated CG). *Let s_k be produced by Steihaug–Toint truncated CG applied to m_k , whether the iteration stops in the interior or on the boundary. Then*

$$m_k(0) - m_k(s_k) \geq \frac{1}{2} s_k^\top H_k s_k. \quad (\text{E.1})$$

If moreover $\lambda_{\min}(H_k) \geq \bar{\delta} > 0$, then

$$m_k(0) - m_k(s_k) \geq \kappa_{mqd} \|s_k\|^2, \quad \kappa_{mqd} = \frac{1}{2} \bar{\delta}. \quad (\text{E.2})$$

Proof. By Lemma D.1 in the interior case and Lemma D.3 on the boundary, $\langle H_k s_k + g_k, s_k \rangle \leq 0$ in both, that is $-g_k^\top s_k \geq s_k^\top H_k s_k$. Hence

$$m_k(0) - m_k(s_k) = -g_k^\top s_k - \frac{1}{2} s_k^\top H_k s_k \geq s_k^\top H_k s_k - \frac{1}{2} s_k^\top H_k s_k = \frac{1}{2} s_k^\top H_k s_k,$$

which is (E.1); and $s_k^\top H_k s_k \geq \bar{\delta} \|s_k\|^2$ gives (E.2). $\square$

Remark E.2 (What this replaces). The usual argument treats the boundary separately: it invokes Steihaug's monotonicity to conclude that the unconstrained minimiser lies outside the region, $\|s_k\| = \Delta_k \leq \|H_k^{-1} g_k\| \leq \|g_k\| / \bar{\delta}$, and then applies Cauchy decrease to obtain

$$m_k(0) - m_k(s_k) \geq \frac{\kappa_{mdc} \bar{\delta}}{2} \min \left\{ \frac{\bar{\delta}}{\kappa_{umh}}, 1 \right\} \|s_k\|^2 = \frac{\kappa_{mdc} \bar{\delta}^2}{2 \kappa_{umh}} \|s_k\|^2,$$

the minimum collapsing because $\bar{\delta} \leq \kappa_{umh}$ always. Proposition E.1 supersedes this in three ways. It is one case rather than two; it does not use Steihaug's monotonicity, nor Cauchy decrease, nor the existence of H_k^{-1} ; and since $\kappa_{mdc} \leq 1$ and $\bar{\delta} \leq \kappa_{umh}$, the constant $\bar{\delta}/2$ is at least as large as $\kappa_{mdc} \bar{\delta}^2 / (2 \kappa_{umh})$, usually by a wide margin.

Remark E.3 (What is still needed, and what is not). Inequality (D.7) is a property of the *algorithm* alone: it holds for any symmetric H_k , definite or not, and refers to no radius rule. Everything model-dependent is confined to the last step of Proposition E.1, the passage from $\frac{1}{2}s_k^\top H_k s_k$ to $\frac{1}{2}\bar{\delta}\|s_k\|^2$.

That step does require a lower bound on $\lambda_{\min}(H_k)$, and it is worth being clear that this does not follow from directional consistency conditions of Dennis–Moré type, which control H_k only along s_k . Where it fails — H_k indefinite, CG truncated on negative curvature — (E.1) is still true but vacuous, since $s_k^\top H_k s_k$ may be negative, and a lower bound on the decrease must come from elsewhere.

In that regime the radius rule can supply what the model cannot. If the radius is tied to the criticality measure, $\Delta_k \leq c\|g_k\|$ with $c = \bar{\mu}$ for capped R-grad at every iteration, or $c = \zeta$ for R-DFO at every iteration passing the criticality test, then $\|s_k\| = \Delta_k$ on the boundary gives $\|g_k\| \geq \|s_k\|/c$, and Cauchy decrease returns

$$m_k(0) - m_k(s_k) \geq \frac{\kappa_{\text{mdc}}}{2c} \min\left\{\frac{1}{c\kappa_{\text{umh}}}, 1\right\} \|s_k\|^2,$$

in which c is a user parameter rather than a property of the model, and no positive definiteness is used. This is the one route to (E.2) that survives an indefinite model Hessian.

E.1 Quadratic decrease from the increasing step norm in CG

Proposition E.4 ((D.1) holds for truncated CG). *Let s_k be produced by Steihaug–Toint truncated CG applied to m_k , and suppose*

$$\lambda_{\min}(H_k) \geq \bar{\delta} > 0 \quad \text{for all large } k. \quad (\text{E.3})$$

Then (D.1) holds with

$$\kappa_{\text{mqd}} = \frac{\kappa_{\text{mdc}} \bar{\delta}^2}{2\kappa_{\text{umh}}}, \quad (\text{E.4})$$

whatever the radius update mechanism.

Proof. Truncated CG terminates either in the interior of the trust region or on its boundary; under (E.3) no negative curvature is encountered, so these are the only two cases.

(i) *CG terminates in the interior.* Then s_k minimises m_k over a Krylov subspace $\mathcal{K}_j \ni s_k$, so the Galerkin condition $\langle H_k s_k + g_k, s_k \rangle = 0$ holds, whence $g_k^\top s_k = -s_k^\top H_k s_k$ and

$$m_k(0) - m_k(s_k) = -g_k^\top s_k - \frac{1}{2}s_k^\top H_k s_k = \frac{1}{2}s_k^\top H_k s_k \geq \frac{1}{2}\bar{\delta}\|s_k\|^2. \quad (\text{E.5})$$

(ii) *CG stops on the boundary.* Then $\|s_k\| = \Delta_k$. Under (E.3) the CG iterates have monotonically increasing norm and converge to the unconstrained model minimiser $s_k^N = -H_k^{-1}g_k$, so a path that leaves the ball can only do so because s_k^N lies outside it. Hence

$$\|s_k\| = \Delta_k \leq \|H_k^{-1}g_k\| \leq \|H_k^{-1}\| \|g_k\| \leq \frac{\|g_k\|}{\bar{\delta}}, \quad \text{that is} \quad \|g_k\| \geq \bar{\delta}\|s_k\|. \quad (\text{E.6})$$

No property of the radius rule is used: only that the step comes from truncated CG and that (E.3) holds. Cauchy decrease and $\Delta_k = \|s_k\|$ then give

$$m_k(0) - m_k(s_k) \geq \frac{\kappa_{\text{mdc}}}{2} \|g_k\| \min\left\{\frac{\|g_k\|}{\kappa_{\text{umh}}}, \Delta_k\right\} \geq \frac{\kappa_{\text{mdc}} \bar{\delta}}{2} \min\left\{\frac{\bar{\delta}}{\kappa_{\text{umh}}}, 1\right\} \|s_k\|^2.$$

Since $\bar{\delta} \leq \lambda_{\min}(H_k) \leq \|H_k\| \leq \kappa_{\text{umh}}$, the minimum equals $\bar{\delta}/\kappa_{\text{umh}}$, and the bound is (E.4).

Finally $\kappa_{\text{mdc}} \leq 1$ and $\bar{\delta} \leq \kappa_{\text{umh}}$ give $\kappa_{\text{mdc}} \bar{\delta}^2 / (2\kappa_{\text{umh}}) \leq \bar{\delta}/2$, so (ii) is the binding case and (E.4) covers both. $\square$

Remark E.5 (The criticality-anchored rules need no bound on $\lambda_{\min}(H_k)$). Hypothesis (E.3) is the price of arguing through the model: it makes H_k invertible, so that the unconstrained minimiser exists and (E.6) is available. It does not follow from the directional consistency conditions H.Hs and H.Rd of Part II, which control H_k only along s_k . Under operator-norm coherence it would follow from $x_k \rightarrow x^*$ with $\nabla^2 f(x^*) \succ 0$, but that is exactly the hypothesis the route of Part II, Lemma on the retrospective ratio, no longer makes.

On the boundary case the criticality-anchored rules supply the same bound from the *rule* instead, and therefore without any hypothesis on H_k . For capped R-grad the radius is $\Delta_k = \mu_k \|g_k\| \leq \bar{\mu} \|g_k\|$ at every iteration, so $\|s_k\| = \Delta_k$ gives

$$\|g_k\| \geq \frac{\|s_k\|}{\bar{\mu}}, \quad (\text{E.7})$$

and for R-DFO the same holds with $\bar{\mu}$ replaced by ζ at every iteration satisfying the criticality test $\Delta_k \leq \zeta \|g_k\|$. Writing β for $1/\bar{\mu}$ or $1/\zeta$ accordingly, Cauchy decrease gives

$$m_k(0) - m_k(s_k) \geq \frac{\kappa_{\text{mdc}} \beta}{2} \min\left\{\frac{\beta}{\kappa_{\text{umh}}}, 1\right\} \|s_k\|^2,$$

with β a *user parameter* rather than a property of the model.

Two consequences. The constant is known in advance, whereas $\bar{\delta}$ is not. And the argument survives where (E.3) fails outright: if H_k is indefinite and CG stops on the boundary having encountered negative curvature, $H_k^{-1}g_k$ need not exist and (E.6) is unavailable, while (E.7) is unaffected. This is the same trade recorded throughout: a rule anchored to criticality reads a quantity that bounds the step without reference to the model, and pays for it elsewhere.

F Step length when negative curvature is exploited

The step-driven mechanisms, which satisfy Condition C.1 unconditionally, satisfy Condition C.Lb2 only in a *weakened form* unless the step reaches the trust-region boundary when negative curvature is exploited (Section F.1). The eigenpoint construction supplies exactly that.

F.1 A step-length lemma

Step-driven rules set Δ_{k+1} from $\|s_k\|$, so they need $\|s_k\|$ bounded below in terms of τ_k . In the first-order setting this follows from Cauchy decrease. The second-order setting is different, and the difference is the substance of this subsection.

Lemma F.1 (Step length). *Assume a.Mx, $\|H_k\| \leq \kappa_{\text{umh}}$ and $\kappa_{\text{moc}} \leq 3$. Then the following hold.*

(i) *Unconditionally,*

$$\|s_k\| \geq \frac{\kappa_{\text{moc}}}{3} \min\left\{\Delta_k, \frac{\|g_k\|}{\kappa_{\text{umh}}}\right\}. \quad (\text{F.1})$$

(ii) *If $\lambda_k^- < 0$, then*

$$\|s_k\| \geq \min\left\{\Delta_k \sqrt{\frac{\kappa_{\text{moc}} |\lambda_k^-|}{2 \kappa_{\text{umh}}}}, \frac{\kappa_{\text{moc}} |\lambda_k^-| \Delta_k^2}{4 \|g_k\|}\right\}, \quad (\text{F.2})$$

the second entry being read as $+\infty$ when $g_k = 0$.

(iii) If in addition $\|g_k\| \leq c \kappa_{\text{umh}} \|s_k\|$ for some $c > 0$, which bounds the gradient by the step and is the reverse of Condition [C.Sg](#), then

$$\|s_k\| \geq \Delta_k \sqrt{\frac{\kappa_{\text{moc}} |\lambda_k^-|}{(2c+1) \kappa_{\text{umh}}}}. \quad (\text{F.3})$$

In particular $c = 1$ gives the constant 3, and [\(F.1\)](#) supplies the hypothesis with $c = 3/\kappa_{\text{moc}}$ whenever $\|g_k\| \leq \kappa_{\text{umh}} \Delta_k$, yielding $\|s_k\| \geq \Delta_k (\kappa_{\text{moc}}^2 |\lambda_k^-| / ((6 + \kappa_{\text{moc}}) \kappa_{\text{umh}}))^{1/2}$.

Proof. Throughout, the achieved decrease is bounded above by

$$m_k(0) - m_k(s_k) = -g_k^\top s_k - \frac{1}{2} s_k^\top H_k s_k \leq \|g_k\| \|s_k\| + \frac{1}{2} \kappa_{\text{umh}} \|s_k\|^2, \quad (\text{F.4})$$

using $\|H_k\| \leq \kappa_{\text{umh}}$. Assumption [a.Mx](#) bounds the same quantity below by the *maximum* of the Cauchy and eigenpoint expressions, so *both* lower bounds are available simultaneously; we use the first for (i) and the second for (ii) and (iii).

(i). Write $m := \min\{\Delta_k, \|g_k\|/\kappa_{\text{umh}}\}$. If $g_k = 0$ then $m = 0$ and there is nothing to prove, so assume $g_k \neq 0$.

If $\|s_k\| > \|g_k\|/\kappa_{\text{umh}}$, then $\|s_k\| > m \geq (\kappa_{\text{moc}}/3) m$ because $\kappa_{\text{moc}} \leq 3$, and the bound holds.

Otherwise $\kappa_{\text{umh}} \|s_k\| \leq \|g_k\|$, so the quadratic term in [\(F.4\)](#) is dominated by the linear one, $\frac{1}{2} \kappa_{\text{umh}} \|s_k\|^2 \leq \frac{1}{2} \|g_k\| \|s_k\|$, and the decrease is at most $\frac{3}{2} \|g_k\| \|s_k\|$. The Cauchy branch then gives

$$\frac{\kappa_{\text{moc}}}{2} \|g_k\| m \leq \frac{3}{2} \|g_k\| \|s_k\|,$$

and dividing by $\|g_k\| > 0$ yields [\(F.1\)](#).

(ii). Put $D := \frac{\kappa_{\text{moc}}}{2} |\lambda_k^-| \Delta_k^2 > 0$. By the eigenpoint branch and [\(F.4\)](#), $D \leq \|g_k\| \|s_k\| + \frac{1}{2} \kappa_{\text{umh}} \|s_k\|^2$, so at least one of the two terms on the right is at least $D/2$.

If $\frac{1}{2} \kappa_{\text{umh}} \|s_k\|^2 \geq D/2$, then $\|s_k\|^2 \geq D/\kappa_{\text{umh}} = \kappa_{\text{moc}} |\lambda_k^-| \Delta_k^2 / (2\kappa_{\text{umh}})$, which after taking square roots is the first entry of [\(F.2\)](#). Otherwise $\|g_k\| \|s_k\| \geq D/2$, which forces $g_k \neq 0$ and gives $\|s_k\| \geq D/(2\|g_k\|) = \kappa_{\text{moc}} |\lambda_k^-| \Delta_k^2 / (4\|g_k\|)$, the second entry.

(iii). The hypothesis gives $\|g_k\| \|s_k\| \leq c \kappa_{\text{umh}} \|s_k\|^2$, so [\(F.4\)](#) becomes

$$m_k(0) - m_k(s_k) \leq \frac{2c+1}{2} \kappa_{\text{umh}} \|s_k\|^2.$$

Combined with the eigenpoint branch,

$$\frac{\kappa_{\text{moc}}}{2} |\lambda_k^-| \Delta_k^2 \leq \frac{2c+1}{2} \kappa_{\text{umh}} \|s_k\|^2,$$

so $\|s_k\|^2 \geq \kappa_{\text{moc}} |\lambda_k^-| \Delta_k^2 / ((2c+1) \kappa_{\text{umh}})$, and taking square roots gives [\(F.3\)](#).

For the last claim, if $\|g_k\| \leq \kappa_{\text{umh}} \Delta_k$ then the minimum in [\(F.1\)](#) equals $\|g_k\|/\kappa_{\text{umh}}$, so $\|s_k\| \geq (\kappa_{\text{moc}}/3) \|g_k\|/\kappa_{\text{umh}}$, that is $\|g_k\| \leq (3/\kappa_{\text{moc}}) \kappa_{\text{umh}} \|s_k\|$. Substituting $c = 3/\kappa_{\text{moc}}$, for which $2c+1 = (6 + \kappa_{\text{moc}})/\kappa_{\text{moc}}$, gives the stated constant. $\square$

Remark F.2 (Reading the three bounds). The constant 3 in [\(F.1\)](#) and the constant $2c+1$ in [\(F.3\)](#) are the same arithmetic: each is the price of absorbing the cross term $\|g_k\| \|s_k\|$ of [\(F.4\)](#) into the quadratic one, and in [\(F.1\)](#) that price is the factor $\frac{3}{2}$ against the factor $\frac{1}{2}$ of the Cauchy branch.

Part (iii) is the sharp form but is conditional, and the hypothesis it needs cannot be supplied by (ii): it bounds $\|g_k\|$ by the very quantity being bounded below. Part (ii) removes the difficulty at a cost of $\sqrt{2}$ in the leading constant. Its second entry is the active one exactly when $\|g_k\|$ is small relative to $|\lambda_k^-| \Delta_k^2$, that is, in the saddle regime this section is concerned with, so (ii) is the stronger statement where it matters and (iii) the more convenient one elsewhere.

Finally, $|\lambda_k^-| \leq \|H_k\| \leq \kappa_{\text{umh}}$, so the factor multiplying Δ_k in [\(F.3\)](#) is at most $\sqrt{\kappa_{\text{moc}}/(2c+1)} < 1$: the bounds are consistent with $\|s_k\| \leq \Delta_k$, as they must be.

Proposition F.3 (Degradation of the step-length bound). *Bound (F.3) is attained up to the constant $\sqrt{3}$: for every $\Delta_k > 0$ and every $\lambda \in (0, \kappa_{\text{umh}}]$ there are model data, necessarily with $g_k \neq 0$, satisfying a.Mx, $\|H_k\| \leq \kappa_{\text{umh}}$ and $\lambda_k^- = -\lambda$, for which*

$$\frac{\|s_k\|}{\Delta_k} = \sqrt{\frac{\kappa_{\text{moc}} |\lambda_k^-|}{\kappa_{\text{umh}}}} = \Theta\left(\sqrt{\frac{|\lambda_k^-|}{\kappa_{\text{umh}}}}\right). \quad (\text{F.5})$$

Consequently, on an unsuccessful iteration at which $\tau_k = \varsigma_k = |\lambda_k^-|$, a rule of the form $\Delta_{k+1} = \gamma \|s_k\|$ can produce

$$\Delta_{k+1} = \Theta(\Delta_k \sqrt{\varsigma_k}) = \Theta(\varsigma_k^{3/2}) = o(\varsigma_k) \quad (\varsigma_k \rightarrow 0), \quad (\text{F.6})$$

so no constant $c > 0$ makes $\Delta_{k+1} \geq c \varsigma_k$ hold uniformly. The step-length route therefore cannot close the inductive step of Condition C.Lb2.

Proof. The construction. The required decrease must be extracted from the gradient rather than from the negative curvature, so $g_k \neq 0$ is necessary; see Remark F.4. Work in $\mathbb{R}^2$ with

$$H_k = \text{diag}(-\lambda, \kappa_{\text{umh}}), \quad 0 < \lambda \leq \kappa_{\text{umh}}, \quad g_k = -\kappa_{\text{umh}} t e_2, \quad s_k = t e_2,$$

where $t > 0$ is fixed below. Then $\lambda_k^- = -\lambda < 0$, $\|H_k\| = \kappa_{\text{umh}}$, $\|g_k\| = \kappa_{\text{umh}} \|s_k\|$, and

$$m_k(0) - m_k(s_k) = -g_k^\top s_k - \frac{1}{2} s_k^\top H_k s_k = \kappa_{\text{umh}} t^2 - \frac{1}{2} \kappa_{\text{umh}} t^2 = \frac{1}{2} \kappa_{\text{umh}} t^2.$$

Note that s_k is the exact minimiser of m_k over $\text{span}(e_2)$, the positive-curvature direction, and ignores e_1 entirely. This is the mechanism of the degradation: a.Mx requires a decrease of a certain size, not that the decrease be obtained along the direction of negative curvature.

Admissibility. Choose

$$t := \Delta_k \sqrt{\frac{\kappa_{\text{moc}} |\lambda_k^-|}{\kappa_{\text{umh}}}} \leq \Delta_k, \quad (\text{F.7})$$

the inequality because $\kappa_{\text{moc}} \leq 1$ and $\lambda \leq \kappa_{\text{umh}}$, so s_k is feasible. a.Mx requires the decrease to dominate the maximum of its two branches, so both must be checked.

For the eigenpoint branch, (F.7) gives

$$\frac{1}{2} \kappa_{\text{umh}} t^2 = \frac{1}{2} \kappa_{\text{umh}} \Delta_k^2 \frac{\kappa_{\text{moc}} |\lambda_k^-|}{\kappa_{\text{umh}}} = \frac{\kappa_{\text{moc}}}{2} |\lambda_k^-| \Delta_k^2,$$

with *equality*: the step realises the eigenpoint decrease exactly, which is what makes the construction sharp rather than merely feasible. For the Cauchy branch, $\|g_k\| / \|H_k\| = t \leq \Delta_k$, so the minimum equals t and

$$\frac{\kappa_{\text{moc}}}{2} \|g_k\| \min\left\{\frac{\|g_k\|}{\|H_k\|}, \Delta_k\right\} = \frac{\kappa_{\text{moc}}}{2} \kappa_{\text{umh}} t^2 \leq \frac{1}{2} \kappa_{\text{umh}} t^2,$$

again because $\kappa_{\text{moc}} \leq 1$. Hence s_k is admissible and (F.5) holds, the implied constants being $\sqrt{\kappa_{\text{moc}}}$. The ratio exceeds (F.3) at $c = 1$ by exactly $\sqrt{3}$, and the construction meets that hypothesis with equality, since $\|g_k\| = \kappa_{\text{umh}} \|s_k\|$: it is a witness for the sharpness of (iii), not an artefact of a mismatched regime. It also shows that (F.2) is attained up to a factor 4.

The consequence. Write $\tau := \lambda$ and let $\Delta_k := C\tau$ with $C > (1 - \eta)/M_k^{(2)}$. By Corollary B.11 this is compatible with the iteration being unsuccessful; the corollary certifies that the regime $\Delta_k = \Theta(\tau)$ is available, and is not used to derive a bound.

From (F.7), $\|g_k\| = \kappa_{\text{umh}} t = C\sqrt{\kappa_{\text{moc}}\kappa_{\text{umh}}} \tau^{3/2}$, so $\|g_k\| \leq \tau$ for all τ small enough, and then $\tau_k = \max\{\|g_k\|, |\lambda_k^-|\} = \tau$. Taking τ below every previously attained value gives $\varsigma_k = \tau$ as well. The rule returns

$$\Delta_{k+1} = \gamma \|s_k\| = \gamma \Delta_k \sqrt{\frac{\kappa_{\text{moc}}\tau}{\kappa_{\text{umh}}}} = \gamma C \sqrt{\frac{\kappa_{\text{moc}}}{\kappa_{\text{umh}}}} \tau^{3/2},$$

which is (F.6), and $\Delta_{k+1}/\varsigma_k = \gamma C \sqrt{\kappa_{\text{moc}}/\kappa_{\text{umh}}} \sqrt{\tau} \rightarrow 0$ as $\tau \rightarrow 0$. Hence $\Delta_{k+1} \geq c \varsigma_k$ fails for every fixed $c > 0$ once τ is small enough. $\square$

Remark F.4 (The degradation is impossible at an exact critical point). If $g_k = 0$ then every step satisfies $m_k(0) - m_k(s_k) = -\frac{1}{2}s_k^\top H_k s_k \leq \frac{1}{2}|\lambda_k^-| \|s_k\|^2$, while **a.Mx** requires $m_k(0) - m_k(s_k) \geq \frac{\kappa_{\text{moc}}}{2} |\lambda_k^-| \Delta_k^2$, whence $\|s_k\| \geq \sqrt{\kappa_{\text{moc}}} \Delta_k$ independently of $|\lambda_k^-|$. The step is then a fixed fraction of the radius, which is **a.Sob** with $\kappa_{\text{sob}} = \sqrt{\kappa_{\text{moc}}}$, and no degradation occurs. What creates the square-root loss is therefore not the negative curvature alone but a gradient large enough to pay for the required decrease over a short step.

Remark F.5 (What the proposition does and does not show). The construction exhibits admissible model data at a single iteration, so it shows that the *estimate* $\Delta_{k+1} \geq \gamma \|s_k\|$ cannot yield a bound linear in ς_k : the inductive step of Condition **C.Lb2** cannot be closed along this route. It is not a counterexample to **R-step** satisfying that condition, which would require a trajectory realising these data at infinitely many iterations of a single run.

The distinction matters for how **a.Sob** should be read. That assumption is not a repair of a loose inequality; it excludes exactly the behaviour constructed above, in which the solver banks its decrease in a positive-curvature direction and returns a step of length $\Theta(\Delta_k \sqrt{|\lambda_k^-|})$. See also Remarks **B.1** and **B.2**.

The remedy is to require what the eigenpoint already delivers.

G The reciprocal bound and the multiplier window

Lemma G.1 (A short step forces a large multiplier). *Let $g_k \neq 0$ and let Condition **C.Inx** hold with $\|H_k\| \leq \kappa_{\text{umh}}$ and $\xi_k < 1$. If $\|s_k\| < \frac{1}{2}\Delta_k$, then*

$$\mu_k > D_k := \frac{2(1 - \xi_k)}{\kappa_{\text{umh}}}. \quad (\text{G.1})$$

Proof. Since $\|s_k\| < \frac{1}{2}\Delta_k < \Delta_k$ the trust-region constraint is inactive, so s_k is an approximate unconstrained minimiser of m_k and Condition **C.Inx** applies to it. Writing $g_k = (H_k s_k + g_k) - H_k s_k$,

$$\|g_k\| \leq \|H_k s_k + g_k\| + \|H_k\| \|s_k\| \leq \xi_k \|g_k\| + \kappa_{\text{umh}} \|s_k\|, \quad \text{so} \quad \|s_k\| \geq \frac{(1 - \xi_k) \|g_k\|}{\kappa_{\text{umh}}}.$$

The hypothesis gives $\|s_k\| < \frac{1}{2}\Delta_k = \frac{1}{2}\mu_k \|g_k\|$, and dividing by $\|g_k\| > 0$ yields (G.1). $\square$

Remark G.2 (The two bounds are not mirror images). Lemma **B.8** compares $f(x_k)$ with $f(x_{k+1})$ and closes the comparison with strong convexity at both ends, which is where $\sqrt{M/m}$ comes from. Lemma **G.1** uses no function values at all, only that an interior step approximately solves $H_k s = -g_k$ and therefore cannot be shorter than $\|g_k\|/\kappa_{\text{umh}}$. Neither m nor a square root appears, and the chain of (B.4) cannot be reversed to produce one, bounding $\|x_{k+1} - x^*\|$ from below requiring $\|s_k\|$ to be large, which is exactly what the hypothesis of Lemma **G.1** denies.

Corollary G.3 (Three regimes for the multiplier). *Let $D := 2(1 - \xi)/\kappa_{\text{umh}}$ with $\xi = \sup_k \xi_k < 1$, and let C be as in (B.3). Then $D \leq \bar{\kappa} \leq C$, and at every $k \geq k_\rho$ with x_k, x_{k+1} in the neighbourhood of Proposition B.3:*

- (i) if $\mu_k \leq D$ then $\|s_k\| \geq \frac{1}{2}\Delta_k$, so the expansion branch fires and $\mu_{k+1} = \gamma_3\mu_k$;
- (ii) if $\mu_k > C$ then the expansion branch does not fire and $\mu_{k+1} = \mu_k$;
- (iii) for $D < \mu_k \leq C$ either may occur.

Consequently the number of expansions occurring after the iterates settle in the neighbourhood is at most $\log(\gamma_3 C / \mu_{k_N}) / \log \gamma_3$.

Proof. Part (i) is the contrapositive of Lemma G.1 and (ii) that of Lemma B.8. For the ordering, $\kappa_{\text{umh}} \geq M \geq m$ gives $D \leq 2/m \leq 4/m = \bar{\kappa}$, and $C \geq 4/m$ reduces to $M \geq m$. For the count, μ is non-decreasing and multiplied by γ_3 at each expansion, while (ii) forbids an expansion once $\mu_k > C$, so no expansion starts above C and none lands above $\gamma_3 C$. $\square$

Remark G.4 (A monotone climb through a narrow window). Corollary G.3 turns the informal description of R-grad as a geometric search for $\bar{\kappa}$ into a statement with content. Below D the search is not probabilistic at all, the expansion branch firing at every iteration, so the multiplier climbs by the factor γ_3 deterministically. Above C it stops. In between either may happen, and the threshold $\bar{\kappa}$ that the capped rule must be given by hypothesis lies inside that window.

The window is narrow. Its width on the multiplicative lattice is $\log_{\gamma_3}(C/D)$, which for $\kappa_{\text{umh}} = M$ is $\log_{\gamma_3}(\frac{M}{m}(1 + \sqrt{M/m}))$, a quantity governed by the conditioning of $\nabla^2 f(x^*)$ alone. On the example of Section G.2, where $m = 2$ and $M = \kappa_{\text{umh}} = 17/4$, one has $D = 0.47$, $\bar{\kappa} = 2$ and $C = 2.46$, so the window is 2.4 lattice steps wide at $\gamma_3 = 2$ and the whole search takes fewer than ten expansions from any reasonable μ_0 .

G.1 The sharpness of $\bar{\kappa}$

Condition C.Sg holds with $\bar{\kappa} = 4/m$ by Lemma B.5, and that constant is what Lemma B.6 turns into the inactivity threshold. Neither is sharp, and saying by how much explains the factor 4 that Section G.2 exhibits on an example.

Lemma G.5 (The step bound on an inactive iteration). *Let x_k lie in the neighbourhood of Proposition B.3, let Condition C.Inx and H.Hs hold, and put $\delta_k := \|(H_k - \nabla^2 f(x_k))s_k\| / \|s_k\|$, so that $\delta_k \rightarrow 0$. If the trust-region constraint is inactive at k and $\delta_k < m$, then*

$$\frac{(1 - \xi_k)}{M + \delta_k} \|g_k\| \leq \|s_k\| \leq \frac{(1 + \xi_k)}{m - \delta_k} \|g_k\|. \quad (\text{G.2})$$

In particular $\|s_k\| \leq (1/m + o(1)) \|g_k\|$, and the constant $1/m$ cannot be improved, being attained in the limit by the exact Newton step.

Proof. Inactivity makes s_k an approximate unconstrained minimiser, so Condition C.Inx applies and $H_k s_k = -g_k + r_k$ with $\|r_k\| \leq \xi_k \|g_k\|$. Only the action of H_k on s_k is needed at either end. For the upper bound,

$$\begin{aligned} (m - \delta_k) \|s_k\|^2 &\leq s_k^\top \nabla^2 f(x_k) s_k - \|(H_k - \nabla^2 f(x_k))s_k\| \|s_k\| \leq s_k^\top H_k s_k \\ &= s_k^\top (-g_k + r_k) \leq (1 + \xi_k) \|g_k\| \|s_k\|, \end{aligned}$$

the first inequality by Proposition B.3. For the lower bound, $\|H_k s_k\| \leq (M + \delta_k) \|s_k\|$ by the same decomposition, and $\|g_k\| \leq \|H_k s_k + g_k\| + \|H_k s_k\|$ gives $(1 - \xi_k) \|g_k\| \leq (M + \delta_k) \|s_k\|$. $\square$

Remark G.6 (Where the factor 4 comes from). Comparing (G.2) with $\bar{\kappa} = 4/m$, the constant of Condition C.Sg is loose by a factor of 4 asymptotically, and the reason is structural rather than technical. Lemma B.5 must cover the active iterations as well, where $\|s_k\| = \Delta_k$ and no relation to the model equation is available, and it obtains its bound from the decrease conditions instead. On an inactive iteration the step solves $H_k s = -g_k$ to within ξ_k , and the sharp constant is the one the solution itself carries, namely $1/\lambda_{\min}(H_k) \rightarrow 1/m$.

This is exactly the gap measured in Section G.2, where the exact threshold on the example is $1/\lambda_{\min}^*$ against the sufficient condition $\theta_k > \bar{\kappa}$. The factor 4 is the price of a criterion that does not ask where the step came from.

Lemma G.7 (Sharp inactivity for a monotone solver). *Suppose $H_k \succ 0$ and s_k is the exact subproblem solution, or the Steihaug–Toint truncated conjugate-gradient step. If $\mu_k > 1/\lambda_{\min}(H_k)$ then the constraint is inactive at k .*

Proof. Under $H_k \succ 0$ the conjugate-gradient iterates have increasing norm and converge to $s_k^N = -H_k^{-1}g_k$, so the path leaves the ball only if $\|s_k^N\| \geq \Delta_k$; the same holds trivially for the exact solution. Hence activity implies $\Delta_k \leq \|H_k^{-1}g_k\| \leq \|g_k\|/\lambda_{\min}(H_k)$, that is $\mu_k \leq 1/\lambda_{\min}(H_k)$. $\square$

Three thresholds therefore separate, and it is worth recording all three.

Proposition G.8 (Necessary, exact and sufficient). *On the neighbourhood of Proposition B.3, and asymptotically in $\xi_k, \delta_k \rightarrow 0$:*

- (i) *inactivity at k requires $\mu_k \geq 1/M + o(1)$;*
- (ii) *for an exact or truncated-CG step with $H_k \succ 0$, $\mu_k > 1/m + o(1)$ is sufficient, and no smaller constant is;*
- (iii) *for an arbitrary step satisfying Condition C.Sg, $\mu_k > \bar{\kappa} = 4/m$ is sufficient.*

Proof. Part (i) is the lower bound of (G.2), inactivity requiring $\|s_k\| < \Delta_k = \mu_k \|g_k\|$. Part (ii) is Lemma G.7 with $\lambda_{\min}(H_k) \rightarrow m$, sharpness following since the exact Newton step has norm $\|g_k\|/m$ in the limit along the leftmost eigenvector. Part (iii) is Lemma B.6. $\square$

Corollary G.9 (The window is the condition number). *Under the hypotheses of Lemma G.5, and asymptotically:*

- (i) *if $\mu_k < 2/M$ then $\|s_k\| \geq \frac{1}{2}\Delta_k$ and the expansion branch fires;*
- (ii) *if $\mu_k > 2/m$ then $\|s_k\| < \frac{1}{2}\Delta_k$ and it does not.*

The multiplier therefore climbs geometrically until it reaches $2/M$, never grows once past $2/m$, and $\mu_\infty \leq 2\gamma_3/m$. The number of forced expansions is at most $\log_{\gamma_3}(M/m)$.

Proof. Both parts follow from (G.2) on dividing by $\Delta_k = \mu_k \|g_k\|$, part (ii) using in addition that $\mu_k > 2/m > 1/m$ makes the iteration inactive by Proposition G.8(ii), so that (G.2) applies. The count follows since μ is non-decreasing and multiplied by γ_3 at each expansion. $\square$

Remark G.10 (What the rule estimates, and what it does not). Corollary G.9 gives the geometric search a precise target. The multiplier is driven upward while below $2/M$, is frozen above $2/m$, and the interval between them has multiplicative width exactly the condition number M/m of $\nabla^2 f(x^*)$. The search therefore costs $\log_{\gamma_3}(M/m)$ expansions, independently of the solver, of κ_{umh} and of every parameter of the rule but γ_3 .

Two readings of this deserve separating. The rule does not find $\bar{\kappa}$: the window lies *below* $\bar{\kappa} = 4/m$, and the multiplier may halt without ever reaching it. What the rule finds is the scale $1/\lambda_{\min}^*$ at which the trust region stops binding, which by Proposition G.8(ii) is the exact threshold and is a factor 4 below $\bar{\kappa}$. Inactivity in Theorem B.9 accordingly comes from the $\frac{1}{2}\Delta_k$ test rather than from a crossing of $\bar{\kappa}$, which is why that proof never mentions the constant.

The same factor is available to the capped rule. Theorem B.10 assumes $\bar{\mu} > \bar{\kappa}$; for an exact or truncated-CG solver Proposition G.8(ii) and Corollary G.9(ii) reduce the requirement to $\bar{\mu} > 2/m$, which is $\bar{\kappa}/2$. The example of Section G.2 shows that nothing below $1/m$ can work, so the remaining gap is a factor of 2.

The problem supplies the constant the user cannot. Corollary G.9 bounds the multiplier from both sides by the spectrum at the limit: it is driven upward while below $2/M$, frozen once past $2/m$, and the forced climb costs at most $\log_{\gamma_3}(M/m)$ expansions. What the user would otherwise have to guess is set by the local geometry, at a cost logarithmic in the conditioning of $\nabla^2 f(x^*)$ and independent of the solver and of every parameter but γ_3 .

G.2 Discussion: the role of ζ and $\bar{\mu}$

The constant $\bar{\kappa}$ of Condition C.Sg governs both side conditions of a.Sc, and equals $4/m$ by Lemma B.5. The following example shows that $\bar{\kappa}$ is unbounded over the minimisers of a single fixed C^∞ function with bounded Hessian, so that no admissible ζ or $\bar{\mu}$ exists a priori, and quantifies what goes wrong below the threshold. All constants are exact.

Let

$$f(x) := \frac{\sin x}{x}, \quad f(0) := 1, \quad (\text{G.3})$$

an even entire function on $\mathbb{R}$ with

$$f'(x) = \frac{x \cos x - \sin x}{x^2}, \quad f''(x) = -\frac{\sin x}{x} - \frac{2 \cos x}{x^2} + \frac{2 \sin x}{x^3}. \quad (\text{G.4})$$

The standing assumptions hold with explicit constants: $\sup_{\mathbb{R}} |f''| = |f''(0)| = 1/3$, so f' is Lipschitz and $\kappa_{\text{umh}} = 1/3$ serves for the exact Hessian, and f is bounded below by its global minimum $-0.2172\dots$. Throughout, χ_j denotes the j -th positive critical point, x_k being reserved for iterates.

Lemma G.11 (Critical points). *The nonzero critical points of f are the roots of $\tan \chi = \chi$. The positive ones form a sequence $\{\chi_j\}_{j \geq 1}$ with exactly one in each interval $(j\pi, A_j)$, $A_j := (j + \frac{1}{2})\pi$, together with their reflections $-\chi_j$. Writing $\chi_j = A_j - \delta_j$,*

$$\delta_j \downarrow 0, \quad \delta_j = \frac{1}{A_j} + \frac{2}{3A_j^3} + \mathcal{O}(A_j^{-5}). \quad (\text{G.5})$$

Proof. For $\chi \neq 0$, $f'(\chi) = 0$ reads $\chi \cos \chi = \sin \chi$; if $\cos \chi = 0$ then $\sin \chi = \pm 1$ and the equation fails, so any nonzero root satisfies $\tan \chi = \chi$. On $((j - \frac{1}{2})\pi, j\pi)$ one has $\tan \chi \leq 0 < \chi$, so there is no root; on $(j\pi, A_j)$ the map $\chi \mapsto \tan \chi - \chi$ has derivative $\tan^2 \chi \geq 0$ and increases from $-j\pi$ to $+\infty$, so it vanishes exactly once.

Put $\delta_j := A_j - \chi_j \in (0, \frac{\pi}{2})$. By π -periodicity of $\tan$ and $\tan(\frac{\pi}{2} - \delta) = \cot \delta$, the equation $\tan \chi_j = \chi_j$ reads $F(\delta_j) = A_j$ with $F(\delta) := \cot \delta + \delta$. Since $F'(\delta) = 1 - \sin^{-2} \delta = -\cot^2 \delta \leq 0$, with $F(0^+) = +\infty$ and $F(\pi^-) = -\infty$, F is a decreasing bijection of $(0, \pi)$ onto $\mathbb{R}$; thus δ_j is its unique root, and $A_{j+1} = A_j + \pi > A_j$ forces $\delta_{j+1} < \delta_j$. If $\delta_j \downarrow \ell > 0$ then $A_j = F(\delta_j) \rightarrow F(\ell) < \infty$, contradicting $A_j \rightarrow \infty$; hence $\delta_j \downarrow 0$. Finally $\cot \delta = \delta^{-1} - \frac{1}{3}\delta + \mathcal{O}(\delta^3)$ gives $F(\delta) = \delta^{-1} + \frac{2}{3}\delta + \mathcal{O}(\delta^3)$, so $\delta_j^{-1} = A_j - \frac{2}{3}A_j^{-1} + \mathcal{O}(A_j^{-3})$, and inverting yields (G.5). $\square$

Table 18: Critical points of (G.3). Columns three and four agree to machine precision, confirming (G.6).

j	χ_j	$f''(\chi_j)$	$(-1)^{j+1}(1 + \chi_j^2)^{-1/2}$	type	$\bar{\kappa}$
1	4.493409	+0.21723363	+0.21723363	minimiser	18.41
2	7.725252	-0.12837455	-0.12837455	maximiser	—
3	10.904122	+0.09132520	+0.09132520	minimiser	43.80
4	14.066194	-0.07091346	-0.07091346	maximiser	—
5	17.220755	+0.05797180	+0.05797180	minimiser	69.00
6	20.371303	-0.04902962	-0.04902962	maximiser	—
7	23.519452	+0.04247962	+0.04247962	minimiser	94.16

Proposition G.12 (Exact curvature at a critical point). *For every $j \geq 1$,*

$$f''(\chi_j) = -\frac{\sin \chi_j}{\chi_j} = \frac{(-1)^{j+1}}{\sqrt{1 + \chi_j^2}}, \quad (\text{G.6})$$

so χ_j is a local minimiser for j odd and a local maximiser for j even.

Proof. At a root, $\sin \chi_j = \chi_j \cos \chi_j$, hence $\cos \chi_j = \sin \chi_j / \chi_j$, and the last two terms of (G.4) cancel identically,

$$-\frac{2 \cos \chi_j}{\chi_j^2} + \frac{2 \sin \chi_j}{\chi_j^3} = -\frac{2 \sin \chi_j}{\chi_j^3} + \frac{2 \sin \chi_j}{\chi_j^3} = 0,$$

which gives the first equality. Next, $\sin \chi_j = \chi_j \cos \chi_j$ with $\cos^2 \chi_j = 1 - \sin^2 \chi_j$ yields $\sin^2 \chi_j (1 + \chi_j^2) = \chi_j^2$, so $|\sin \chi_j| = \chi_j / \sqrt{1 + \chi_j^2}$ and $|f''(\chi_j)| = (1 + \chi_j^2)^{-1/2}$. Finally $\chi_j \in (j\pi, (j + \frac{1}{2})\pi)$ gives $\text{sign} \sin \chi_j = (-1)^j$, whence $\text{sign} f''(\chi_j) = (-1)^{j+1}$. $\square$

Corollary G.13 ($\bar{\kappa}$ along the minimisers). *At the minimiser χ_j , j odd,*

$$\lambda_{\min}^* = \frac{1}{\sqrt{1 + \chi_j^2}}, \quad \bar{\kappa} = \frac{4}{m} = \frac{8}{\lambda_{\min}^*} = 8\sqrt{1 + \chi_j^2} = 8(j + \tfrac{1}{2})\pi + \mathcal{O}(j^{-1}), \quad (\text{G.7})$$

the second equality under the concrete choice (B.1). Thus $\bar{\kappa}$ grows linearly in the index of the minimiser and is unbounded over the minimisers of f , whatever admissible $m \leq \lambda_{\min}^*$ is used.

Proof. Proposition G.12 gives $\lambda_{\min}^* = f''(\chi_j)$, positive for j odd. For the rate, $\sqrt{1 + \chi_j^2} = \chi_j + \frac{1}{2}\chi_j^{-1} + \mathcal{O}(\chi_j^{-3})$, and $\chi_j = A_j - \delta_j$ with (G.5) gives $\sqrt{1 + \chi_j^2} = A_j - \frac{1}{2}A_j^{-1} + \mathcal{O}(A_j^{-3})$. $\square$

G.2.1 The exact inactivity threshold

For the model $m_k(s) = f(x_k) + g_k s + \frac{1}{2}a_k s^2$ with $a_k := f''(x_k) > 0$, the subproblem $\min_{|s| \leq \Delta_k} m_k(s)$ is solved exactly by

$$s_k = \begin{cases} -g_k/a_k, & |g_k|/a_k \leq \Delta_k \quad (\text{inactive}), \\ -\Delta_k \text{sign}(g_k), & \text{otherwise} \quad (\text{active}). \end{cases} \quad (\text{G.8})$$

Hence, with $\theta_k = \Delta_k / |g_k|$ as in (B.2),

$$\text{the trust region is active at } k \iff \theta_k < 1/a_k. \quad (\text{G.9})$$

Compare Lemma B.6, which gives inactivity when $\theta_k > \bar{\kappa}$. As $x_k \rightarrow \chi_j$ the exact threshold in (G.9) tends to $1/\lambda_{\min}^*$, which under (B.1) is $\bar{\kappa}/8$. The gap between the sufficient condition of the theory and the exact one is therefore a factor 8 here, of which 4 is intrinsic to Lemma B.5 and 2 is the price of bounding the Hessian on a neighbourhood rather than at x^* . By Remark B.4 the second factor may be brought arbitrarily close to 1 by shrinking Δ_* , so the intrinsic gap is the factor 4.

Theorem G.14 (Capped R-grad below the threshold). *Let $\{x_k\}$ converge to a minimiser χ_j , so that $\lambda_{\min}^* = f''(\chi_j) = (1 + \chi_j^2)^{-1/2}$, and run capped R-grad with the exact Hessian and the exact subproblem solution. If*

$$\bar{\mu} < \frac{1}{\lambda_{\min}^*}, \quad (\text{G.10})$$

then the trust region is active at every sufficiently large k ; in particular the conclusion of Theorem B.7 fails.

Proof. By construction $\theta_k = \mu_k \leq \bar{\mu}$. Since $x_k \rightarrow \chi_j$ and f'' is continuous, $a_k \rightarrow \lambda_{\min}^*$, so $1/a_k \rightarrow 1/\lambda_{\min}^*$. Taking $\epsilon = \frac{1}{2}(1/\lambda_{\min}^* - \bar{\mu}) > 0$ gives $\theta_k \leq \bar{\mu} < 1/\lambda_{\min}^* - \epsilon < 1/a_k$ for all large k , and (G.9) applies. $\square$

G.2.2 Why a small ζ traps R-DFO

R-DFO does not fix the ratio θ_k , so the argument must follow its dynamics. We work with the exact quadratic, that is with f replaced by its second-order Taylor model at χ_j , so that $a_k \equiv \lambda_{\min}^*$ and $\rho_k = 1$ at every iteration; both branch conditions then reduce to the ζ test. The trapping is that R-DFO leaves a whole interval of ratios below the inactivity threshold forward invariant.

Theorem G.15 (R-DFO below the threshold). *In the setting of Theorem G.14 with $a_k \equiv \lambda_{\min}^*$, run R-DFO with the exact Hessian and the exact subproblem solution. If*

$$\zeta \leq \frac{1 - \gamma_2}{\lambda_{\min}^* (\gamma_3 + 1 - \gamma_2)} \quad \left(= \frac{1}{5\lambda_{\min}^*} = \bar{\kappa}/40 \text{ for } \gamma_2 = \frac{1}{2}, \gamma_3 = 2 \right), \quad (\text{G.11})$$

and $\theta_{k_1} \leq (1 - \gamma_2)/\lambda_{\min}^$ at some k_1 , then $\theta_k \leq (1 - \gamma_2)/\lambda_{\min}^* < 1/\lambda_{\min}^*$ for all $k \geq k_1$, and the trust region is active at every $k \geq k_1$.*

Proof. Since $\gamma_3 > 1$, (G.11) gives $\zeta \lambda_{\min}^* \leq (1 - \gamma_2)/(\gamma_3 + 1 - \gamma_2) < 1$, so $\zeta < 1/\lambda_{\min}^*$.

On an active iteration (G.8) gives $x_{k+1} - \chi_j = (x_k - \chi_j)(1 - \theta_k \lambda_{\min}^*)$ and hence $|g_{k+1}| = |g_k|(1 - \theta_k \lambda_{\min}^*)$, the factor being positive because activity means $\theta_k < 1/\lambda_{\min}^*$. The R-DFO update is $\Delta_{k+1} = \gamma_2 \Delta_k$ if $\theta_k > \zeta$ and $\gamma_3 \Delta_k$ otherwise, so

$$\theta_{k+1} = \frac{\gamma \theta_k}{1 - \theta_k \lambda_{\min}^*}, \quad \gamma = \begin{cases} \gamma_2, & \theta_k > \zeta, \\ \gamma_3, & \theta_k \leq \zeta. \end{cases} \quad (\text{G.12})$$

Suppose $\theta_k \leq (1 - \gamma_2)/\lambda_{\min}^*$, so that $1 - \theta_k \lambda_{\min}^* \geq \gamma_2$. If $\theta_k > \zeta$ then $\theta_{k+1} = \gamma_2 \theta_k / (1 - \theta_k \lambda_{\min}^*) \leq \theta_k \leq (1 - \gamma_2)/\lambda_{\min}^*$. If $\theta_k \leq \zeta$ then, $h(\theta) := \gamma_3 \theta / (1 - \theta \lambda_{\min}^*)$ being increasing on $[0, 1/\lambda_{\min}^*)$,

$$\theta_{k+1} \leq h(\zeta) = \frac{\gamma_3 \zeta}{1 - \zeta \lambda_{\min}^*} \leq \frac{1 - \gamma_2}{\lambda_{\min}^*},$$

the last inequality being $\gamma_3 \zeta \lambda_{\min}^* \leq (1 - \gamma_2)(1 - \zeta \lambda_{\min}^*)$, which is (G.11) rearranged. So $[0, (1 - \gamma_2)/\lambda_{\min}^*]$ is forward invariant, and $(1 - \gamma_2)/\lambda_{\min}^* < 1/\lambda_{\min}^*$ with (G.9) gives activity throughout. $\square$

The hypothesis $\theta_{k_1} \leq (1 - \gamma_2)/\lambda_{\min}^*$ is needed and marks a real difference between the two rules. R-DFO propagates the radius multiplicatively, so a large Δ_0 can leave the region inactive from the outset, the Newton step is taken, and θ_k escapes upward before the contraction branch can bite. R-grad recomputes $\Delta_k = \mu_k |g_k|$ from scratch and forgets Δ_0 after one step. Thus R-grad's behaviour is determined by its parameter, R-DFO's by its parameter and its initial radius.

G.2.3 What the example establishes

1. $\bar{\kappa} = 8\sqrt{1 + \chi_j^2}$ is unbounded over the minimisers of a single fixed C^∞ function with bounded Hessian and Lipschitz gradient, growing linearly with slope 4π . No universal ζ or $\bar{\mu}$ exists, even for one problem.
2. [a.Sc](#) is necessary up to a constant. For R-grad the exact threshold is $1/\lambda_{\min}^*$ (Theorem [G.14](#)); for R-DFO permanent activity is provable for $\zeta \leq 1/(5\lambda_{\min}^*)$ (Theorem [G.15](#)). Under [\(B.1\)](#) these are $\bar{\kappa}/8$ and $\bar{\kappa}/40$.
3. Failure is not a slowdown but a change of regime: below the threshold the constraint binds not merely infinitely often but at every sufficiently large iteration, so the method converges linearly and in practice exhausts the available precision while ρ_k looks healthy and $\|g_k\|$ is small but not small enough.
4. The two mechanisms fail differently, R-grad's activity being decided by $\bar{\mu}$ alone and R-DFO's by ζ together with Δ_0 .

Part III reports the observed thresholds, which exceed the provable ones, and extends Theorem [G.15](#) numerically to the true f rather than its quadratic model.

H Figures

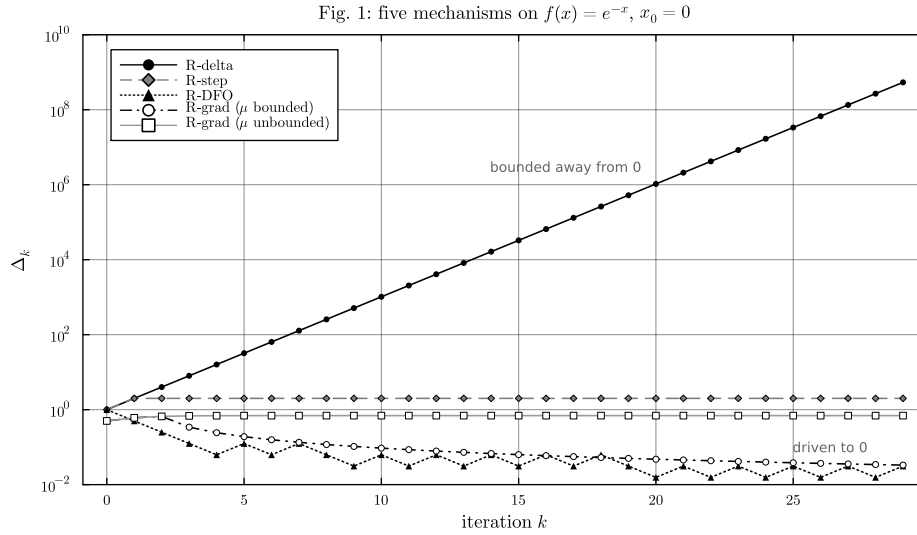


Figure 11: Radius trajectories of the four mechanisms of this section, with R-grad in both variants, on $f(x) = e^{-x}$ with $x_0 = 0$, on a logarithmic scale. R-delta and R-step are unanchored and grow without bound; R-DFO and R-grad are anchored to a criticality measure and decay. The twelve decades of vertical range compress the decaying curves near the axis; Figure [12](#) resolves them.

Fig. 2: the criticality-anchored rules

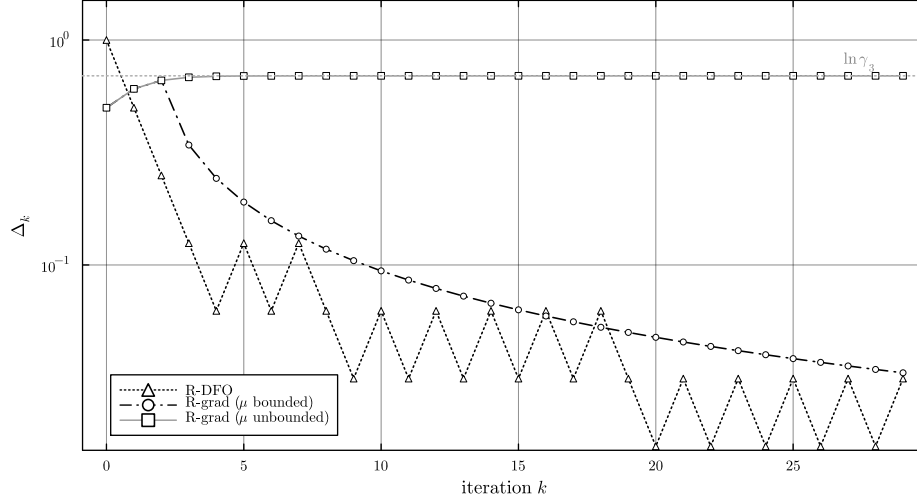


Figure 12: Detail of the two criticality-anchored mechanisms on $f(x) = e^{-x}$. R-grad with μ bounded decays smoothly, while R-DFO exhibits a sawtooth confined to a band of width $\Theta(1/k)$: both are $\Theta(1/k)$, but only the former converges after rescaling.

Table 19: Partial sums over $N = 1000$ iterations on $f(x) = e^{-x}$, $x_0 = 0$. Every rule has $\sum_k \Delta_k = +\infty$: the iterates must travel an infinite distance to drive the gradient to zero. For the criticality-anchored rules (R-DFO and R-grad with μ bounded) the radius decays harmonically, so the divergence is only logarithmic and $\sum_k \Delta_k^2$ converges. For R-delta the radius reaches $\approx 5 \times 10^{300}$ by $k = 999$, so $\sum_k \Delta_k^2$ overflows double precision: the entry $+\infty$ is a floating-point overflow, not a proof of divergence (though the series does diverge).

Rule	$\sum_{k < N} \Delta_k$	$\sum_{k < N} \Delta_k^2$
R-delta	1.072×10^{301}	$+\infty$
R-step	1.999×10^3	3.997×10^3
R-DFO	6.290×10^0	1.431×10^0
R-grad (μ bounded)	7.604×10^0	1.430×10^0
R-grad (μ unbounded)	6.928×10^2	4.800×10^2

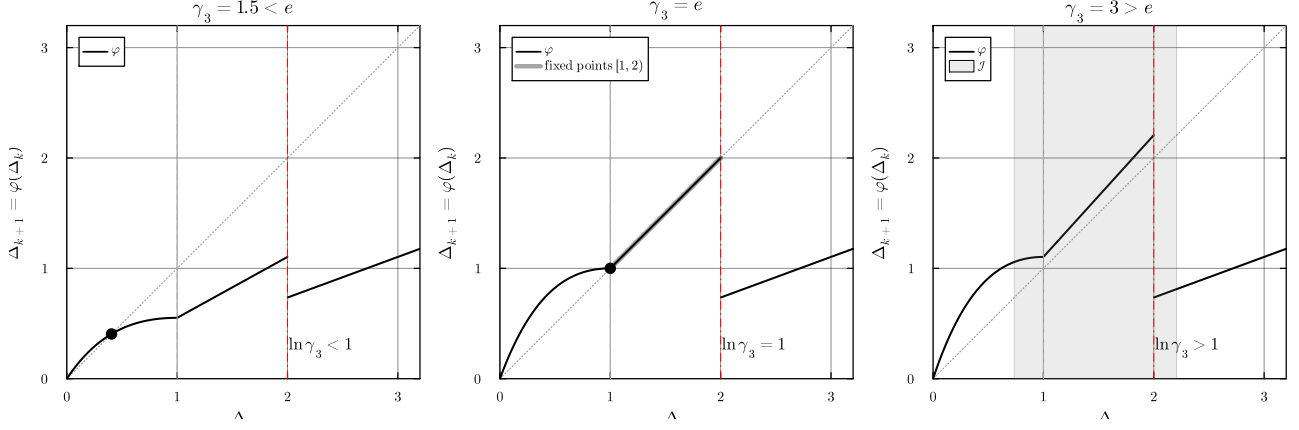


Figure 13: The map φ of (B.5) in the three regimes, with the diagonal $\varphi(\Delta) = \Delta$ shown dotted so that fixed points appear as crossings. Left, $\gamma_3 < e$: a unique attracting fixed point at $\ln \gamma_3$, towards which every trajectory converges. Centre, $\gamma_3 = e$: the expansion factor and the gradient decay over a unit step cancel, and every point of $[1, 2)$ is fixed. Right, $\gamma_3 > e$: no fixed point below the threshold, φ lies strictly above the diagonal there, and the discontinuity at $\Delta = 2$ returns the radius to the shaded invariant band $\mathcal{I}$, producing a persistent oscillation.

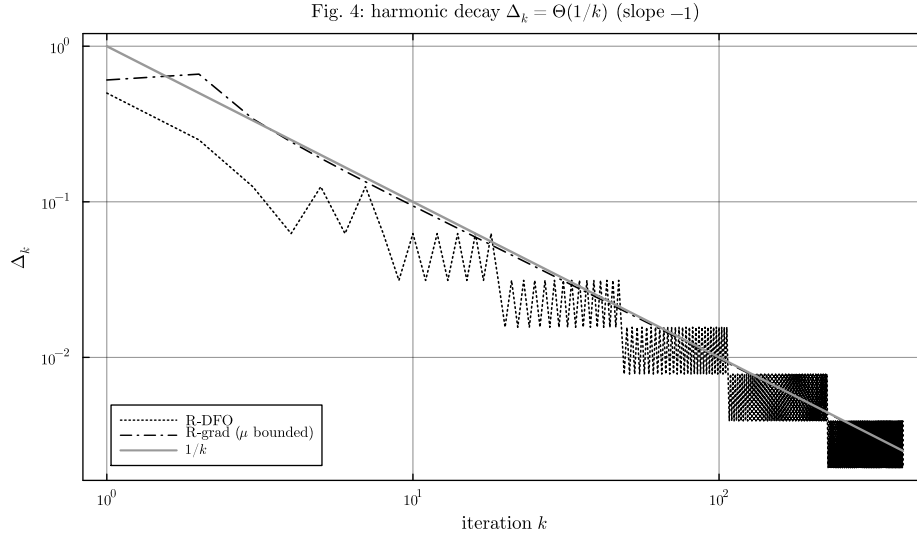


Figure 14: R-grad with μ bounded, on $f(x) = e^{-x}$, $x_0 = 0$. The radius decays harmonically, $\Delta_k = 1/k + o(1/k)$, so that $\sum_k \Delta_k = +\infty$ while $\sum_k \Delta_k^2 < +\infty$.

Fig. 5: $k\Delta_k \rightarrow 1$ for R-grad; $\Theta(1/k)$ band for R-DFO

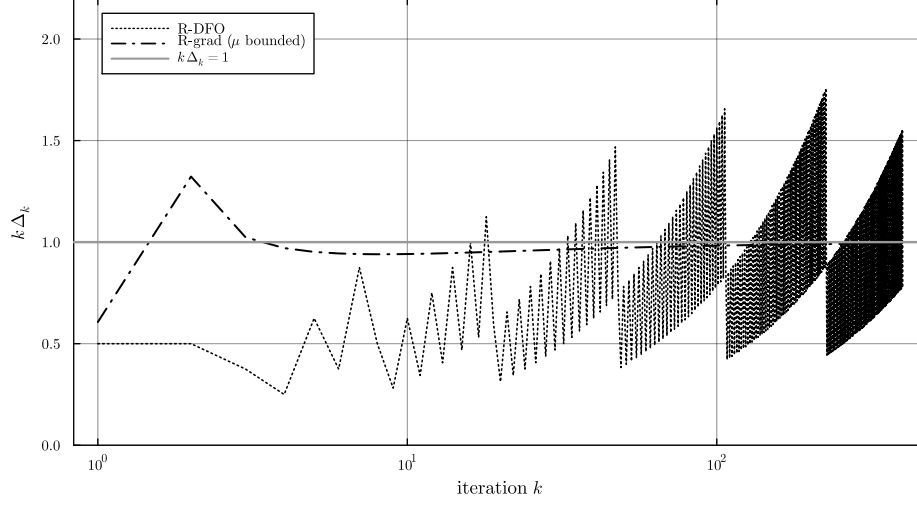


Figure 15: The rescaled radius $k\Delta_k$ on $f(x) = e^{-x}$. For R-grad the sequence converges to the limit 1 of Proposition B.12; for R-DFO it oscillates persistently within a bounded band, so $\Delta_k = \Theta(1/k)$ holds without $k\Delta_k$ having a limit.

Fig. 3: local regime, $f(x) = \frac{1}{2}x^2$, $H_k = 4$

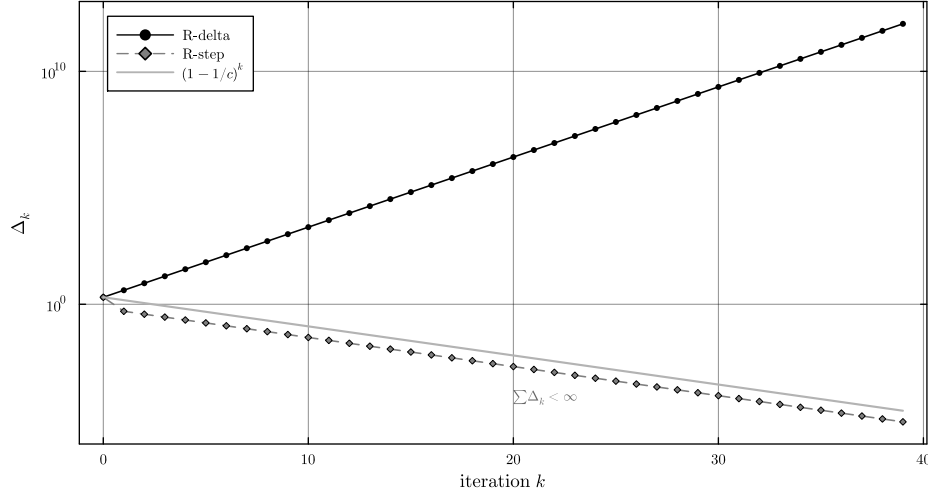


Figure 16: Local regime on $f(x) = \frac{1}{2}x^2$ with model Hessian $H_k = c = 4$ and $x_0 = 1$. R-step inherits the geometric contraction of the iterates at rate $1 - 1/c$, so $\sum_k \Delta_k = \Delta_0 + \gamma_3 x_0$ is finite; R-delta expands by γ_3 at every very successful iteration and diverges, leaving the trust-region constraint permanently inactive.